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USING LARGE-SCALE DATASETS TO ENHANCE DIAGNOSIS IN LIMITED MULTI-SPECTRAL SKIN IMAGING

BACHELOR THESIS

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ABSTRACT

Keywords: Computer Vision, Multimodal Learning, Transfer Learning, Medical Image Analysis, Multilayer Perceptron (MLP), Skin Lesion Classification.

Deep learning has shown remarkable progress in dermatological image analysis, but its clinical deployment is constrained by the limited size of available training data. Large public datasets such as HAM10000 consist almost exclusively of standard white-light RGB dermoscopic images, while proprietary clinical datasets, captured by multispectral devices under narrow-band illumination, remain comparatively small and difficult to scale.

This bachelor thesis addresses this gap through two complementary strategies for a four-class skin lesion classification task covering basal cell carcinoma, benign keratosis, melanoma, and melanocytic nevus. The first is a multimodal learning framework, MultiModalSkinNet, which integrates an EfficientNet-B0 image backbone with a lightweight metadata branch processing patient age, sex, and lesion localisation. The second is a multistage transfer learning strategy that progressively adapts a single backbone from ImageNet, through HAM10000, to the proprietary multispectral dataset.

The framework was evaluated using 5-fold stratified group cross-validation on both datasets. The results confirm that clinical metadata contributes a measurable and stable improvement over image-only baselines, and that domain-relevant intermediate pretraining is an effective strategy for transferring knowledge from large RGB skin datasets to smaller multispectral datasets.

The main part of this bachelor thesis comprises 76 pages and it includes 1 formula, 39 tables, 117 figures, 26 appendixes and 64 sources of reference.

ANOTĀCIJA

Atslēgvārdi: datorredze, multimodālā apmācība, pārnesamā apmācība, medicīnisko attēlu analīze, daudzslāņu perceptrons (MLP), ādas bojājumu klasifikācija.

Dziļā mācīšanās ir panākusi ievērojamu progresu dermatoloģisko attēlu analīzē, taču tās klīnisko ieviešanu ierobežo pieejamo mācību datu nelielais apjoms. Lielie publiski pieejamie datu kopumi, piemēram, HAM10000, sastāv gandrīz vienīgi no standarta RGB dermoskopijas attēliem, kas uzņemti baltajā gaismā, savukārt patentētie klīniskie datu kopumi, kas iegūti ar multispektrālajām ierīcēm šaurjoslas apgaismojumā, joprojām ir salīdzinoši nelieli un grūti paplašināmi.

Šajā bakalaura darbā šī nepilnība tiek novērsta, izmantojot divas savstarpēji papildinošas stratēģijas četru klases ādas bojājumu klasifikācijas uzdevumam, kas aptver bazālo šūnu karcinomu, labdabīgo keratozi, melanomu un melanocītisko nevusu. Pirmā ir multimodālā mācīšanās platforma MultiModalSkinNet, kas apvieno attēlu pamatstruktūru EfficientNet-B0 ar vieglāku metadatu atzaru, kas apstrādā pacienta vecumu, dzimumu un bojājuma lokalizāciju. Otrā ir daudzpakāpju pārneses mācīšanās stratēģija, kas pakāpeniski pielāgo vienu pamatstruktūru no ImageNet, izmantojot HAM10000, līdz patentētajai multispektrālajai datu kopai.

Modelis tika novērtēts, izmantojot 5-kārtīgu stratificētu grupu savstarpējo validāciju abos datu kopumos. Rezultāti apstiprina, ka klīniskie metadati nodrošina ievērojamu un stabilu uzlabojumu salīdzinājumā ar modeļiem, kas balstās tikai uz attēliem, un ka nozarei atbilstoša starpposma iepriekšēja apmācība ir efektīva stratēģija, lai pārnestu zināšanas no lieliem RGB ādas datu kopumiem uz mazākiem multispektrāliem datu kopumiem.

Šī bakalaura darba kopējais apjoms pašlaik ir 76 lappuses, un tajā ir iekļauta 1 formula, 39 tabulas, 117 attēli, 26 pielikumi un 64 atsauces.

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INTRODUCTION

This thesis focuses on an important issue in skin lesions analysis. Skin lesions are areas of skin that differ from the surrounding tissue in colour, size, shape, or texture, and although most of them are benign, they can in some cases be the first visible sign of skin cancer (Cleveland Clinic, 2022). Skin cancer affects all demographics regardless of nationality, socioeconomic status, geographical region, or age (Elgamal, 2013), and its incidence shows no signs of plateauing. It currently ranks as the 17th most frequently diagnosed cancer worldwide, and its detection at an early stage remains a central public health objective.

Because dermatology is a strongly visual specialty that relies on the identification of distinctive morphological characteristics and patterns observed in clinical and dermoscopic images, it has become a pioneering area for the application of artificial intelligence in diagnostic support (Bajaj, Marchetti et al., 2016; Møllersen, Kirchesch et al., 2015). Medical imaging is itself essential to modern dermatologic practice: it aids in clinician-to-clinician communication, provides documentation of skin diseases, helps assess treatment responses and plays a crucial role in the detection, diagnosis and monitoring of skin cancer (Hibler, Qi et al., 2016).

Despite this favourable visual nature of the problem, the deployment of deep learning systems for skin lesion analysis in clinical practice remains constrained by two persistent challenges related to the data on which the models are trained. The first challenge is the dominance of standard white-light RGB dermoscopic imaging in publicly available datasets. Large-scale public collections such as the ISIC Archive contain more than 500,000 dermoscopic images, but these are almost exclusively captured under broadband visible illumination and stored as conventional three-channel RGB tensors. While this modality is well-suited for surface-level visual inspection, it provides limited insight into the deeper biological characteristics of skin tissue, such as fluorophore distribution or sub-surface vascularisation, which are clinically relevant for distinguishing benign from malignant lesions. Multispectral narrow-band imaging, which records the lesion at several specific wavelengths including green, red, infrared, and ultraviolet excitation, addresses this limitation by capturing complementary spectral information beyond the visible RGB range. However, multispectral skin imaging datasets remain comparatively scarce, are typically collected manually in research settings, and rarely reach the scale needed to train deep learning models from scratch.

The second challenge is the size and homogeneity of the proprietary multispectral dataset available for this research. The dataset comprises images captured by a specialised multispectral imaging device under four narrow spectral bands including green (526 nm), red (663 nm), near-infrared (964 nm), and autofluorescence under 405 nm UV excitation, and contains approximately 1,774 lesion cases, which is several orders of magnitude smaller than the public RGB collections used in mainstream skin lesion research. Training a high-capacity deep learning model directly on a dataset of this size is known to produce unstable and easily overfitting classifiers.

Together, these two challenges motivate a transfer learning approach that bridges the gap between abundant public RGB data and the smaller proprietary multispectral dataset, while at the same time exploiting the structured clinical metadata available in public datasets to enrich the diagnostic decision. The central research question of this bachelor thesis can therefore be formulated as follows: how can knowledge learned from large public white-light RGB skin lesion datasets, such as HAM10000, be effectively transferred and adapted to improve the diagnostic performance of deep learning models trained on a smaller multispectral dataset, in the context of a four-class skin lesion classification task covering basal cell carcinoma, benign keratosis, melanoma, and melanocytic nevus.

The goal of the bachelor thesis:

Develop and evaluate a deep learning framework that combines multimodal learning and multistage transfer learning in order to improve diagnostic performance on a small proprietary multispectral skin imaging dataset, by leveraging knowledge learned from large public RGB dermoscopic datasets.

The tasks of the bachelor thesis:

1) Review dermatological imaging modalities, public and proprietary skin lesion datasets, and existing deep learning architectures relevant to skin lesion classification.
2) Compare existing AI models and methods relevant to skin lesion classification, with a focus on convolutional neural network and multilayer perceptron architectures, transfer learning strategies, and

multimodal fusion techniques applicable to small medical imaging datasets.

3) Design two deep learning architectures for skin lesion classification: MultiModalSkinNet, a multimodal architecture, and ImageOnlySkinNet, an image-only counterpart that serves as an ablation baseline to quantify the contribution of the metadata branch.
4) Train the proposed framework on the HAM10000 dataset and subsequently transfer it to the proprietary multispectral dataset using stratified group cross-validation.
5) Compare the experimental results obtained at each stage of the framework between the image-only and multimodal architectures on HAM10000, and between the ImageNet-only and the HAM10000-pretrained initialisations on the proprietary multispectral dataset, and validate HAM10000 proposed frameworks against existing literature.

Chapters:

This bachelor thesis is organised into four chapters. Chapter 1 provides an overview of dermatological imaging modalities, the application of deep learning to medical image and tabular data, the main challenges in existing skin lesion datasets, and the principles of transfer learning and multimodal learning relevant to this work. Chapter 2 reviews the public HAM10000 dataset and the proprietary multispectral dataset used in this thesis, together with the convolutional neural network and multilayer perceptron architectures considered as candidate building blocks for the proposed framework. Chapter 3 presents the methodology, including the baseline model design and training on HAM10000, the preprocessing pipeline, and the multistage transfer learning procedure applied to the proprietary multispectral dataset. Chapter 4 reports the experimental results obtained on both datasets, including 5-fold cross-validation, confusion matrix analysis, ROC analysis, and per-class metric analysis, and discusses the contribution of multimodal fusion and multistage transfer learning to the final diagnostic performance.

1. OVERVIEW OF SKIN LESION ANALYSIS AND AI IN DERMATOLOGY

In this chapter, the author of this bachelor thesis presents key concepts of dermatological imaging modalities (RGB imaging, and multispectral imaging). The chapter also discusses the application of deep learning in the analysis of medical images and tabular data. It discusses main challenges in existing datasets, and introduces approaches such as multistage transfer learning and multimodal learning that are relevant to improving diagnostic performance.

1.1. Dermatological Imaging Modalities

The thesis focuses on two different dermoscopic imaging modalities, including the standard white-light RGB imaging and the multispectral narrow-band imaging. The RGB imaging is able to capture the information of the visible spectrum through three color channels, while the multispectral imaging extends beyond the visible range to include more wavelengths that provide additional spectral information for skin lesion analysis.

Standard White-Light RGB Dermoscopic Images are captured under normal white light and stored as three-channel RGB images. RGB image is also called a true color image. In a true color image, each pixel is defined by three numerical values representing its red, green, and blue components of the pixel scalar (Kumar & Verma, 2010). Major dermatological datasets, such as the HAM10000 dataset, predominantly consist of RGB dermoscopic images.

Multispectral Narrow-Band Dermoscopic Images records image at different specific wavelengths commonly including green, red, infrared, and ultraviolet. Instead of one RGB image, it creates a series of grayscale images, with each one corresponding to an individual spectral channel (Nicolis & Gonzalez, 2021). Multispectral imaging has a lot of potential in medicine. Through spectral analysis, MSI can enhance image clarity and contrast in certain clinical applications. In practice, MSI can be used to capture the detailed spectral information of each pixel in an image which can precisely represent real objects regardless of the type of device or illumination change (Levenson & Mansfield, 2006). The proprietary multispectral imaging dataset used in this study was developed by Jumutc et al., 2026 and is described in detail in Section 2.2.

Nishibori et al. suggest that traditional color imaging, which relies on just three-color channels, often lacks the accuracy needed for reliable medical diagnosis. In specialty - dermatology, highly accurate color representation is essential, RGB imaging is unable to consistently achieve this level of precision because of differences in illumination. Multispectral imaging is an approach to overcoming these color-related limitations and has the potential to improve many areas of medical practice (Nishibori et al., 2004).

1.2. Deep Learning in Clinical Data Analysis

Deep learning is transforming healthcare (Naylor, 2018). Deep learning has been applied in many clinical sectors (Shaffie et al., 2018). This thesis focuses on two primary data modalities including medical images and structured tabular data, each playing distinct yet complementary roles in clinical decision support.

Medical image analysis represents the most common application of deep learning in healthcare. Robust medical image analysis requires large datasets (Kiryati & Landau, 2021). The deep learning revolution in 2012 has changed the landscape of medical image analysis especially for classification problems (Litjens et al., 2017). This revolution emphasizes the need for large-scale image datasets as deep learning models need large amounts of data to learn complex patterns and to achieve high levels of generalization capabilities. However, as research standards evolve, datasets that were sufficient in the past are no longer sufficient in today's standards, leading to limitations in model performance and reliability (Kiryati & Landau, 2021). This highlights a challenge in medical imaging including data availability and scale remain critical to the success of deep learning systems.

Tabular data is structured information organized into rows and columns. Each row represents a single subject, and each column corresponds to a specific feature (Krones et al., 2025). This format is widely used because of its simplicity and compatibility with various analytical methods.

In healthcare, tabular data are essential for structuring and analyzing patient records. For example, demographic data, including age, gender, and ethnicity, is often provided in tabular format; these attributes provide crucial contextual details that inform individualized diagnoses and therapeutic interventions (Ali et al., 2026; Xu et al., 2020).

The importance of tabular data extends across a wide range of medical applications. They are commonly used for disease prediction (Pölsterl et al., 2021), patient risk assessment (Bagheri et al., 2020), and outcome forecasting (Vanguri et al., 2022). Furthermore, the incorporation of tabular data with diverse data modalities frequently serves to improve diagnostic accuracy. As an illustration, tabular data has been integrated with MRI data to facilitate prostate cancer detection (Reda et al., 2018), with demographic and audio data for the detection of heart murmurs, and with demographic, imaging and time-series data for the diagnosis of Alzheimer's disease (El-Sappagh et al., 2020). In addition, tabular data has been combined with genomic, CT, and pathological imaging data to forecast treatment response in lung cancer (Vanguri et al., 2022).

1.3. Skin Lesion Classification and Challenges in Skin Lesion Datasets

Skin lesions are of paramount importance in the early diagnosis and subsequent treatment of various dermatological disorders. The recent progress in computer vision and machine learning has generated considerable interest in learning-based methodologies for the analysis of skin lesions. Employing machine learning and deep learning techniques, skin lesion classification endeavors to accurately determine the specific type of skin lesion presented in either dermoscopic or clinical images (Alquran & Girma Debelee, 2023). However, skin lesion datasets present several important challenges that significantly impact the development and performance of automated diagnostic systems.

A major issue lies in the difficulty of developing reliable image analysis methods for the detection of skin diseases. Accurately detecting, classifying, and segmenting skin lesions in dermatoscopic images remains complex due to differences in lesion appearance and imaging conditions (Esteva et al., 2017; Li et al., 2022). In addition, the lack of standardized and well-annotated datasets limits the ability to train and fairly evaluate machine learning models/algorithms, making dataset quality a crucial factor in models' performance (Nanni et al., 2023; Rasmussen et al., 2022; Tschandl et al., 2018). A significant challenge lies in the integration of heterogeneous clinical data, including patient histories and reported symptoms. While this data is advantageous for enhancing diagnostic precision, its effective incorporation into current models presents a considerable challenge (Esteva et al., 2017).

Conversely, numerous dermatological datasets suffer from intrinsic biases, specifically the under-representation of particular skin types and infrequent diseases, potentially resulting in inequitable and imprecise predictions (Rezk et al., 2022).

1.4. Transfer Learning

The author of this bachelor thesis has divided this section into three subsections to clearly describe the principles of transfer learning relevant to this work. The following subsections present the advantages and limitations of transfer learning in the medical imaging domain, the multistage transfer learning paradigm and its three-phase structure, and the specific transfer learning strategies and domain adaptation techniques adopted in this thesis.

1.4.1. Advantages and Limitations of Transfer Learning in Medicine

Medical image classification tasks are often confronted with data scarcity, class imbalance, and complex structures in specific domains. These challenges are indicative of the limitations of the traditional methods in medical image classification. Transfer learning is an efficient method to address these problems and improve the performance of medical image classification models (Das et al., 2023).

Transfer learning has great advantages in classifying medical images. First, it addresses the challenge of small medical datasets. Models pre-trained on large datasets have learned general features such as edges and textures which can be used in medical images to improve the generalization of the model.

Secondly, transfer learning can help to avoid overfitting. It supplies stable initial weights, which enhances model robustness when trained on small datasets and decreases the danger of overfitting.

Finally, transfer learning facilitates the handling of complex image features. Pre-trained models learn rich visual representations from large data sets and can be adapted to the various structures in medical images and thus improve the classification accuracy.

However, transfer learning is not without its constraints. A primary challenge arises from the domain disparity between natural and medical images; specifically, medical data demonstrate unique feature distributions. Consequently, the features learned from natural images might not always transfer well. While pre-training on medical datasets has been shown to improve performance in some studies, access to these datasets is often limited.

1.4.2. Multistage Transfer Learning (MSTL) in Medicine

Multistage Transfer Learning (MSTL) addresses some limitations of traditional transfer learning, such as domain difference and lack of domain specialization (Ayana et al., 2024).

Multistage transfer learning is used to enhance the efficiency of deep learning models in medical image analysis by using pretrained models (Ayana, Park, Jeong, et al., 2022). This method applies pretrained models on large nonmedical datasets, and then adapts the model to a specific medical imaging application, thus solving the common problem of limited annotated medical data.

While the traditional transfer learning is carried out by using a one-step transfer learning algorithm, multistage transfer learning broadens this approach by adding several sequential transfer stages (at least two steps of transfer learning), which enables knowledge adaptation to be more efficient (Ayana et al., 2024). A multistage transfer learning framework is divided into three phases, including pretraining, intermediate transfer, and final transfer. Each phase was designed to address a specific aspect of the targeted medical imaging task (Samala et al., 2017). Initially, in the pretraining phase, a deep learning model is trained with a large and diverse dataset, often sourced from general image repositories such as ImageNet to learn general features and patterns (Christodoulidis et al., 2017). Then, the intermediate transfer learning phase is the fine-tuning of the pretrained model on a smaller dataset more relevant to the medical field, helping to extract domain-specific features (Ayana, Park, & Choe, 2022). Intermediate datasets, which are related to medical images, provide initial weights for the next training phase, enriching the model with features similar to those of the target medical image dataset (Meng et al., 2022). Finally, the final transfer phase is the further fine-tuning of the model on the target medical dataset, with the goal of optimizing performance for the specific diagnostic task (Ayana, Park, & Choe, 2022; Meng et al., 2022).

The multistage architecture of the model allows a progressive evolution from general features to specific medical features. In particular, the weights learned on large datasets are progressively reused and updated across the stages, thus enhancing the representation of the features and reducing the risk of overfitting. This approach allows a better and progressive adaptation to the particularities of the medical images contrary to the classical single-stage transfer learning, capturing general and task-specific features and therefore improving the global performance and generalization (Ayana & Choe, 2022).

This thesis presents a multimodal framework that integrates a multistage transfer learning strategy within the image branch, starting with ImageNet pretraining, progressing through intermediate fine-tuning on the HAM10000 dataset, and culminating in final adaptation on the internal multispectral dataset.

The three-stage progression of this multistage transfer learning pipeline is illustrated in Fig. 1.1, which shows how knowledge is transferred from the original task on a large natural image dataset (ImageNet), through the intermediate task on dermoscopic RGB images (HAM10000), to the target task on the proprietary multispectral dataset acquired under narrow-band illumination.

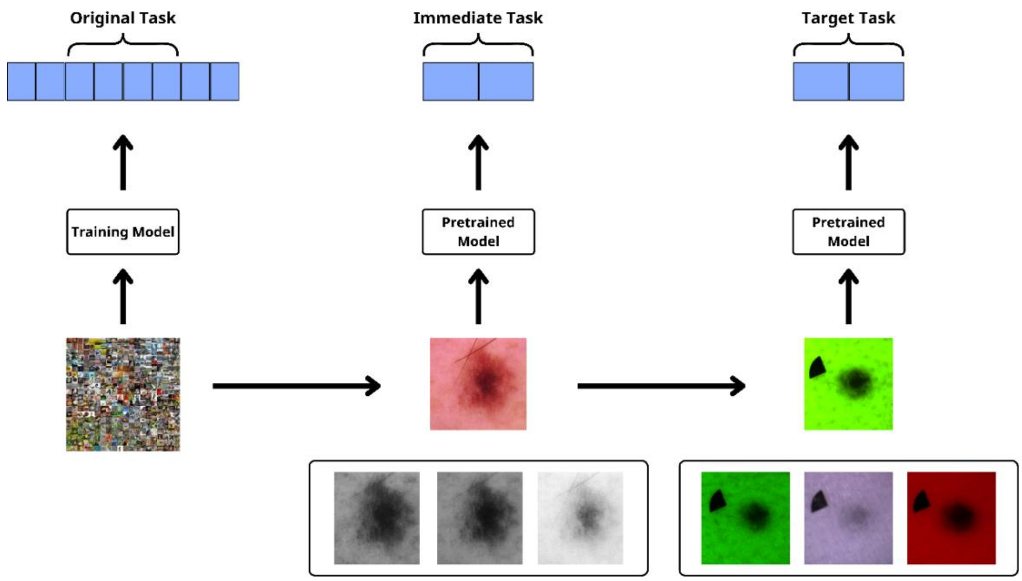

Fig. 1.1. Cross-Domain Knowledge Transfer from Natural Images to Medical Imaging

The concrete instantiation of this pipeline adopted in the present work is illustrated in Fig. 1.2. Model A is trained on the ImageNet-1K dataset for 1000-class natural image classification, producing Weight A. Model B is then initialised with Weight A and fine-tuned on the HAM10000 white-light RGB dermoscopy dataset for skin lesion classification, producing Weight B. Finally, Model C is initialised with Weight B and fine-tuned on the proprietary multispectral narrow-band dataset for the final skin cancer classification task, so that the dermatology-specific representations learned at the intermediate stage are progressively adapted to the multispectral acquisition modality of the target domain.

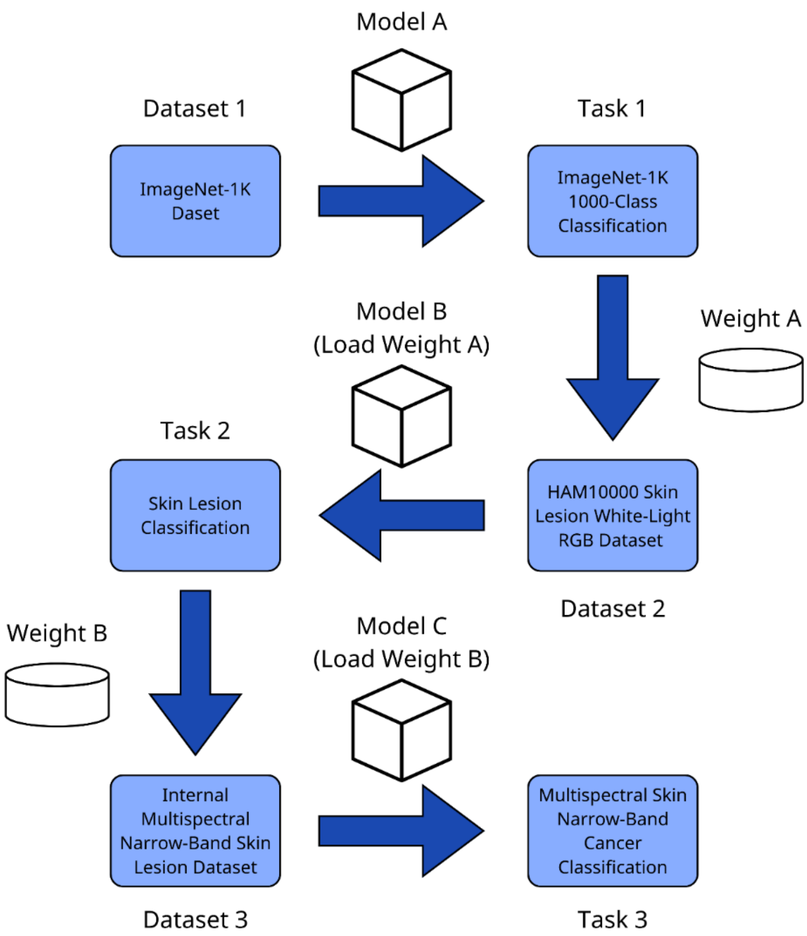

Fig. 1.2. Multistage Transfer Learning for Skin Lesion Classification

1.4.3. Transfer Learning Strategies and Domain Adaptation Techniques

Transfer learning requires designing an appropriate strategy to transfer previously acquired knowledge to the target task. This involves figuring out which layers need to be fine-tuned or trained and how to adjust the weights to fit the new task (Dawud et al., 2019).

To apply a pre-trained model for a new target task, the original classifier is first removed and replaced with a new classifier suited to the target task, followed by fine-tuning the model. Based on how similar the source and target datasets are, as well as the size of the target dataset, in this thesis, a two-phase progressive fine-tuning approach is applied across the multistage transfer learning pipeline.

1.5. Multimodal Learning in Medical AI

By learning from multimodal data, models can identify intermodal relationships, thus enabling a more comprehensive understanding.

Machine learning applications in healthcare have to date been limited to single modality datasets, which does not allow them to imitate the practice of clinicians who integrate different sources of information to improve their decision-making processes. Clinicians generally rely on different sources of data, such as demographic data, laboratory results, vital sign measurements, and imaging to support their clinical judgments (Krones et al., 2025).

To address this limitation, recent machine learning approaches have made it possible to integrate multimodal healthcare data, combining imaging data with clinical metadata such as demographic and clinical data to better reflect real-world clinical judgments (Krones et al., 2025).

Despite its potential, multimodal machine learning introduces several fundamental challenges. A first key challenge lies in learning how to effectively represent and summarize multimodal data in a way that leverages both the complementarity and redundancy across modalities. The heterogeneous nature of multimodal data poses challenges for constructing such representations (Baltrusaitis et al., 2017). As an illustration, tabular clinical metadata is generally symbolic and structured, while dermoscopic images are represented as pixel signals.

A second obstacle involves the integration of multimodal data to achieve precise predictions. Nevertheless, modalities can exhibit considerable disparities in predictive power, noise profiles, and data accessibility, with at least one of the modalities potentially having missing data (Baltrusaitis et al., 2017).

To address these issues, multimodal models typically learn a joint representation that integrates the information from multiple modalities. One common strategy is joint representation in which features from different modalities are fused into a single representation:

$$x_m = f(x_{image}, x_{metadata})$$

x_m: multimodal representation

f: fusion function as a multi-layer perceptron

$x_{image} \in R^d_{image}$: CNN-extracted image features

$x_{metadata} \in R^d_{metadata}$: Processed clinical metadata features

In the intermediate transfer stage (HAM10000), both image and metadata features are used. The joint representation is formed by concatenating CNN-extracted image features with processed clinical metadata, which are then mapped to the target class space through a classification head.

In the final transfer stage (Multispectral dataset), due to the absence of clinical metadata, only image features are utilized for classification. The complete architecture for both stages is detailed in Section 3.1.4 and Section 3.2.4.

2. ANALYSIS OF DATASETS AND EXISTING DEEP LEARNING APPROACHES

In this chapter, there is an analysis of the datasets and the state-of-the-art deep learning techniques relevant to this work. It begins with a description of the public dataset used for training, the HAM10000 dataset and then it describes the internal multispectral skin imaging dataset. This chapter analyzes the deep learning architectures used in this thesis, namely Convolutional Neural Networks and Multi-Layer Perceptrons. In particular, it focuses on EfficientNet-B0 as a base for the image processing part and explains why a multimodal learning approach has been used.

2.1. Public Datasets For Skin Lesion Analysis - HAM10000 (Human Against Machine With 10000 Training Images)

The HAM10000 dataset utilized in this thesis is accessible to the public. The official version can be found in the Harvard Dataverse. Additionally, a version that is easier to use and integrate into machine learning workflows is available on Kaggle.

The HAM10000 dataset, which includes a large collection of dermoscopic images of common pigmented skin lesions, was gathered from various sources. According to the documentation for the HAM10000 dataset, the images are collected from a wide range of populations and represent all major diagnostic categories of pigmented skin lesions (Tschandl et al., 2018). However, one study points out that the HAM10000 dataset suffers from imbalance in its representation of skin lesions, especially due to the limited number of images from individuals with darker skin tones (Morales-Forero et al., 2024). The HAM10000 dataset contains 10,015 images, each with a size of 600×450 pixels (Alam et al., 2022). Over 50% of lesions have been validated through pathology, while the ground truth for the remaining instances was established via follow-up, expert consensus, or in-vivo confocal microscopy confirmation (Tschandl et al., 2018). HAM10000 dataset has a highly imbalanced class distribution (Mikołajczyk & Grochowski, 2018; Sabanci et al., 2022). To clarify this imbalance, Table 2.1 summarizes the seven diagnostic categories included in HAM10000, together with their clinical meanings, lesion-level counts, and image-level counts.

Table 2.1

Overview of the Diagnostic Classes (dx) in the HAM10000 Dataset with corresponding Clinical Meanings and Sample Counts

Class	**Meaning**	**Number of Lesions (%)**	**Number of Images (%)**
akiec	Actinic keratosis and intraepithelial carcinoma / Bowen's disease	228 (3.05%)	327 (3.27%)
bcc	Basal cell carcinoma	327 (4.38%)	514 (5.13%)
bkl	Benign keratosis (solar lentigines / seborrheic keratoses and lichen-planus like keratoses)	727 (9.73%)	1099 (10.97%)
df	Dermatofibroma	73 (0.98%)	115 (1.15%)
mel	Melanoma	614 (8.22%)	1113 (11.11%)
nv	Melanocytic nevus	5403 (72.33%)	6705 (66.95%)
vasc	Vascular lesion (angiomas, angiokeratomas, pyogenic granulomas, and hemorrhage)	98 (1.31%)	142 (1.42%)
Total		**7470 (100%)**	**10015 (100%)**

Representative dermoscopic images of the seven diagnostic classes in the HAM10000 dataset are shown in Fig. 2.1.

Fig. 2.1. Representative dermoscopic images of the seven diagnostic classes in the HAM10000 dataset: bcc, df, akiec, mel, bkl, vasc, and nv

The HAM10000 dataset contains multiple dermoscopic images for some lesions, captured from different perspectives and zoom levels, which necessitates lesion-level data splitting to prevent data leakage. An example of this multi-view characteristic is shown in Fig. 2.2.

Fig. 2.2. Multiple dermoscopic images of the same lesion (HAM_0000002) captured from different angles and zoom levels

Although the original dataset consists of seven classes (Tschandl et al., 2018), this thesis focuses on four classes: mel, nv, bcc, and bkl. This selection is motivated by two main reasons. Firstly, these classes have relatively higher sample counts leading to a more balanced class distribution for training than the excluded classes (akiec: 327 images, df: 115 images, vasc: 142 images). Secondly, they fit better with the diagnostic classes of the internal multispectral dataset.

In the original HAM10000 dataset, there are 10,015 samples corresponding to 7,470 unique lesions. After choosing the target diagnostic labels (dx) limited to 4 classes including basal cell carcinoma (bcc), benign keratosis (bkl), melanoma (mel), and melanocytic nevus (nv), the dataset is reduced to 9,431 samples representing 7,071 unique lesions. This filtering step not only aligns the dataset with the specific classification objectives of this study but also improves class balance and consistency with the internal multispectral dataset. The effect of this class filtering on the image-level distribution is shown in Fig. 2.3. The corresponding lesion-level distribution is shown in Fig. 2.4.

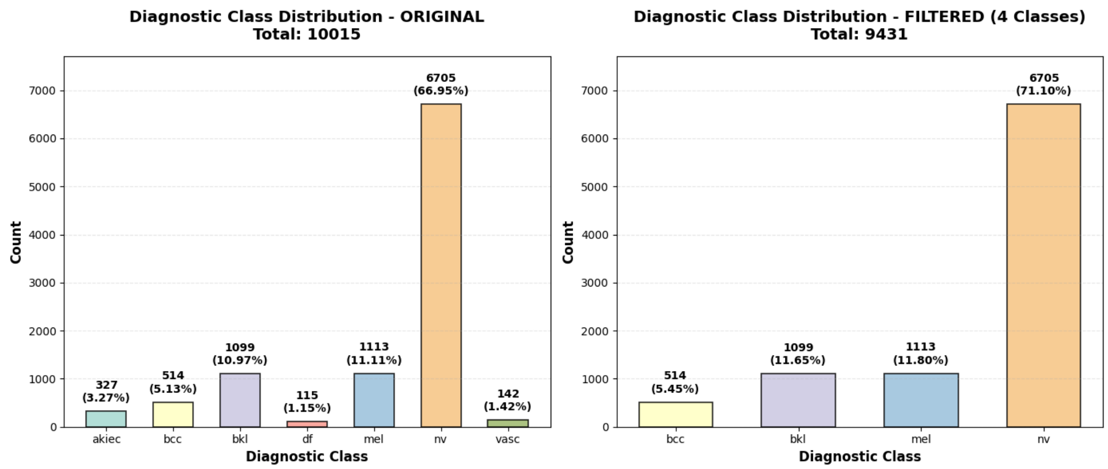

Fig. 2.3. Diagnostic Class Distribution Before and After Class Filtering in the HAM10000 Dataset by Image Count

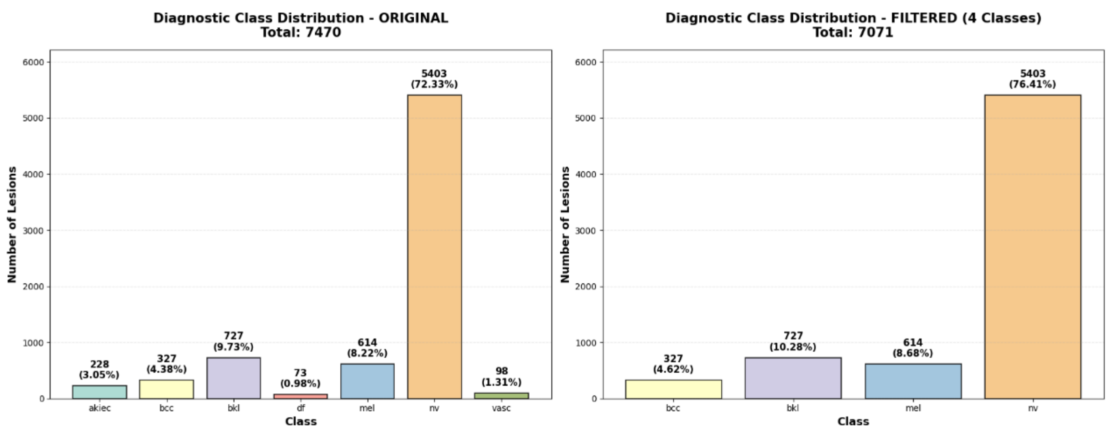

Fig. 2.4. Diagnostic Class Distribution Before and After Class Filtering in the HAM10000 Dataset by Lesion Count

Detailed demographic and anatomical distribution analyses of the HAM10000 dataset before and after class filtering are provided in Appendix 2 (Figures A.2.1-A.2.4).

For the metadata of HAM10000, it comprises seven features. Each attribute represents different aspects of the lesion and associated patient information (Appendix 1 - Table A.1.1).

Regarding the image data, the HAM10000 dataset consists of dermoscopic images acquired under standard white-light conditions, stored as RGB images.

2.2. Proprietary Multispectral Skin Imaging Dataset

The multispectral dataset used in this study was provided by the Institute of Applied Computer Systems, Riga Technical University, Latvia, and the Institute of Atomic Physics and Spectroscopy, University of Latvia, Latvia (Jumutc et al., 2026). The dataset is continuously updated with new lesions. Each lesion was captured using a portable multispectral device equipped with narrow-band LEDs centered at 405 nm (autofluorescence excitation), 526 nm (green reflectance), 663 nm (red reflectance), and 964 nm (near-infrared reflectance). A long-pass filter was placed to block reflected 405 nm excitation light from the skin surface.

The author of this bachelor thesis has divided this section into three subsections to clearly describe the main aspects of the proprietary multispectral dataset. The following subsections present the imaging modalities, the metadata structure provided together with the images, and the final dataset composition used for model training.

2.2.1. Imaging Modalities

The device captures images across the following spectral bands:

G (Green, 526 nm): A grayscale image captured under green light, sensitive to superficial skin structures.

IR (Infrared, 964 nm): A grayscale image captured under infrared light, penetrating deeper skin layers.

R (Red, 663 nm): A grayscale image captured under red light, providing complementary spectral information to G and IR.

White: A white light image representing standard visible light illumination.

UV and UV10-UV29 (405 nm): A series of ultraviolet images captured at sequential time intervals following UV exposure, representing the fluorescence decay process of skin tissue over time.

BAD_G, BAD_R, BAD_IR: Quality-flagged or corrupted captures in Green, Red, and Infrared bands, typically excluded from processing.

Visual examples of the five spectral bands (G, IR, R, UV, White) and the fluorescence decay sequence (UV10-UV29) are shown in Appendix 3, Figures A.3.1 and A.3.2, respectively.

For the purpose of this study, only the G, IR, and R channels are used as model inputs, as they represent the primary spectral modalities of the device. The UV fluorescence decay frames and White light images are noted as potential features for future work.

To minimize optical artifacts, the imaging system utilized a cross-polarization configuration, with illumination source polarization orthogonal to the camera lens polarizer. This setup suppressed specular reflection from the skin surface, enabling reliable measurement of diffusely reflected and fluorescent signals. All images were captured using an IDS uEye color camera (Jumutc et al., 2026).

2.2.2. Metadata Structure

Metadata was provided in two formats: metadata.csv/metadata2.csv and search-res-info.csv, stored within lesion-specific folders (Appendix 5, Table A.5.2).

Metadata CSV Files

Metadata folders were present for 1,371/1,774 lesions (77.3%) (Appendix 5, Table A.5.3). Each folder contained one or two CSV files:

metadata.csv: Found in 1,371/1,371 lesions (100% of metadata folders)

metadata2.csv: Found in 752/1,371 lesions (54.9% of metadata folders)

Total metadata CSV files: 2,123

Both metadata files share an identical 27-column structure (Appendix 5, Table A.5.5), covering measurement information, device details, patient records, and clinical annotations.

Metadata was excluded from training for three reasons. First, metadata coverage was incomplete: only 1,371/1,774 lesions (77.3%) had metadata folders. Second, critical diagnostic fields had extremely high missing rates: histology codes (82.28%), clinical codes (80.38%), and staging information (>91%) were largely absent (Appendix 5, Table A.5.5). Third, available complete fields lacked diagnostic value: the columns with complete or near-complete data, such as measurementId, deviceId, deviceName, and measurement timestamps, contained only technical identifiers and device metadata with no clinical or morphological information useful for lesion classification. Even fields with moderate completeness, such as patientDescription and analyzeDoctorComment, consisted of unstructured free-text annotations without standardized diagnostic labels. Consequently, metadata.csv and metadata2.csv were excluded from the training pipeline.

Search-Resolved Information Files

All 1,371 metadata folders contained search-res-info.csv, a 12-identical-column file (Appendix 5, Table A.5.4).

Available complete fields lacked diagnostic value: the four columns with <10% missing data: codes (ICD-10 disease category), new name (lesion identifier), exp type in md1 (metadata format flag), and typo (quality control flag) provided no clinical or morphological information useful for lesion classification. Even fields with moderate completeness, such as patient desc (23.41% missing) and analyze doc comm (8.61% missing), suffered from inconsistent free-text formatting. Consequently, search-res-info.csv was excluded from training.

2.2.3. Dataset Composition

The dataset comprises 1,774 lesions across four diagnostic classes: bcc (basal cell carcinoma), bkl (benign keratosis), mel (melanoma), and nevus (melanocytic nevus), corresponding to the bcc, bkl, mel, and nv categories in HAM10000. Class distribution and channel availability are shown in Appendix 5 - Table A.5.1.

Lesions lacking at least one of the three primary channels (R, G, IR) were excluded from training. Files marked with the prefix BAD_ indicated acquisition or image-quality issues and were treated as missing channels. Although some BAD-flagged images remained visually interpretable, they were excluded to maintain consistent data quality. This filtering removed 9 lesions (1,774 to 1,765).

Subsequently, manual inspection identified 3 additional lesions with severe image quality issues (Appendix 4, Figure A.4.1 - A.4.3), including complete channel dropout or extreme noise. After removing these cases, 1,762 lesions (99.32% of the original dataset) were retained for training.

2.3. Existing Deep Learning Architectures for Skin Lesion Diagnosis

This thesis presents a multimodal deep learning framework for diagnosing skin lesions. It includes two connected parts: an image branch and a metadata branch. The image branch, which analyzes dermoscopic images, uses a Convolutional Neural Network (CNN), specifically EfficientNet-B0. At the same time, the metadata branch processes structured clinical data, employing a Multi-Layer Perceptron (MLP). The final classification result is obtained by combining the outputs from both branches. The following subsections offer a comprehensive account of each architectural element.

2.3.1. Convolutional Neural Network - CNN

CNNs are comprised of neurons that self-optimize through learning (O'Shea & Nash, 2015). A CNN, frequently referred to as a ConvNet, is characterized by its deep feed-forward architecture and demonstrates a remarkable capacity for generalization (Indolia et al., 2018; Neubauer, 1998).

This architectural design allows for the learning of highly abstract features, which helps in identifying objects effectively. Therefore, CNNs are widely used in pattern recognition, particularly in image processing. This approach facilitates the integration of image-specific characteristics into the network's design, thereby enhancing its suitability for image-centric tasks and simultaneously decreasing the number of parameters needed for model initialization (O'Shea & Nash, 2015).

CNNs are especially effective in image classification, a core computer vision task. They have exhibited superior performance on datasets such as ImageNet.

In the domain of medical image analysis, CNNs can be trained using large datasets of medical images, to detect subtle patterns and irregularities indicative of disease, thus improving diagnostic accuracy and efficiency.

CNNs are generally composed of multiple layer types including convolutional layers for feature extraction, pooling layers for spatial dimensionality reduction, and fully connected layers for classification (Afshine Amidi & Shervine Amidi, 2018).

The pre-trained models are mostly selected from deep learning models trained on ImageNet-1K. ImageNet-1K is a dataset of 1.2 million images of 1,000 different classes. Selecting a model that performs well in the pre-training task and learns general features is important. Selecting the right pre-trained model can affect the performance of transfer learning, because good initial weights can speed up the learning process of model on the target task (Chen et al., 2025). Besides that, the choice of a transfer learning model is also influenced by task complexity, dataset size, computational resources, and domain characteristics (Chen et al., 2025).

While conventional CNN architectures commonly scale only one of three dimensions (depth, width, or image resolution), arbitrary scaling of multiple dimensions requires tedious manual tuning and often yields sub-optimal accuracy and efficiency (Tan & Le, 2019). To address this limitation, this thesis adopts the EfficientNet architecture, which balances all three dimensions simultaneously through a principled compound scaling method, achieving both accuracy and efficiency with a constant scaling ratio (Tan & Le, 2019).

The selection of a suitable backbone architecture for transfer learning in the field of medical image classification is a trade-off between the model capacity, computational efficiency and the task-specific demands. The author selects EfficientNet-B0 as a candidate architecture for this thesis based on the following criteria:

EfficientNet-B0 is a good trade-off option between the model complexity and the computational requirements. It has a total of 5.3M parameters, which allows to perform faster training iterations and hyper-parameter optimization with competitive performance. This efficiency is particularly important for multi-stage transfer learning pipelines and resource-constrained clinical deployment scenarios.

EfficientNet architectures pre-trained on ImageNet have demonstrated strong feature extraction ability for texture and pattern recognition which are important features for dermoscopic image analysis. The efficiency of EfficientNet in medical image tasks has been confirmed in previous study. Mahbod et al. (2020) achieved 96.30% accuracy on ISIC datasets with an ensemble approach combined EfficientNet-B0, EfficientNet-B1 and SeResNeXt-50 effectively handling data imbalance and limited sample sizes that are also challenges in HAM10000.

In preliminary experiments, the author of this thesis compared the performance of EfficientNet-B0 and EfficientNet-B3 for the same training configurations. EfficientNet-B3 (12M parameters, 300×300 input) outperformed EfficientNet-B0 (5.3M parameters, 224×224 input) by a margin of approximately 4% in terms of metrics (weighted F1-score, accuracy). This improvement is indicative of B3's improved capacity to extract richer features from dermoscopic images, especially in the context of processing separated channels and clinical metadata. However, training EfficientNet-B3 with the multi-stage transfer learning pipeline (ImageNet - HAM10000 - target multispectral dataset) exceeded available GPU memory constraints. Considering the moderate gap in performance and resource constraints, EfficientNet-B0 was chosen as the final architecture.

The selection of a backbone architecture for transfer learning depends on the complexity of the task, the nature of the dataset and the computational power at disposal. For medical imaging, lightweight architectures like EfficientNet, DenseNet121, ResNet18 and MobileNet proved successful with simpler feature structures (e.g. X-ray classification) and deeper models like VGG, ResNet50 and DenseNet201, GoogLeNet are preferable for detection of complex boundaries in MRI/CT data (Chen et al., 2025). Skin lesion classification, where dermoscopic images display two-dimensional surface patterns (texture, color, morphology) unlike complex volumetric structures. Lighter architectures are capable of effectively capturing these discriminative features based on texture, without the computational cost of deeper networks.

EfficientNet-B0 offers a practical trade-off among performance, trainability, and deployment feasibility for our multi-stage transfer learning approach, which makes it the appropriate choice for this thesis.

The modified EfficientNet-B0 architecture adopted in this thesis, with its first convolutional layer adapted to accept a single-channel grayscale input, is illustrated in Fig. 2.5.

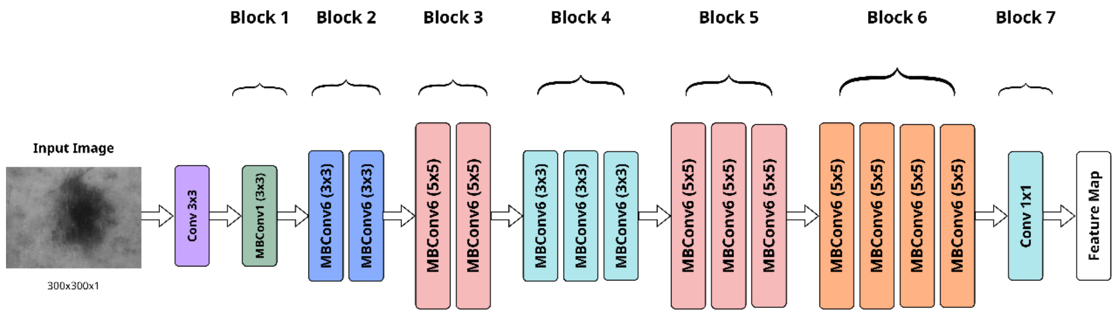

Fig. 2.5. Modified EfficientNet-B0 Architecture for Single-Channel Input Processing

2.3.2. Multi-Layer Perceptron - MLP

An MLP is a feedforward neural network. The MLP consists of several layers of interconnected nodes where each node is fully connected to all nodes in the next layer.

In this thesis, the MLP is used to process and integrate structured metadata about patient and lesion attributes such as age, sex and the lesion’s anatomical location, that are commonly used in dermatological assessments and practices. The metadata are essential since their absence could reduce the model’s ability to generalize and correctly classify rare/edge cases.

The features (sex, age, localization) are mapped to a meaningful embedding by applying multiple fully connected layers including Layer Normalization after the first linear layer, rectified linear unit (ReLU) activation and dropout regularization. The resulting embedding thus preserves important clinical relationships leading to accurate skin cancer classification when combined with image features (Pacheco & Krohling, 2021).

Unlike the image branch which utilizes a pretrained EfficientNet-B0 backbone, the metadata branch is trained from scratch, as no pretrained model exists for structured clinical tabular data. The specific architecture details, including layer dimensions and fusion strategy, are presented in Chapter 3.

The classification output in this thesis is computed using CrossEntropyLoss that has the softmax function built in, and it works for multi-class classification. The training of the network is based on the backpropagation method, and the gradient descent optimization method that adjusts the step size in the training process.

This thesis uses the Adam optimizer (Kingma & Ba, 2014), an often used adaptive step size optimization method.

3. TRANSFER LEARNING METHODOLOGY FOR MULTISPECTRAL SKIN DIAGNOSIS

This chapter presents the transfer learning methodology adopted in the multispectral skin lesion diagnosis setting. First, the chapter describes in detail the training process carried out on the HAM10000 public dataset, starting from data preparation and preprocessing techniques, to the two model architectures developed (MultiModalSkinNet and ImageOnlySkinNet) and the training strategy adopted. Then, the chapter describes the process of transferring to the internal multispectral dataset, exploiting a multistage transfer learning approach to progressively adapt the pretrained representations from ImageNet through HAM10000 to the target multispectral dataset.

3.1. Baseline Model Training on HAM10000

This section provides an overview of the pipeline that used the HAM10000 dataset to train the model. This section is organised into five subsections, each describing one stage of the baseline training pipeline on HAM10000. The objective of this step is to build a robust and generalizable baseline model using traditional dermoscopic RGB images, which can be subsequently transferred and adapted to the internal multispectral dataset. The central component of the system is the proposed multimodal architecture, MultiModalSkinNet. It combines visual features from dermoscopic images with structured clinical metadata. A shared EfficientNet-B0 backbone is used to extract deep image representations from decomposed RGB channels, while a separate multilayer perceptron (MLP) processes metadata features. These two modalities are fused through a late fusion strategy to produce a unified representation for classification.

The complete training pipeline on the HAM10000 dataset, covering data collection, image and metadata preprocessing, 5-fold stratified group splitting, model architecture, two-phase training, and evaluation, is illustrated in Fig. 3.1.

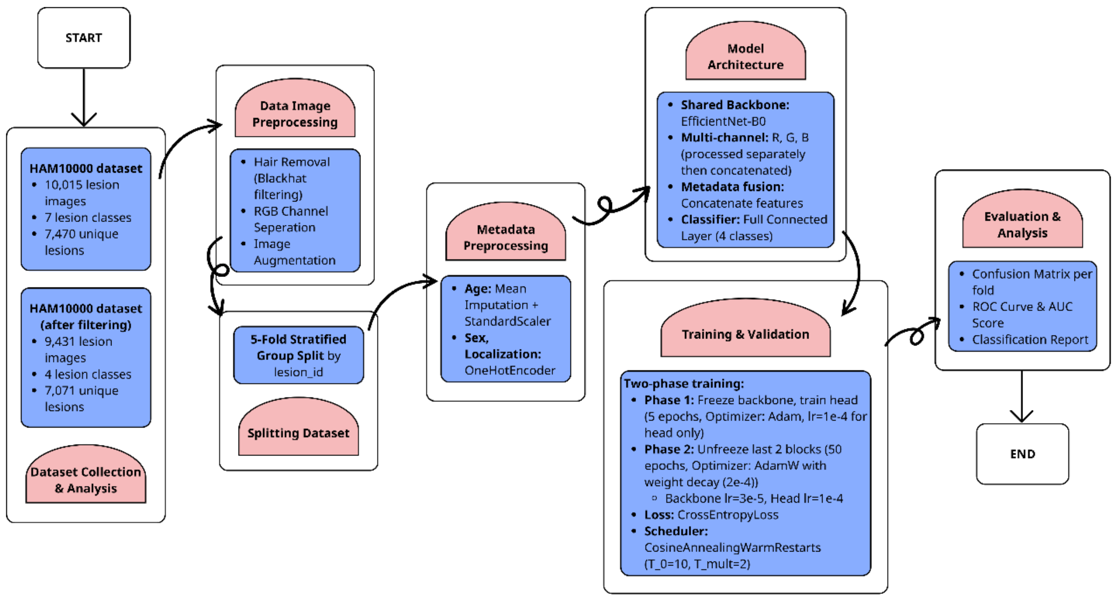

Fig. 3.1. Complete Pipeline for Skin Lesion Classification Using HAM10000 Dataset

3.1.1. Training Setup

The source model was developed and trained using Google Colab, a cloud-based Jupyter notebook environment providing free access to GPU resources. This platform was selected for initial model development due to its accessibility and availability of high-performance computing resources for publicly available datasets.

The cloud infrastructure provided an NVIDIA A100-SXM4-80GB GPU with 85.1 GB of memory, CUDA version 12.8, and compute capability 8.0. The system was equipped with an Intel Xeon CPU at 2.20GHz with 12 cores, supported by 167.05 GB of total RAM with 163.61 GB available during training. The operating system was Linux kernel 6.6.122+ running on x86_64 architecture (Linux-6.6.122+-x86_64-with-glibc2.35).

3.1.2. Data Preparation

Following one of the common splitting protocols for HAM10000 in recent dermoscopic lesion studies, this thesis uses a 5-fold cross-validation method. In each fold, about 80% of the lesions were used for training, and the remaining 20% were used for validation (APURVA, 2023).

To assess performance, Stratified 5-Fold Cross-Validation was employed. This technique preserves the proportional distribution of classes across all folds, which is essential for datasets with class imbalance to reduce evaluation bias and improve model generalizability (Abuhammad et al., 2025).

A single fixed stratified group-wise splitting methodology was implemented, leveraging the StratifiedGroupKFold function from the scikit-learn library. To ensure experimental uniformity, a constant random seed (random_state = 42) was utilized, thus guaranteeing the reproducibility of the splits throughout all experimental trials.

To prevent data leakage, the dataset was split at the group level using lesion_id as the grouping variable. Given that a single lesion might be represented by multiple dermoscopic images, all images sharing the same lesion_id were aggregated and assigned a singular diagnostic label. Stratified sampling was used at the lesion level to maintain the same class distribution across the subsets.

To summarize the resulting split sizes, Tables 3.1 and 3.2 report the overall distribution of samples and unique lesions across the five folds. Table 3.1 presents the image-level distribution, where the number of training and validation samples varies slightly across folds because some lesions are represented by multiple images. Table 3.2 presents the lesion-level distribution, showing that the group-wise splitting procedure maintained a nearly equal number of unique lesions in each validation fold.

Table 3.1.

Distribution of samples across training sets, validation sets across five folds in HAM10000 dataset

Dataset	Fold 1	Fold 2	Fold 3	Fold 4	Fold 5
Training Set	7585	7507	7531	7560	7541
Validation Set	1846	1924	1900	1871	1890

Table 3.2.

Distribution of lesions across training sets, validation sets across five folds in HAM10000 dataset

Dataset	Fold 1	Fold 2	Fold 3	Fold 4	Fold 5
Training Set	5656	5656	5657	5657	5658

Validation Set	1415	1415	1414	1414	1413

Detailed per-class distributions for both training and validation sets across all folds are provided in Appendix 6 and Appendix 7, respectively.

3.1.3. Data Preprocessing Pipeline

The data preprocessing pipeline consists of two parallel streams, including metadata preprocessing and image preprocessing, which are customized to the specific characteristics and requirements of the two modalities.

Metadata Preprocessing

Three metadata features were selected based on the results of similar studies and the exploratory analysis of the dataset (Adebiyi et al., 2024; Atiq & Fattah, 2025) including age, sex and localization. The dx feature was used as the target label for classification.

Image Path Mapping and Dataset Cleaning:

To ensure data integrity in the multimodal dataset, an image path mapping and dataset cleaning procedure was conducted. The “image_id” attribute was used to retrieve the corresponding dermatoscopic images from two separate directories. A mapping function was designed to verify the existence of each image file and assign its path to a newly created “image_path” column in the metadata table. Entries for which no valid image file could be found were excluded from further analysis. This step guaranteed alignment between image data and metadata, thereby reducing inconsistencies and preventing potential errors during model training (Naveen & Khanam, 2025).

Label Encoding:

The diagnostic categorization of each lesion, the target feature (dx) was transformed into numerical format using label encoding.

A mapping dictionary was defined to assign each skin lesion category a unique integer value, as follows:

- bcc: 0
- bkl: 1
- mel: 2
- nv: 3

One-Hot Encoding:

The categorical features (sex and localization) were transformed by one-hot encoding for compatibility with the machine learning models (Furqan, Katuk et al., 2026).

The encoder was fitted only on the training fold and then applied to the validation fold within each fold to prevent data leakage across folds. In particular, the sex feature was encoded into two binary variables: sex_female and sex_male. Also, the localization variable was expanded into several binary variables, each representing a different anatomical site. The binary variables were localization_abdomen, localization_acral, localization_back, localization_chest, localization_ear, localization_face, localization_foot, localization_genital, localization_hand, localization_lower extremity, localization_neck, localization_scalp, localization_trunk, and localization_upper extremity.

This transformation resulted in the expansion of the three original features into a total of seventeen features. The final feature set includes age, and all the encoded binary variables that were derived from sex and localization.

The resulting metadata preprocessing flow and the corresponding two-layer MLP that transforms the encoded features into a 32-dimensional metadata representation are illustrated in Fig. 3.2.

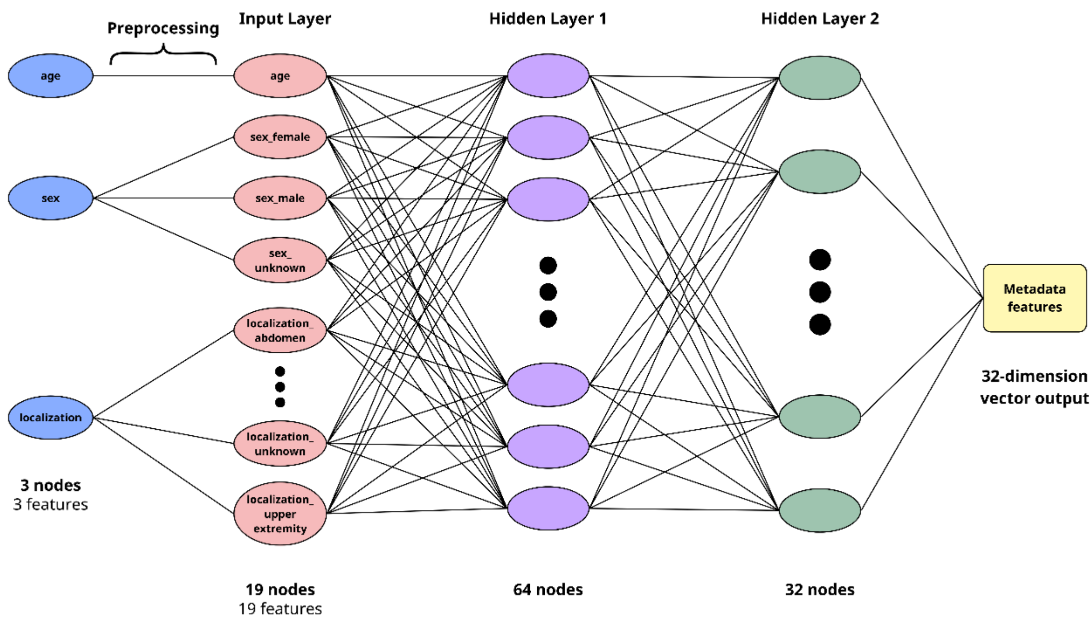

Fig. 3.2. Architecture of the Metadata-Based Neural Network for Feature Extraction (HAM10000 Dataset)

Handling Missing Values:

Mean imputation was employed; this method entails calculating the relevant statistical measure from the non-missing values in each column and subsequently replacing the missing entries. Mean imputation is frequently utilized for numerical features (Zhou et al., 2024). However, it is important to acknowledge that this simple imputation method depend on basic statistical assumptions. They might not accurately represent the underlying data distribution or the complex relationships between variables (Zhou et al., 2024).

The age variable exhibited a minor degree of missing data, specifically 57 instances within a total of 10,015 observations. Given this minimal proportion, the impact on the overall data distribution is negligible, and mean imputation was considered sufficient for this thesis (Pavel et al., 2025).

Normalization - Standard Scaler:

The age feature in the HAM10000 dataset is a numerical scalar which usually has a value range from 0 to 100. To have proper training for the model, this feature is normalized using the StandardScaler. The data is normalized to a z-score distribution such that the mean is 0 and the variance or standard deviation is 1.

StandardScaler is helpful in stabilizing the gradient updates during training, which is a very useful property for neural networks like MLPs. Compared to Min Max normalization which normalizes data in the [0, 1] interval, z-score standardization is less sensitive to data compression and preserves the original data distribution more accurately.

Image Preprocessing

The image preprocessing pipeline addresses dermoscopic image-specific challenges including hair artifact removal, multispectral simulation via RGB channel decomposition, and data imbalance mitigation through augmentation strategies.

Hair Artifact Removal:

The presence of hair can create false positives (incorrectly identifying hair as part of the lesion) or false negatives (missing portions of the lesion because they are hidden under hair). These errors can significantly reduce the accuracy of melanoma detection segmentation procedure (Elena & Ghiță, 2025).

Hair removal was applied as a preprocessing step to reduce noise caused by hair artifacts in dermoscopic images, which can obscure lesion boundaries and interfere with feature extraction (Lee et al., 1997).

The methodology consisted of a four-step process. The first step was to convert each image to grayscale, which was a precondition for the next morphological operations to be performed. The next step was to perform a Black-Hat morphological transformation, using a rectangular structuring element of size 7x7 pixels, which would highlight dark, elongated structures such as hair strands on the lighter background. The output of the Black-Hat transformation was then thresholded to binary with a threshold value of 20, to produce a hair mask, which would isolate the pixels that were likely to be hair.

To avoid unnecessary processing, a hair pixel threshold of 500 was used. Images with fewer than 500 detected hair pixels were considered to have minimal or no hair. These images were kept unchanged, without inpainting. This decision was made to prevent potential distortion of lesion textures and a decrease in image quality that could result from applying inpainting to images without hair.

Finally, for images exceeding this threshold, the Navier-Stokes inpainting method (cv2.INPAINT_NS) was applied with a radius of 5 pixels to reconstruct the regions identified as hair by propagating surrounding pixel information into the masked areas.

Visual examples demonstrating the effectiveness of this hair removal procedure on images with varying degrees of hair artifacts are presented in Appendix 9.

Multispectral Simulation via RGB Channel Decomposition:

Each preprocessed image was converted from BGR to RGB format after the hair removal. The image was then split into its three constituent channels, red (R), green (G) and blue (B), to simulate the narrow-band nature of the internal multispectral dataset. Each of the channels was saved separately as a single-channel grayscale image. This channel splitting simulates the internal multispectral dataset narrow-band imaging characteristics and forms the input for the shared image backbone.

Although the internal dataset utilizes infrared (IR, >700nm) instead of blue (~450-495nm), the model is designed to learn generalizable structural and textural features from single-channel representations. The blue channel serves as a proxy during pretraining, with the expectation that the backbone will adapt to IR-specific features during fine-tuning on the target dataset.

An example of this RGB channel decomposition applied to a melanocytic nevus lesion, showing the original image and the resulting R, G, and B grayscale channels, is illustrated in Fig. 3.3.

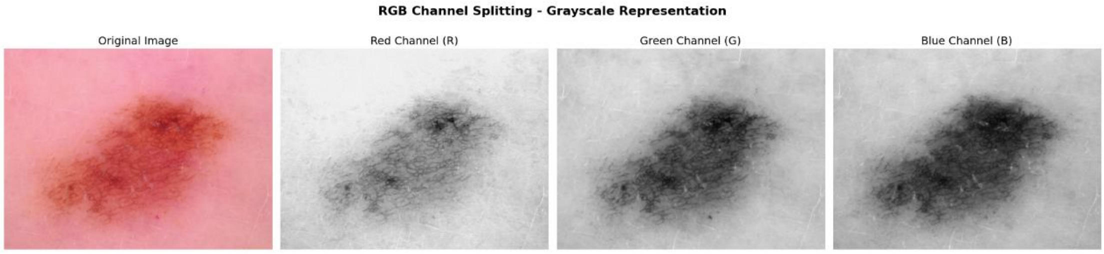

Fig. 3.3. Grayscale RGB Channel Decomposition of Melanocytic Nevus (nv) to reflect single-channel intensity as used in multispectral simulation training

Handle Data Imbalance - Data Augmentation

Data augmentation was employed as a key strategy to enhance model robustness, improve generalization, and mitigate overfitting (Tschandl et al., 2018).

Data augmentation was applied in an online manner during training, meaning that transformations were performed dynamically when images were loaded, rather than generating and storing augmented images beforehand. This approach does not increase the dataset size but enhances data diversity by exposing the model to different variations of the same images across training epochs.

As part of the augmentation process, Gaussian noise was introduced to simulate real-world imaging scenarios, including sensor noise from imaging devices and variations caused by inconsistent lighting (Shen et al., 2022). Gaussian noise, characterized by a mean of 0.0 (indicating zero-centered noise), was added to the input image tensor. For base augmentation, the noise's standard deviation was randomly sampled from the interval 0.01 to 0.03 with application probability 0.3. This probability ensures that not all images are augmented, thus preventing excessive distortion and maintaining the integrity of the original data distribution. For minority augmentation, a more intensive noise regime was applied with standard deviation sampled from 0.01 to 0.05 and probability 0.4, thereby introducing greater variation in noise intensity (Priyadarsini et al., 2026). Subsequently, following the addition of noise, pixel values were clipped to the [0, 1] range to maintain valid image intensities.

Each original image in HAM10000 was of size 600×450 pixels. Prior to applying augmentation, all images were resized to match the input requirements of different deep learning models. For EfficientNetB0, the images were resized to 224×224 pixels (Adebiyi, Abdalnabi, et al., 2024).

During the image preprocessing phase, distinct transformation pipelines were established for both the training and evaluation/test sets. For the training set, two augmentation strategies were defined:

A base augmentation pipeline was applied to the general training images, including random resized cropping (scale 0.85-1.0), horizontal and vertical flipping with probability 0.5, slight brightness and contrast adjustment (±15%), custom Gaussian noise injection (std 0.01-0.03, p=0.3), tensor conversion, and normalization using a single-channel mean of 0.449 and standard deviation of 0.226, approximated by averaging the ImageNet RGB statistics to accommodate grayscale input (Priyadarsini et al., 2026).

To address data imbalance and augment sample diversity, a more robust augmentation pipeline was implemented for minority-class samples (bcc, bkl, mel) within the training set. This enhanced pipeline, which builds upon the base augmentation pipeline, incorporates supplementary transformations. These include random perspective distortion (distortion scale 0.15, p=0.2), enhanced brightness (±20%) and contrast variation with additional saturation (±10%) and hue adjustment (±5%), and more intensive Gaussian noise (std 0.01-0.05, p=0.4) through a wider standard deviation range and higher application probability. Additionally, Random Erasing (p=0.3, scale 0.02-0.1, ratio 0.3-3.3) was applied with a probability of 0.3, randomly masking small rectangular regions of the image to simulate occlusion and further improve model robustness. The minority augmentation pipeline was applied with a probability of 0.7, ensuring stochastic variation while avoiding over-augmentation.

Examples of the base augmentation pipeline applied to a melanocytic nevus sample are shown in Fig. 3.4, while examples of the stronger minority-class augmentation pipeline applied to a benign keratosis sample are shown in Fig. 3.5.

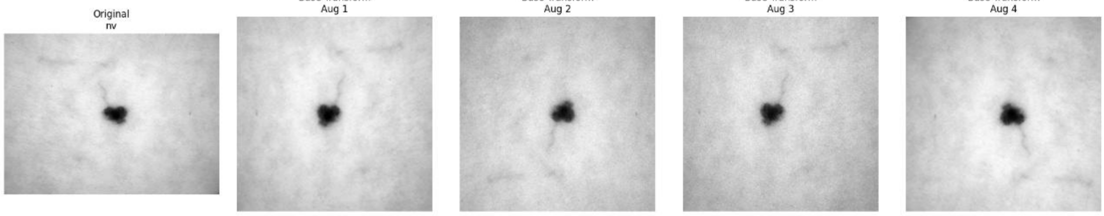

Fig. 3.4. Sample Image with Augmented Variants (Base Transform) - HAM10000 Dataset

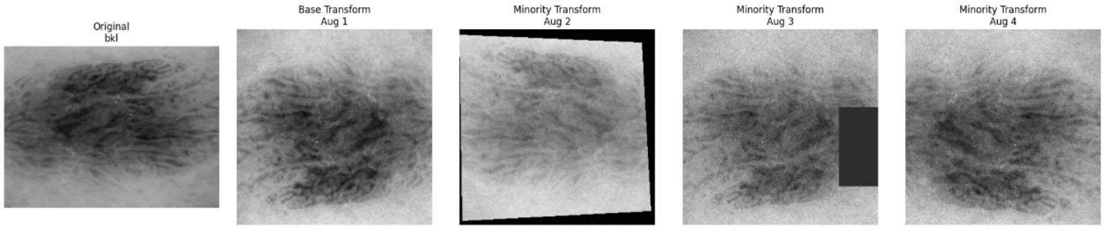

Fig. 3.5. Sample Image with Augmented Variants (Stronger Transform) - HAM10000 Dataset

During training, image augmentation was applied only to the dermoscopic images. The augmented images remained associated with the same metadata features and class labels, since augmentation changes only the visual representation and not the underlying clinical information (Shetty et al., 2022).

In contrast, no augmentation was applied to the validation and test sets. The images were only resized, converted to tensors, and normalized. This ensured a fair and consistent evaluation of how well the model performed.

Handle Data Imbalance - WeightedRandomSampler:

The WeightedRandomSampler is operating on sample level, meaning that each selected sample includes both the image and its associated metadata. Each training sample was assigned a sampling weight inversely proportional to the class frequency. The sampler then randomly draws samples with replacement based on these weights, where samples from minority classes have higher probability of being selected multiple times, while majority class samples may be selected less frequently or not at all within a single epoch. The sampler samples exactly one epoch worth of samples (num_samples = len(sample_weights) × 1), so that the total number of samples per epoch is equal to the original training set size, but with resampling according to class weights to achieve a balanced class distribution.

The original imbalanced class distributions across training folds are documented in Appendix 6, while the rebalanced distributions achieved through WeightedRandomSampler are presented in Appendix 8.

3.1.4. Design AI Model Architecture

Two architectures were developed for the HAM10000 pretraining phase: MultiModalSkinNet, a multimodal architecture that combines dermoscopic images with clinical metadata, and ImageOnlySkinNet, an image-only variant designed as an ablation baseline to quantify the contribution of metadata features to classification performance. Both architectures share the same image processing backbone but differ in their input modalities and classifier configurations.

MultiModalSkinNet

The complete architecture of MultiModalSkinNet, including the shared EfficientNet-B0 backbone processing the three decomposed channels, the two-layer MLP metadata branch, and the late fusion classifier, is illustrated in Fig. 3.6.

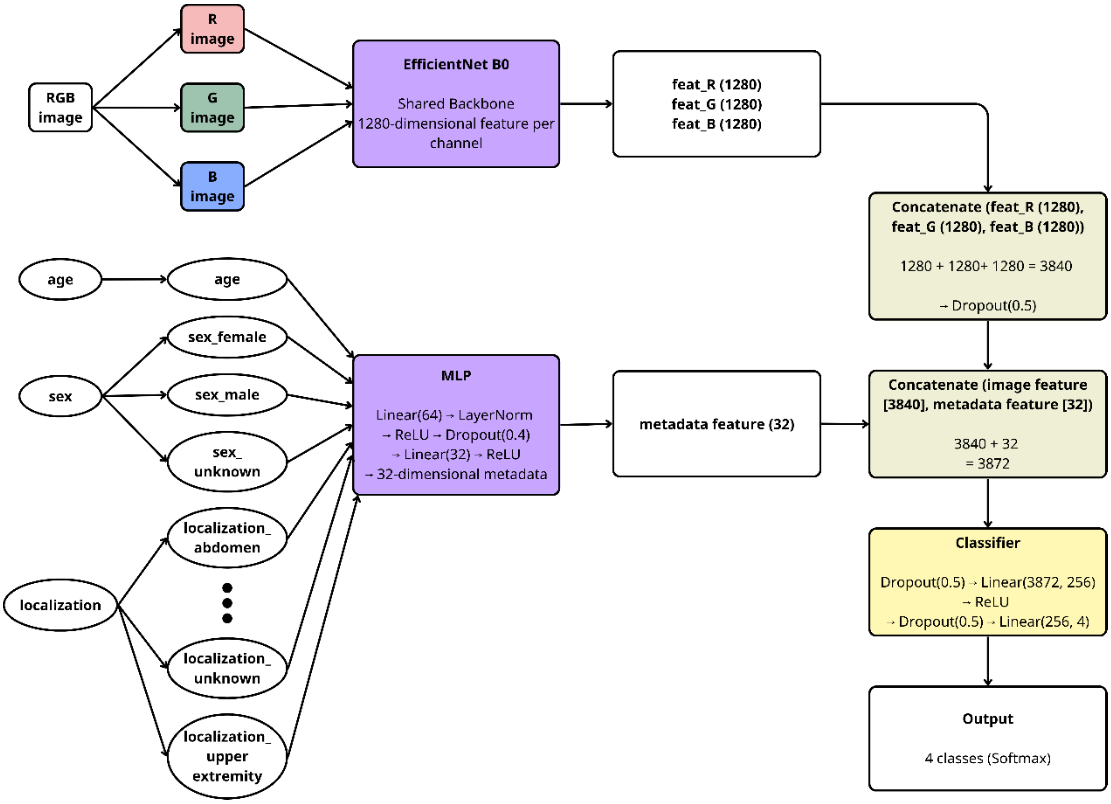

Fig. 3.6. Architecture of the proposed MultiModalSkinNet combining three single-channel image branches and metadata features

MultiModalSkinNet, the proposed model, is a multimodal architecture designed to classify skin lesions by combining dermoscopic images with structured clinical data.

Image Branch: To accommodate single-channel grayscale inputs, the first convolutional layer of EfficientNet-B0 was changed from a 3-channel to a 1-channel input. The pretrained weights of this initial convolutional layer were preserved by averaging the weights across the three input channels into a single channel, thereby maintaining the learned representations acquired during ImageNet-1K pretraining. A single shared EfficientNet-B0 backbone independently processes each of the three grayscale channels (R, G, B). Global Average Pooling is applied to each channel's output, producing a 1280-dimensional feature vector per channel. These three vectors are subsequently concatenated to create a 3840-dimensional image feature representation, followed by dropout regularization (0.5).

Metadata Branch: The structured clinical metadata is processed through a two-layer MLP trained from scratch. The first linear layer maps the input to 64 dimensions, followed by Layer Normalization, ReLU activation, and dropout regularization (0.4). The second linear layer then reduces the representation to 32 dimensions, followed by ReLU activation.

Fusion and Classification: The 3840-dimensional image features and the 32-dimensional metadata features are concatenated, resulting in a 3872-dimensional joint representation. To integrate both image-based features and structured metadata, a late fusion strategy is adopted, resulting in a model that produces more context-aware and robust classification outcomes (Pavan, Bhanusree et al., 2025). This fused vector is then processed by a classifier comprising dropout (0.5), a linear layer (3872 to 256), ReLU activation, dropout (0.5), and a final linear layer (256 to 4), which generates logits corresponding to the four diagnostic classes: bcc, bkl, mel, and nv.

ImageOnlySkinNet

The corresponding ImageOnlySkinNet architecture, which preserves the same shared backbone and channel decomposition but omits the metadata branch and uses an image-only classifier head, is illustrated in Fig. 3.7.

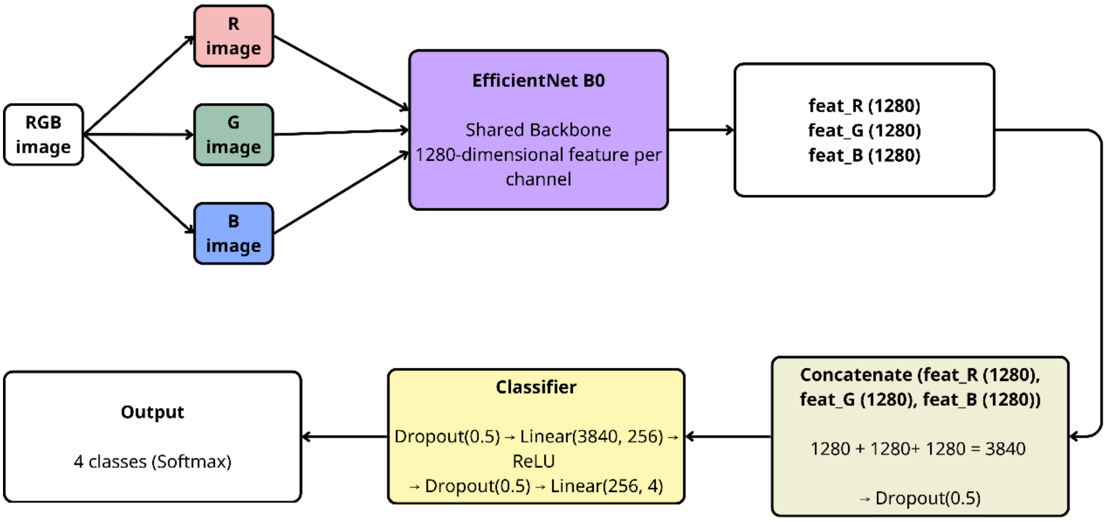

Figure. 3.7. Architecture of the proposed ImageOnlySkinNet combining three single-channel image branches

ImageOnlySkinNet is a single modality skin lesion classification architecture from dermoscopic images without clinical metadata.

Image Branch: Similar to MultiModalSkinNet, the first convolutional layer of EfficientNet-B0 was changed from 3-channel to 1-channel input for gray-scale inputs by averaging the weights of the three input channels into a single channel, while retaining the pretrained weights. A single shared EfficientNet-B0 backbone is used to process each of the three gray-scale channels (R, G, B) independently. Global Average Pooling is applied to the output of each channel resulting in a 1280-dimensional feature vector for each channel. These three vectors are then concatenated into a 3840-dimensional image feature representation and followed by dropout regularization (0.5).

Classification: Unlike MultiModalSkinNet which employs late fusion with a metadata branch, ImageOnlySkinNet utilizes only the 3840-dimensional image features. This vector is processed by a classifier comprising dropout (0.3), a linear layer (3840 to 128), ReLU activation, dropout (0.3), and a final linear layer (128 to 4), which generates logits corresponding to the four diagnostic classes: bcc, bkl, mel, and nv.

ImageOnlySkinNet serves as an ablation study to quantify the contribution of clinical metadata to classification performance. To ensure a fair comparison, the image branch weights are directly transferred from the trained MultiModalSkinNet model.

3.1.5. Train AI Model - Training Configuration

The training process was structured in two phases. This approach was designed to stabilize the learning and, importantly, to avoid overfitting that could occur prematurely.

In the first phase, the entire backbone was frozen and only the metadata branch and classifier were trained for 5 epochs using the Adam optimizer with a learning rate of 1e-4. This phase allowed the randomly initialised head to adapt to the task before any backbone weights were changed.

In the second phase, the final two blocks of the shared EfficientNet-B0 backbone were unfrozen and fine-tuned for up to 50 epochs. To avoid too aggressive update of the backbone, the author of this thesis used different learning rates: a lower learning rate of 3e-5 for the backbone parameters and a higher learning rate of 1e-4 for the metadata branch and classifier. CosineAnnealingWarmRestarts scheduler was used with T_0=10, T_mult=2 and eta_min=1e-7 , which resets the learning rate periodically to encourage exploration of different local minima.

Early stopping was applied during Phase 2 with a patience of 9 epochs and a minimum improvement delta of 0.001, based on validation macro F1-score. The best model state was saved and then restored at the end of each fold.

The Adam optimizer was used in both training phases. This ability of the optimizer to adapt the learning rate means that each parameter can have a different step size, which is very useful in training deep neural networks, particularly those working with sparse or noisy gradients.

The loss function used was CrossEntropyLoss, which internally combines log-softmax and negative log-likelihood loss, and is appropriate for multi-class classification tasks. Despite the large class imbalance in the HAM10000 dataset, explicit class weighting was not added to the loss function. Instead, class imbalance was addressed separately using WeightedRandomSampler and minority-class data augmentation, which together provided a more stable balancing strategy than direct loss reweighting.

In Phase 2, early stopping and model checkpointing were used to prevent overfitting and to choose the best model performance during training. Early stopping tracked the validation macro F1-score over the training epochs. Training was stopped early if the score did not improve more than 0.001 for nine consecutive epochs. The best model state, which is the epoch that achieves the highest validation macro F1-score, was saved and restored at the end of training for each fold.

3.2. Model Training in Proprietary Multispectral Dataset

Following a structure similar to that of Section 3.1, this section is divided into five subsections. This section shows the application of the transfer learning pipeline to the proprietary multispectral dataset, which is the last step of the proposed multistage transfer learning framework. This section explores the baseline model trained on HAM10000 to transfer the learned feature representations from the dermoscopic domain to the target multispectral imaging modality. The transfer strategy employs the weights of the best MultiModalSkinNet checkpoint trained on HAM10000 to initialize the ImageOnlySkinNet architecture for training on the proprietary dataset. Finally, the section reports a comparative evaluation of the two initialization strategies, ImageNet-only baseline vs. HAM10000-pretrained multistage transfer, to empirically confirm the hypothesis that domain-specific pretraining on dermoscopic images improves the feature transferability and classification performance on the target multispectral dataset.

The complete training pipeline on the proprietary multispectral dataset, covering data collection and filtering, image preprocessing, 5-fold stratified group splitting, model architecture, two-phase training, and evaluation, is illustrated in Fig. 3.8.

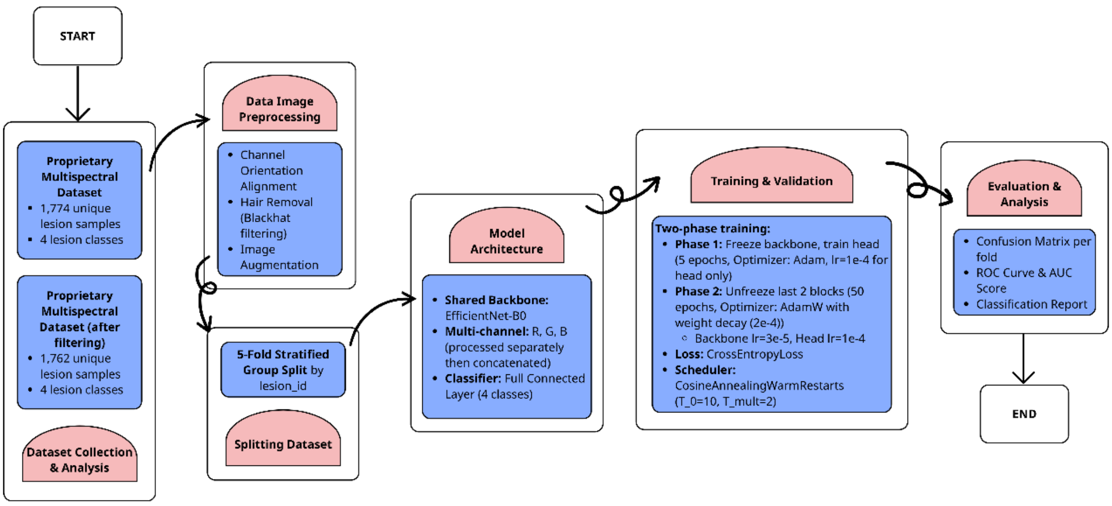

Fig. 3.8. Complete Pipeline for Skin Lesion Classification Using Proprietary Multispectral Dataset

3.2.1. Computing Environment

The training infrastructure consisted of a high-performance workstation equipped with an NVIDIA GeForce RTX 4090 GPU featuring 25.4 GB of memory, CUDA version 12.8, and compute capability 8.9. The system was powered by a 13th Generation Intel Core i5-13500 processor with 20 cores, supported by 62.57 GB of total RAM with 56.76 GB available during training. The operating system was Linux with kernel version 6.8.0-90-generic running on x86_64 architecture. The platform configuration was Linux-6.8.0-90-generic-x86_64-with-glibc2.35.

3.2.2. Data Preparation

The target multispectral dataset followed the same data preparation protocol as the HAM10000 dataset (Section 3.1.2), employing Stratified 5-Fold Cross-Validation with StratifiedGroupKFold (random_state=42) and lesion-level grouping to prevent data leakage.

Since each lesion may be represented by multiple channel images (R, G, IR), all images belonging to the same lesion_id were grouped together and assigned a single diagnostic label during splitting.

To summarize the resulting fold sizes, Table 3.3 presents the distribution of training and validation samples/lesions across the five folds of the proprietary multispectral dataset.

Table 3.3.

Distribution of samples/ lesions across training sets, validation sets across five folds in proprietary multispectral dataset

Dataset	Fold 1	Fold 2	Fold 3	Fold 4	Fold 5
Training Set	1409	1409	1410	1410	1410
Validation Set	353	353	352	352	352

Detailed per-class distributions for training and validation sets across all folds are provided in Appendix 10 and Appendix 11, respectively.

3.2.3. Data Preprocessing

Channel Orientation Alignment:

The provided dataset contains lesion images with three spectral channels (R, G, IR). Upon the initial inspection of the data it was noticed that not all channels share the same spatial orientation. Some of the IR and R channel images were found to be rotated by 90°, 180°, or 270° with respect to the corresponding G channel images.

In order to solve this problem an automated alignment algorithm based on the Structural Similarity Index (SSIM) was developed by the author of this thesis. The algorithm uses the G channel as the reference orientation and determines the optimal rotation angle for IR and R channels through the following procedure:

6) For each image triplet (G, IR, R), the G channel remains fixed as the reference
7) For each target channel (IR or R):
- Generate four rotation candidates: 0°, 90°, 180°, and 270°
- Calculate SSIM between the G channel and each rotated candidate
- Select the rotation angle yielding the highest SSIM score
8) Save the aligned images to a new directory structure

A representative example of this misalignment and its correction, showing the original rotated IR and R channels alongside their SSIM-aligned counterparts for a basal cell carcinoma sample, is illustrated in Fig. 3.9.

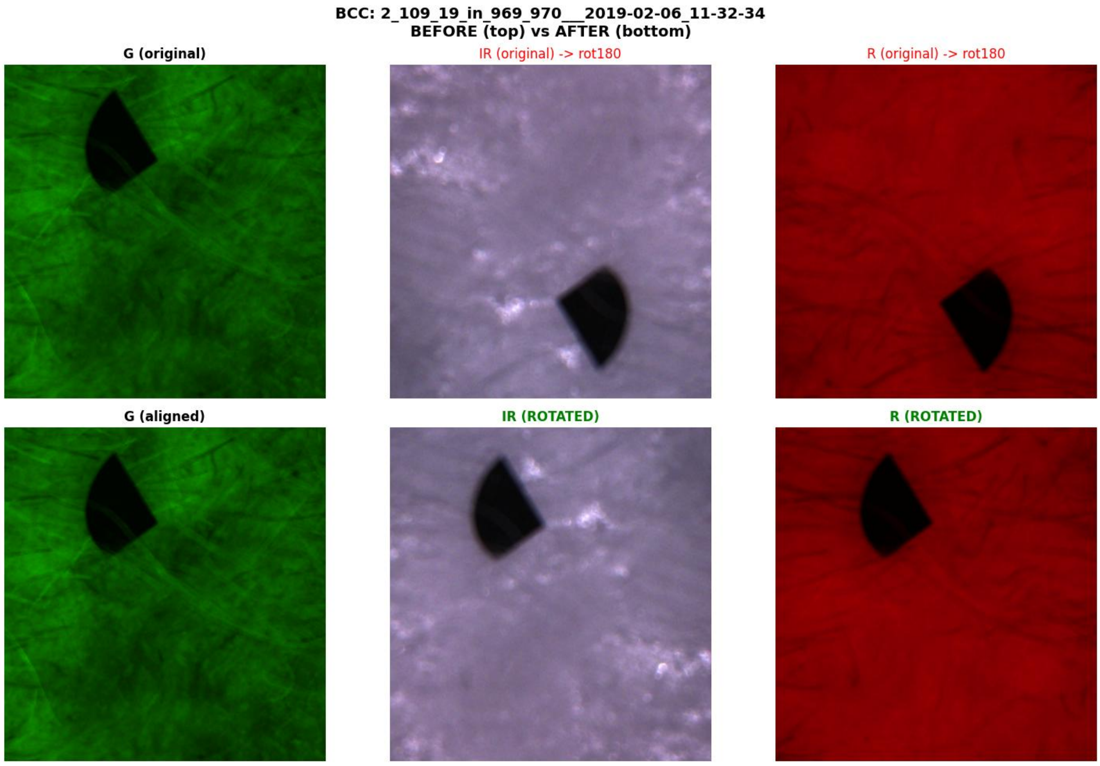

Fig. 3.9. Example of channel misalignment correction - Original misaligned channels (top) and SSIM-aligned channels (bottom) for bcc sample

SSIM was chosen as the similarity metric because it captures structural information (luminance, contrast, and spatial patterns) rather than raw pixel intensity differences. This makes it robust to the inherent intensity variations between spectral channels while remaining sensitive to spatial structure, the key indicator of correct orientation.

The alignment algorithm processed all 1,765 lesion images in the dataset (450 bcc, 250 bkl, 538 mel, 527 nevus). Statistical analysis revealed that 5.0% (88/1765) of IR channels and 0.6% (11/1765) of R channels required rotation correction. In total, 94 lesions (5.3%) needed at least one channel corrected, with 5 lesions requiring correction in both channels simultaneously. The distribution of corrected lesions across classes was: bcc (28 lesions), bkl (4 lesions), mel (53 lesions), and nevus (9 lesions). After automated alignment, the author of this thesis performed manual verification on a whole dataset - 1,765 images and found the algorithm achieved near-perfect accuracy. A small number of edge cases were corrected manually.

Hair Artifact Removal:

Hair removal was performed independently on each spectral channel (R, G, IR) with channel-specific parameters tuned to the characteristics of each wavelength.

The methodology applied Black-Hat morphological transformation followed by binary thresholding was used to detect hair-like structures. For the R channel a 9 × 9 rectangular kernel with threshold 6 was used to capture fine hair details, with no minimum area filtering applied to preserve detection sensitivity. The G channel used the same kernel size but a higher threshold (16) with additional removal of small objects (minimum area 150 pixels) to reduce false positives caused by sensor noise. The IR channel, which exhibits lower hair contrast, used an 11×11 kernel, threshold 25, and stricter object filtering (minimum area 200 pixels).

Connected component analysis identified discrete white regions in the binary mask to identify separate white regions, removing false positives resulting from sensor noise and image artifacts. The mask was then dilated for 2 iterations to expand detected hair regions beyond the original detected regions to ensure the complete coverage of hair strands including semitransparent edges that may have been missed by thresholding.

The minimum hair coverage criterion for each channel was then used to determine whether inpainting was necessary. Images with total mask area of less than 2500 pixels (R), 3000 pixels (G), or 7000 pixels (IR) were classified as having minimal or no significant hair and were left unprocessed. This threshold avoided unnecessary inpainting of near hairless images that could lead to texture distortion or quality degradation. Only images above these thresholds were inpainted by Telea algorithm with propagation radii of 5 pixels (R, G) and 6 pixels (IR) which fills in the regions occluded by hair by diffusing color and texture information from surrounding non-hair pixels into the masked regions.

This per-channel approach accounts for the varying hair visibility across spectral bands, applying more aggressive detection to channels where hair is prominent (R) while maintaining conservative thresholds for channels where hair contrast is naturally lower (IR).

Representative before-and-after examples demonstrating the effectiveness of this preprocessing across all three spectral channels (R, G, IR) are presented in Appendix 13.

Handle Data Imbalance - Data Augmentation:

The data augmentation method for the target model was the same as the source model (Section 3.1.3).

Handle Data Imbalance - WeightedRandomSampler:

The WeightedRandomSampler setting was similar to the HAM10000 dataset (Section 3.1.3) for weighted random sampling at the sample level with inverse frequency weighting. However, the sampler was set to draw twice the number of samples per epoch (num_samples = len(sample_weights) × 2). This means if the training set has 1,409 samples, it will randomly draw 2,818 samples (with replacement based on class weights) in each epoch. As a result, the average number of times each sample in the dataset will appear per epoch is approximately twice, thus increasing exposure during training.

This setting was used to account for the much smaller dataset size (1,409 training samples vs 7,585 in HAM10000) to ensure enough exposure to minority class samples while also training stably.

The original imbalanced class distributions across training folds are documented in Appendix 10, while the rebalanced distributions achieved through WeightedRandomSampler are presented in Appendix 12.

3.2.4. Transfer Learning

The ImageOnlySkinNet architecture is adopted as the target model. The dataset contains clinical metadata, but the data quality assessment led to the exclusion of the metadata, and the model is trained solely on dermoscopic image features as the primary diagnostic modality.

Although the model uses the ImageOnlySkinNet architecture, the model backbone is initialized with weights from the MultiModalSkinNet trained on HAM10000. This decision is based on the better feature learning demonstrated during the multimodal training.

During the multimodal training, the presence of the metadata branch acts as an implicit regularizer that prevents the backbone from overfitting to image-only patterns. The backbone is forced to learn visual features that can be combined with clinical metadata effectively, as opposed to relying on spurious visual shortcuts. This makes the learning more generalizable, capturing clinically relevant dermoscopic patterns instead of dataset-specific artifacts.

Conversely, the ImageOnlySkinNet trained on HAM10000 may learn feature representations optimized for image-only classification, and exploit spurious visual correlations instead of learning clinically robust features.

The best-performing MultiModalSkinNet checkpoint (validation F1: 76.52% on HAM10000) is used as the initialization source. In particular, the shared EfficientNet-B0 backbone, the global average pooling layer and the image dropout layer are transferred, while the metadata branch and fusion classifier are removed. The classifier head is randomly initialized and trained from scratch on the proprietary multispectral dataset, enabling the model to learn classification decisions only from image features.

To empirically validate the benefits of multistage transfer learning, two initialization strategies were compared:

1. Baseline (ImageNet): ImageOnlySkinNet initialized with ImageNet-pretrained EfficientNet-B0 weights, bypassing HAM10000 transfer learning entirely.

2. Multistage Transfer (HAM10000): The ImageOnlySkinNet is initialized with the best MultiModalSkinNet checkpoint from the HAM10000 training; this preserves the dermoscopic feature representations learned in Stage 2.

Both models were trained with identical hyperparameters on the proprietary multispectral dataset.

3.2.5. Train AI Model - Training Configuration

The model architecture was simplified to an image-only variant without the metadata branch. The backbone remained the shared EfficientNet-B0 processing R, G, and IR channels separately, but the classifier was adapted to process only the concatenated image features (3840 dimensions). To accommodate the smaller data set size and to mitigate overfitting, the author of this thesis adjusted the classifier architecture from the HAM10000 ImageOnlySkinNet: the hidden layer dimension was increased from 128 to 256 and the dropout rate was increased from 0.3 to 0.5. The resulting classifier processes the 3840-dimensional concatenated image features through a dropout layer (p=0.5), a linear transformation to 256 dimensions, ReLU activation, another dropout layer (p=0.5), and a final linear layer for 4-class classification.

The training process followed a two-phase progressive unfreezing strategy similar to the source model, designed to stabilize learning and to mitigate premature overfitting on the smaller proprietary dataset (approximately 1,409 training samples per fold compared to 7,585 in HAM10000).

In Phase 1, the entire backbone was frozen while only the classifier head was trained for 5 epochs using Adam optimizer with a learning rate of 1e-3. This helps the randomly initialized head to adapt before any update of backbone.

In Phase 2, the entire backbone was unfrozen and fine-tuned for up to 50 epochs, employing differential learning rates via AdamW optimizer: 1e-4 for the backbone and 3e-4 for the classifier head, with weight decay of 1e-3 and beta parameters (0.9, 0.999). A CosineAnnealingWarmRestarts scheduler (T_0=10, T_mult=2, eta_min=1e-6) provided periodic learning rate resets to escape local minima. Early stopping was applied with a patience of 12 epochs (increased from 9 in HAM10000 to allow for longer convergence on the smaller dataset) and minimum improvement delta of 0.001, monitoring validation macro F1-score across both phases combined.

CrossEntropyLoss, model checkpointing, and best-weight restoration were applied consistently with the source model training.

4. EXPERIMENTAL RESULTS AND PERFORMANCE EVALUATION

In this chapter, there is a thorough evaluation of the proposed deep learning approach for skin lesion classification on two different datasets. The chapter begins with the evaluation framework. The evaluation is carried out in two phases: first, the performance of the baseline model is evaluated using 5-fold cross-validation on the public HAM10000 dataset, where the proposed MultiModalSkinNet architecture is compared to ImageOnlySkinNet to quantify the discriminative value of including clinical metadata. The confusion matrices, ROC curves and per-class metrics are analyzed in detail across the best and worst performing folds, revealing systematic classification patterns and challenges with the minority classes. Second, the transfer learning performance is evaluated on the proprietary multispectral dataset, where the initialization pretrained on HAM10000 is compared against the ImageNet-only baseline to validate the hypothesis that domain-specific pretraining improves feature transferability for the target classification task. The performance is also compared with the existing literature to contextualize the proposed channel decomposition approach within the broader landscape of CNN-based skin lesion classification methods.

4.1. Evaluation Metrics

After training and fine-tuning the model, it is important to evaluate the performance of the model for confirming the effectiveness of the model for skin lesion detection. The thesis uses accuracy, precision, recall, F1 score, confusion matrix, and AUC-ROC for the performance evaluation of the model.

Accuracy is the ratio of the correctly classified instances to the total instances. However, accuracy can be misleading in imbalanced datasets such as HAM10000. A model that classifies all the samples as the majority class can give high accuracy but does not classify any minority classes correctly.

To address these limitations in imbalanced classification, this thesis evaluates complementary metrics that measure different aspects of model performance. Precision is the ratio of true positive predictions (correctly predicted malignant cases) to all positive predictions (all cases that the model predicted as positive/malignant) (Al Mamun et al., 2025). High precision helps to minimize false positive results and to avoid unnecessary medical interventions.

Recall is the ratio of true positive predictions (correctly predicted malignant cases) to the total number of positive instances (all actual malignant cases) (Al Mamun et al., 2025). High recall ensures that true skin cancer cases are detected as early and as comprehensively as possible.

Both precision and recall are useful measures although they tend to have a trade-off. The primary metric for this evaluation is the F1 score. This combines recall and precision into one metric giving a balanced measure of model performance . Due to the class imbalance of HAM10000 and the proprietary multispectral dataset, the macro F1 score is chosen as the main metric as it gives a balanced measure across all classes without being biased against the majority class. Unlike weighted F1 or raw accuracy which can appear artificially high by correctly predicting only the majority class, macro F1 treats all classes equally regardless of their frequency, providing a balanced assessment that prevents majority class bias and ensures minority class performance is not overlooked.

Beyond aggregate scalar metrics, understanding where the model fails is important for clinical deployment. A confusion matrix is a tabular representation that compares the actual target values with the predicted target values to assess the performance of a classification model and provides a detailed view of the model prediction and types of errors it makes. In this thesis we use a 4x4 confusion matrix encoding the four diagnostic categories (bcc, bkl, mel, and nv). Each row represents the actual class label, and each column represents the predicted class label. The diagonal elements are correctly classified instances, while the off-diagonal elements are the misclassified instances. This enables a detailed look at the most common confusion between classes, revealing systematic misclassification patterns that are important in a clinical setting.

The matrix reveals misclassification patterns at a fixed decision threshold, but clinical application may require operating at different points depending on the context of screening. The area under the ROC curve (AUC) gives an overall measure of model's ability to discriminate between classes at different decision thresholds, allowing to determine operating points suitable for clinical needs. In multiclass scenarios, the AUC is calculated using a One-vs-Rest (OvR) strategy, where each class is considered against all other classes separately . This method allows to evaluate the discriminative performance for each class separately, which is especially important in imbalanced datasets where the accuracy metric can be misleading.

4.2. Baseline Model Performance (HAM10000)

In this section, it evaluates the baseline performance of MultiModalSkinNet and ImageOnlySkinNet on the HAM10000 dataset through a comprehensive analysis of 5-fold cross-validation. The author of this bachelor thesis has divided this section into four subsections to clearly describe each evaluation aspect of the baseline model performance on HAM10000. The following subsections present the 5-fold cross-validation overview results, the confusion matrix analysis for the best and worst performing folds, the ROC analysis, and the per-class metrics analysis, comparing MultiModalSkinNet and ImageOnlySkinNet across all four evaluation dimensions.

4.2.1. 5-Fold Cross-Validation Overview Results

Table 4.1 reports the 5-fold cross-validation results of MultiModalSkinNet and ImageOnlySkinNet on the HAM10000 dataset, covering the validation accuracy and macro validation F1-score per fold along with the mean and standard deviation across all folds.

Table 4.1

5-Fold Cross-Validation Results of MultiModalSkinNet and ImageOnlySkinNet on HAM10000, reporting Validation Accuracy, and Macro Validation F1-score per fold alongside mean performance across all folds

	ImageOnlySkinNet			MultiModalSkinNet		
Fold	Best Epoch	Validation Accuracy	Validation F1	Best Epoch	Validation Accuracy	Validation F1
1	18	0.8147	0.6826	29	0.8559	0.7565
2	6	0.7859	0.6646	14	0.8358	0.7306
3	47	0.7974	0.6754	19	0.8453	0.7389
4	33	0.8300	0.7191	15	0.8626	0.7607
5	18	0.7963	0.6825	19	0.8587	0.7585

	Mean F1: 0.6848 ± 0.0183	Mean F1: 0.7491 ± 0.0120
	Mean Accuracy: 0.8049 ± 0.0156	Mean Accuracy: 0.8517 ± 0.0098

Some important observations on the 5-fold cross validation results for the Baseline Model Performance (MultiModalSkinNet and ImageOnlySkinNet) section are:

MultiModalSkinNet performed better than ImageOnlySkinNet (mean F1-score of 0.7491 ± 0.0120 vs. 0.6848 ± 0.0183 and mean accuracy of 0.8517 ± 0.0098 vs. 0.8049 ± 0.0156), indicating the significant discriminative value of clinical metadata for skin lesion classification on HAM10000.

MultiModalSkinNet showed better consistency with 34-37% lower standard deviations on both metrics (F1 score and accuracy respectively), indicating more stable learning and better generalization. Higher variance of ImageOnlySkinNet indicates higher sensitivity to fold-specific characteristics when only visual features are considered.

MultiModalSkinNet converged faster and more consistently (best epochs 14-29, median ~19) than ImageOnlySkinNet's erratic pattern (6-47, median ~18 with Fold 3 requiring 47 epochs). The stability of the multimodal architecture indicates that metadata features provide an additional learning signal that helps guide optimization more effectively, but all folds converged within budget with no severe underfitting or overfitting.

For both models, Fold 4 was the peak and Fold 2 was the most difficult. MultiModalSkinNet ranged from 0.7306-0.7607 F1 (0.0301 spread) and 0.8358-0.8626 accuracy (0.0268 spread). ImageOnlySkinNet was more variable: 0.6646-0.7191 F1 (0.0545 spread) and 0.7859-0.8300 accuracy (0.0441 spread). The narrower range of the multimodal approach suggests more consistent performance irrespective of the composition of the data subsets. The low variance (std < 0.012) of MultiModalSkinNet confirms a reliable generalization across all data subsets without outliers.

Both models presented significant gaps reflecting the class imbalance in HAM10000 dataset. The gap was 10.26% (85.17% accuracy vs 74.91% F1) for MultiModalSkinNet and 12.01% (80.49% vs 68.48%) for ImageOnlySkinNet. Accuracy rewards making correct predictions on the majority classes while F1-score gives a more conservative estimate as it penalizes poor performance on minority classes. Smaller gap in MultiModalSkinNet indicates that metadata is particularly useful for minority class predictions in addition to overall accuracy.

4.2.2. Confusion Matrix Analysis: Best vs Worst Performing Folds

MultiModalSkinNet (Fold 2 vs Fold 4):

The confusion matrices of MultiModalSkinNet on the worst performing fold (Fold 2, F1: 0.7306) and the best performing fold (Fold 4, F1: 0.7607) are shown in Fig. 4.1 and Fig. 4.2, respectively.

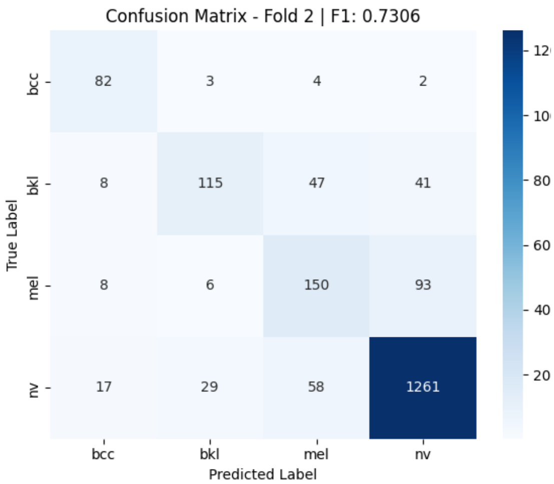

Fig. 4.1. Confusion matrix showing MultiModalSkinNet performance on HAM10000 fold 2

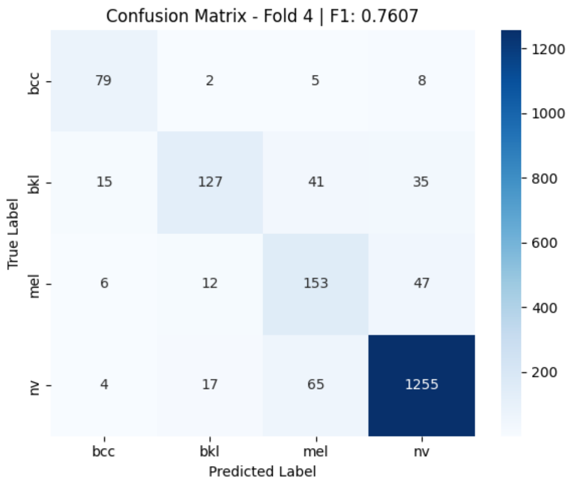

Fig. 4.2. Confusion matrix showing MultiModalSkinNet performance on HAM10000 fold 4

For the majority class (nv - melanocytic nevi), both folds showed strong performance with high true positives (1261 in Fold 2, 1255 in Fold 4). The critical difference lies in minority class (bcc, bkl, mel) performance. Fold 2 struggled significantly with bkl (benign keratosis-like lesions), achieving only 115/211 correct predictions (54.5% recall) with substantial confusion to mel (47 cases) and nv (41 cases). On the other hand, bkl performed better on Fold 4 (127/218, 58.37% recall) with less class confusion. Similarly, mel (melanoma) performed better on Fold 4 (153/218 correct) than on Fold 2 (150/257 correct), both folds showing significant nv-mel confusion, highlighting the clinical difficulty of distinguishing these lesion types. The bcc (basal cell carcinoma) class was interesting for its high recall in both folds (90.11%

recall: 82/91 on Fold 2; 84.04% recall: 79/94 on Fold 4), suggesting good feature learning for this distinct lesion type with small variations.

ImageOnlySkinNet (Fold 2 vs Fold 4):

The confusion matrices of ImageOnlySkinNet on the worst performing fold (Fold 2, F1: 0.6646) and the best performing fold (Fold 4, F1: 0.7191) are shown in Fig. 4.3 and Fig. 4.4, respectively.

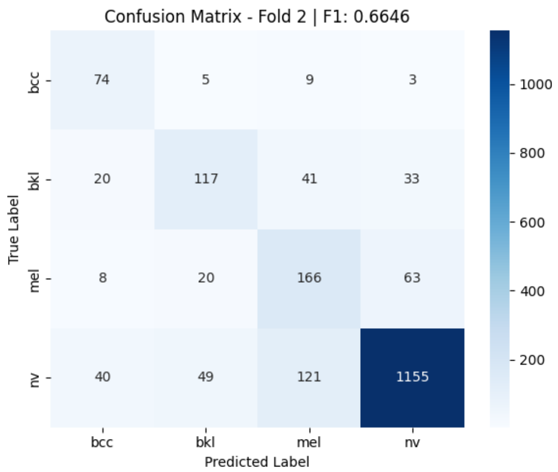

Fig. 4.3. Confusion matrix showing ImageOnlySkinNet performance on HAM10000 fold 2

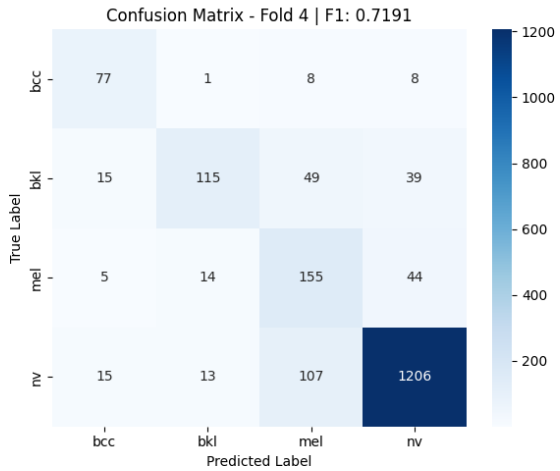

Fig. 4.4. Confusion matrix showing ImageOnlySkinNet performance on HAM10000 fold 4

Fold 2 exhibited severe classification instability: bkl collapsed to 117/211 correct (55.45% recall) with heavy confusion to mel (41) and nv (33); mel achieved only 166/257 correct (64.59% recall) with 63 false negatives to nv, indicating the model's difficulty distinguishing these visually similar lesions without clinical context. The nv majority class, while maintaining 1155/1365 true positives, showed scattered misclassification across all other classes (40 bcc, 49 bkl, 121 mel). Fold 4 with the results: bkl recall remained challenging with 115/218 true positives (52.75%), mel improved to 155/218 (71.10%), and nv achieved 1206/1341 true positives with reduced inter-class confusion. However, even in the best fold, ImageOnlySkinNet's confusion patterns remained more diffuse than MultiModalSkinNet's, particularly for the clinically critical mel class where misclassification as nv poses diagnostic risk.

Complete confusion matrices for all five folds are available in Appendix 14 (ImageOnlySkinNet) and Appendix 15 (MultiModalSkinNet).

4.2.3. ROC Analysis: Best vs Worst Performing Folds

MultiModalSkinNet (Fold 2 vs Fold 4):

The ROC curves of MultiModalSkinNet across the best and worst performing folds are depicted in Fig. 4.5 (Fold 2, worst performing) and Fig. 4.6 (Fold 4, best performing).

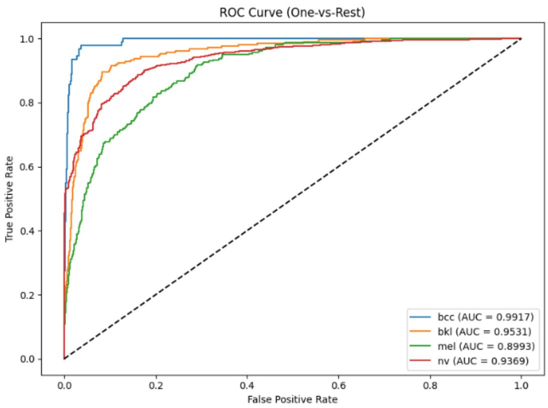

Fig. 4.5. ROC curves showing MultiModalSkinNet performance on HAM10000 fold 2

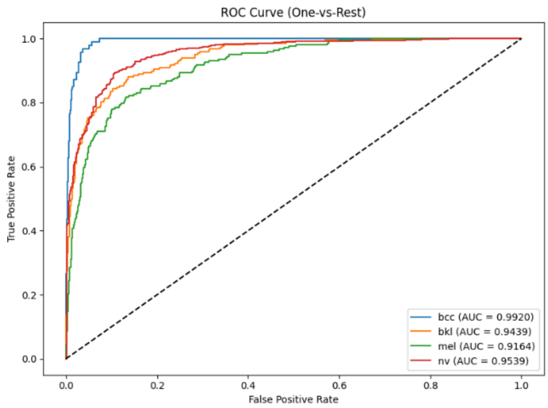

Fig. 4.6. ROC curves showing MultiModalSkinNet performance on HAM10000 Fold 4

In Fold 2, bcc achieved the highest AUC (0.9917), followed by bkl (0.9531), mel (0.8993), and nv (0.9369). Fold 4 showed overall improvement with bcc maintaining excellent discrimination (AUC: 0.9920), bkl slightly decreasing to 0.9439, mel substantially improving to 0.9164, and nv reaching 0.9539. The most notable improvement occurred in mel and nv classification, where mel gained +0.0171 and nv improved +0.0170. Despite mel having the lowest AUC in both folds, all classes achieved AUC > 0.89, indicating strong discriminative ability across the entire HAM10000 class spectrum. The consistent bcc superiority (AUC > 0.98 in both folds) confirms this lesion type's distinct visual and clinical features make it the most separable class.

ImageOnlySkinNet (Fold 2 vs Fold 4):

Similarly, the ROC curves of ImageOnlySkinNet across the best and worst performing folds are presented in Fig. 4.7 (Fold 2, worst performing) and Fig. 4.8 (Fold 4, best performing).

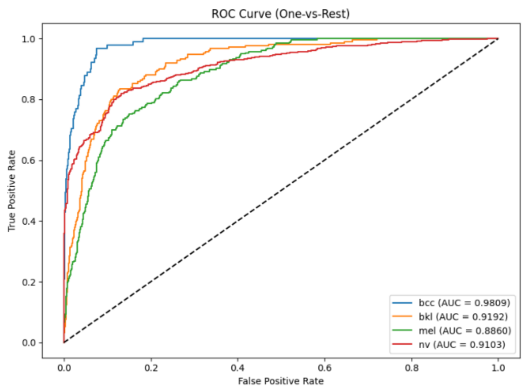

Fig. 4.7. ROC curves showing ImageOnlySkinNet performance on HAM10000 fold 2

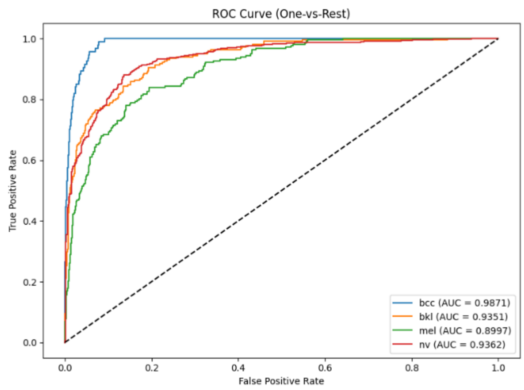

Fig. 4.8. ROC curves showing ImageOnlySkinNet performance on HAM10000 fold 4

ImageOnlySkinNet exhibited lower absolute AUC values and greater inter-fold variability compared to the multimodal approach. Fold 2 showed: bcc (0.9809), bkl (0.9192), mel (0.8860), nv (0.9103), while Fold 4 improved across all classes: bcc (0.9871), bkl (0.9351), mel (0.8997), nv (0.9362). The performance gap between folds was more pronounced than in MultiModalSkinNet, with nv showing the largest improvement (+0.0259), followed by bkl (+0.0159), mel (+0.0137), and bcc (+0.0062). However, mel remained the most challenging class, achieving only 0.8997 AUC even in the best fold.

Full ROC curves across all five cross-validation folds can be found in Appendix 16 (ImageOnlySkinNet) and Appendix 17 (MultiModalSkinNet).

4.2.4. Per-class Metrics Analysis: Best vs Worst Performing Folds

MultiModalSkinNet (Fold 2 vs Fold 4):

The per-class classification reports of MultiModalSkinNet across the best and worst performing folds are illustrated in Fig. 4.9 (Fold 2, worst performing) and Fig. 4.10 (Fold 4, best performing).

Fold 2 – Per-class metrics:

	precision	recall	f1-score	support
bcc	0.7130	0.9011	0.7961	91
bkl	0.7516	0.5450	0.6319	211
mel	0.5792	0.5837	0.5814	257
nv	0.9026	0.9238	0.9131	1365
accuracy			0.8358	1924
macro avg	0.7366	0.7384	0.7306	1924
weighted avg	0.8339	0.8358	0.8324	1924

Fig. 4.9. Per-class Classification Report showing MultiModalSkinNet performance on HAM10000 Fold 2

Fold 4 – Per-class metrics:

	precision	recall	f1-score	support
bcc	0.7596	0.8404	0.7980	94
bkl	0.8038	0.5826	0.6755	218
mel	0.5795	0.7018	0.6349	218
nv	0.9331	0.9359	0.9345	1341
accuracy			0.8626	1871
macro avg	0.7690	0.7652	0.7607	1871
weighted avg	0.8681	0.8626	0.8625	1871

Fig. 4.10. Per-class Classification Report showing MultiModalSkinNet performance on HAM10000 Fold 4

The per-class metrics reveal distinct discrimination challenges across lesion types. In both folds, nv (melanocytic nevi) achieved the highest F1-scores (0.9131 in Fold 2, 0.9345 in Fold 4) due to high precision (0.9026, 0.9331) and recall (0.9238, 0.9359), benefiting from its 70% class representation. bcc (basal cell carcinoma) maintained strong performance with F1-scores of 0.7961 (Fold 2) and 0.7980 (Fold 4), showing balanced precision-recall trade-offs (precision: 0.7130, 0.7596; recall: 0.9011, 0.8404).

Minority Class Challenges:

bkl (benign keratosis) exhibited moderate performance with F1-scores of 0.6319 (Fold 2) and 0.6755 (Fold 4). The consistent pattern of high precision (0.7516, 0.8038) but low recall (0.5450, 0.5826) indicates the model's conservative predictions, it correctly identifies bkl when predicted, but misses many actual cases, likely confusing them with mel or nv as shown in confusion matrices.

mel (melanoma) proved most challenging despite clinical importance, achieving F1-scores of only 0.5814 (Fold 2) and 0.6349 (Fold 4). The balanced precision-recall (0.5792 in precision and 0.5837 in recall for Fold 2, improving to 0.5795 precision and 0.7018 recall in Fold 4) shows the model struggles equally with false positives and false negatives. Fold 4's improvement came primarily from recall gain (+0.1181: 0.5837 to 0.7018), indicating better detection of actual melanoma cases, though precision remained similar. This mel-specific recall improvement directly correlates with the +0.171 AUC gain and higher overall F1-score in Fold 4.

Precision-Recall Trade-offs: Fold 4's superior performance (macro F1: 0.7607 vs 0.7306) stemmed from improved recall across minority classes, particularly mel (+0.1181) and bcc (-0.0607 decline in recall but +0.0466 precision gain suggesting better specificity), while maintaining or improving precision. The macro vs weighted average gap (~0.009 in both folds) confirms class imbalance impact: weighted metrics benefit from strong nv performance, while macro averages expose minority class difficulties.

ImageOnlySkinNet (Fold 2 vs Fold 4):

Likewise, the per-class classification reports of ImageOnlySkinNet across the best and worst performing folds are reported in Fig. 4.11 (Fold 2, worst performing) and Fig. 4.12 (Fold 4, best performing).

Fold 2 – Per-class metrics:

	precision	recall	f1-score	support
bcc	0.5211	0.8132	0.6352	91
bkl	0.6126	0.5545	0.5821	211
mel	0.4926	0.6459	0.5589	257
nv	0.9211	0.8462	0.8820	1365
accuracy			0.7859	1924
macro avg	0.6368	0.7149	0.6646	1924
weighted avg	0.8111	0.7859	0.7943	1924

Fig. 4.11. Per-class Classification Report showing ImageOnlySkinNet performance on HAM10000 Fold 2

Fold 4 – Per-class metrics:

	precision	recall	f1-score	support
bcc	0.6875	0.8191	0.7476	94
bkl	0.8042	0.5275	0.6371	218
mel	0.4859	0.7110	0.5773	218
nv	0.9298	0.8993	0.9143	1341
accuracy			0.8300	1871
macro avg	0.7269	0.7393	0.7191	1871
weighted avg	0.8513	0.8300	0.8344	1871

Fig. 4.12. Per-class Classification Report showing ImageOnlySkinNet performance on HAM10000 Fold 4

ImageOnlySkinNet showed substantially lower per-class performance compared to MultiModalSkinNet, with greater instability across folds. nv (melanocytic nevi) remained the strongest class with F1-scores of 0.8820 (Fold 2) and 0.9143 (Fold 4), though notably lower than MultiModalSkinNet's 0.9131-0.9345. bcc exhibited moderate performance: 0.6352 (Fold 2) improving to 0.7476 (Fold 4), with Fold 4's +0.1124 gain driven by precision improvement (0.5211 to 0.6875, +0.1664) while maintaining similar recall (~0.81).

Minority Class Challenges:

For bkl, the F1-scores of only 0.5821 (Fold 2) and 0.6371 (Fold 4) are much lower than the MultiModalSkinNet’s 0.6319-0.6755. Moderate precision (0.6126, 0.8042) but poor recall (0.5545, 0.5275) suggest that the model misses the majority of bkl cases, and the Fold 4 precision improvement (+0.1916) was offset by the recall decline (-0.0270), indicating that the model became more conservative.

For mel (melanoma), the model’s F1-scores of 0.5589 (Fold 2) and 0.5773 (Fold 4) are below the MultiModalSkinNet’s 0.5814-0.6349. The low precision (0.4926, 0.4859) indicates a high false positive rate, where the model misclassifies many non-melanoma lesions as melanoma. Although the recall is reasonable (0.6459, 0.7110), the poor precision undermines the clinical utility.

Detailed per-class classification reports for all five cross-validation folds are provided in Appendix 18 (ImageOnlySkinNet) and Appendix 19 (MultiModalSkinNet).

4.3. Comparison with Existing Model Performance

A comparison of the proposed MultiModalSkinNet and ImageOnlySkinNet against representative baseline CNN architectures (MobileNetV2, VGG11, ResNet18) reported in (Liu et al., 2024) on the same HAM10000 dataset, in terms of accuracy, F1-score, precision, and recall, is provided in Table 4.2.

Table 4.2

Model Performance Comparison (Baseline RGB vs. Channel Decomposition Architectures)

	Accuracy	**F1**	**Precision**	**Recall**
MobileNetV2	79.84%	77.94%	76.97%	79.84%
VGG11	80.51%	78.46%	77.43%	80.51%
ResNet18	80.24%	78.46%	77.56%	80.24%
ImageOnlySkinNet (EfficientNetB0)	80.49%	68.48%	68.62%	70.85%
MultiModalSkinNet	85.17%	74.91%	76.89%	74.08%

The MobileNetV2, VGG11, and ResNet18 results are reproduced from (Liu et al.), who evaluated these architectures on the same HAM10000 dataset but with standard RGB input processing and the full 7-class classification task. The traditional CNN architectures (MobileNetV2, VGG11, ResNet18) demonstrate balanced performance across all metrics, with accuracy-F1 gaps ranging from 1.78% to 2.05%. These models achieve approximately 80% accuracy with proportional F1 scores (77.94-78.46%), indicating consistent performance across all classes.

The proposed models employing channel decomposition exhibit significantly larger metric discrepancies. ImageOnlySkinNet achieves 80.49% accuracy but only 68.48% F1 score (12.01% gap), while MultiModalSkinNet reaches the highest accuracy at 85.17% with 74.91% F1 (10.26% gap). This phenomenon (bigger gap between F1 score and accuracy) is characteristic of the channel decomposition approach rather than specific architecture choices.

The channel decomposition strategy, while designed to simulate multispectral narrow-band imaging, inadvertently introduces two key architectural limitations. First, the late fusion approach lacks cross-channel attention mechanisms, processing R, G, B independently prevents the model from learning diagnostic patterns based on inter-channel relationships (e.g., color variegation in melanoma). Minority classes, whose signatures depend on such cross-channel correlations, suffer disproportionately compared to majority classes with single-channel-dominant patterns. Second, the shared backbone creates gradient conflicts when different channels signal conflicting weight updates, leading to unstable training.

MultiModalSkinNet demonstrates the value of clinical metadata integration, achieving both the highest accuracy (85.17%) and substantially improved F1 score (+6.43% over ImageOnlySkinNet). The metadata branch incorporating age, sex, and lesion localization provides complementary discriminative information independent of image features.

Direct comparison with baseline models should be interpreted cautiously due to fundamental architectural differences. Unlike standard RGB input processing, the proposed models decompose channels into separate grayscale images, fundamentally altering the input representation. Additionally, this thesis uses a 4-class formulation compared to the original 7-class task in related work, affecting both task difficulty and class distribution patterns.

4.4. Transfer Learning Performance (Multispectral Dataset)

This section evaluates the impact of domain-specific pretraining on transfer learning performance to the internal multispectral dataset. The author of this thesis compare two initialization strategies: (1) ImageNet1K-only weights representing generic visual features, and (2) ImageNet1K to HAM10000 two-stage pretrained weights representing dermatology-specific features. The central hypothesis is that models pretrained on dermatological images (HAM10000) will have learned more transferable feature representations for skin lesion classification than generic ImageNet features alone. The following subsections present the 5-fold cross-validation overview results, the confusion matrix analysis for the best and worst performing folds, the ROC analysis, and the per-class metrics analysis, comparing the ImageNet1K-only and HAM10000-pretrained initialisation strategies across all four evaluation dimensions.

4.4.1. 5-Fold Cross-Validation Overview Results

The 5-fold cross-validation results comparing the two initialisation strategies on the proprietary multispectral dataset, reporting validation accuracy, and macro validation F1-score per fold together with the mean and standard deviation across all folds, are presented in Table 4.3.

Table 4.3

Performance Comparison Between ImageNet1K-Pretrained Weights and HAM10000-Finetuned Weights Across 5-Fold Cross-Validation

	With EfficientNetB0 Weights			**With HAM10000 Pretrained Model Weights**		
Fold	**Best Epoch**	**Validation Accuracy**	**Validation F1**	**Best Epoch**	**Validation Accuracy**	**Validation F1**
1	29	0.7875	0.7641	25	0.8102	0.7824
2	24	0.8329	0.8227	21	0.8555	0.8428
3	21	0.7869	0.7780	12	0.8267	0.8148
4	10	0.7898	0.7636	9	0.7841	0.7674
5	10	0.7585	0.7352	32	0.7983	0.7814
	Mean F1: 0.7727 ± 0.0286			Mean F1: 0.7978 ± 0.0274		
	Mean Accuracy: 0.7911 ± 0.0238			Mean Accuracy: 0.8150 ± 0.0246		

HAM10000 pretraining demonstrates moderate but meaningful gains over ImageNet-only initialization, achieving mean F1-score of 0.7978 ± 0.0274 versus 0.7727 ± 0.0286 (+0.0251) and mean accuracy of 0.8150 ± 0.0246 versus 0.7911 ± 0.0238 (+0.0239). This validates the hypothesis that domain-specific pretraining on dermatological images (HAM10000) provides more transferable feature representations than generic ImageNet features alone for the target skin lesion classification task.

Both initialization strategies exhibit comparable cross-fold variance (±0.0274 vs ±0.0286 for F1-score; ±0.0246 vs ±0.0238 for accuracy), indicating that HAM10000 pretraining enhances mean performance without compromising robustness. The similar standard deviations suggest that both approaches generalize consistently across data subsets.

Best epochs vary considerably for both approaches (HAM10000: 9-32, median ~21; ImageNet-only: 10-29, median ~21), reflecting inherent difficulty differences between data splits rather than systematic convergence patterns. The wide epoch range for HAM10000-pretrained models (especially Fold 5 at epoch 32) suggests domain-relevant initialization enables continued learning on challenging subsets rather than premature overfitting. Both methods ultimately achieve stable convergence within the training budget without severe underfitting or overfitting.

HAM10000-pretrained models demonstrate robust improvements across 4 out of 5 folds. Performance ranged from 0.7674-0.8428 F1 (0.0754 spread) and 0.7841-0.8555 accuracy (0.0714 spread). ImageNet-only initialization showed comparable variation: 0.7352-0.8227 F1 (0.0875 spread) and 0.7585-0.8329 accuracy (0.0744 spread). The largest improvements were found for Fold 2 (+0.0201 F1) and Fold 3 (+0.0368 F1), and the smallest for Fold 4 (+0.0038 F1), indicating that the performance improvements are sensitive to some fold properties (e.g., class distribution or image quality). Despite the differences at the fold level, consistent improvements in 4 out of 5 folds confirm the generalizability of the domain-specific pretraining benefits.

4.4.2. Confusion Matrix Analysis: Best vs Worst Performing Folds

Best-Case Performance - Fold 2 (ImageNet1K Weight vs HAM10000 Pretrained Model Weight):

The confusion matrices on the best performing fold (Fold 2) for the ImageNet1K-initialised model (F1: 0.8227) and the HAM10000-pretrained model (F1: 0.8428) are shown in Fig. 4.13 and Fig. 4.14, respectively.

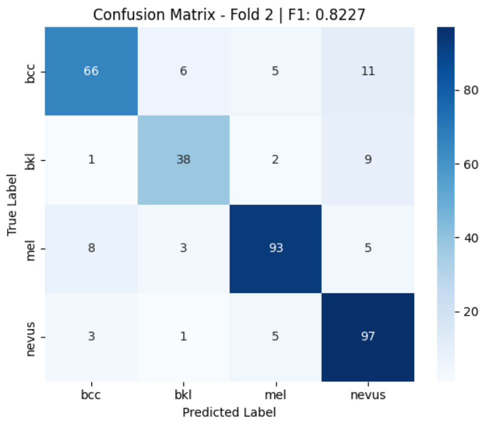

Fig. 4.13. Fold 2 Confusion Matrix - ImageNet1K Weight Model Performance on Target Dataset

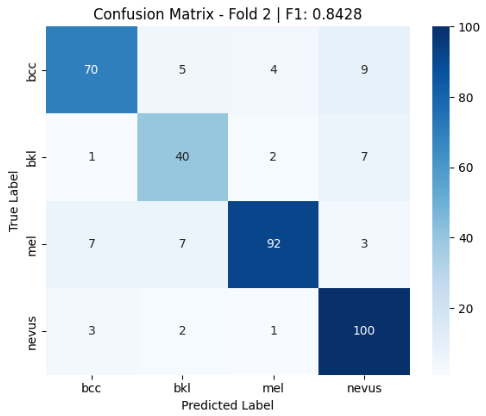

Fig. 4.14. Fold 2 Confusion Matrix - HAM10000 Pretrained Model Weight Performance on Target Dataset

HAM10000-pretrained models achieved superior discrimination across all classes. nv class maintained strong performance with 100/106 true positives. mel class showed robust predictions: achieved 92/109 correct with only 7 misclassification to nv. bkl class reached 40/50 correct with minimal confusion to mel (2 cases) and nv (7 cases). bcc class showed strong performance with 70/88 true positives, though with some confusion to nevus (9 cases). HAM10000 pretraining notably sharpened mel-nv boundaries and improved bkl discrimination.

In contrast, ImageNet-only initialization in Fold 2 showed degraded minority class performance: bkl collapsed to 38/50 correct with increased confusion to mel (2 cases) and nv (9 cases); mel achieved 93/109 correct with 8 nv misclassification; bcc maintained 66/88 correct but with higher diffuse confusion (11 to nv, 6 to bkl). The performance gap is most evident in bkl discrimination, where HAM10000 pretraining reduced confusion patterns.

Worst-Case Performance - Fold 4 (ImageNet1K Weight vs HAM10000 Pretrained Model Weight):

Conversely, the confusion matrices on the worst performing fold (Fold 4) for the ImageNet1K-initialised model and the HAM10000-pretrained model are depicted in Fig. 4.15 and Fig. 4.16, respectively.

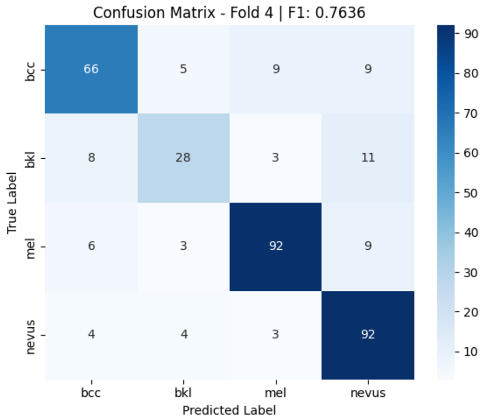

Fig. 4.15. Fold 4 Confusion Matrix - ImageNet1K Weight Model Performance on Target Dataset

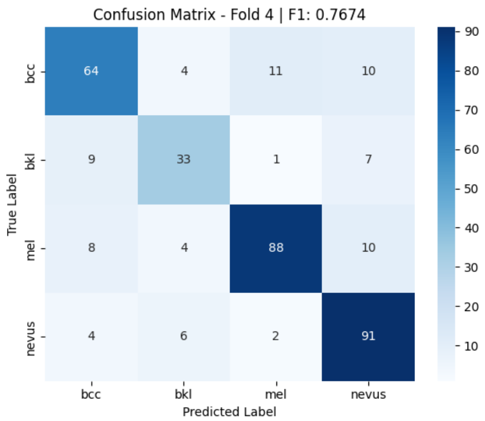

Fig. 4.16. Fold 4 Confusion Matrix - HAM10000 Pretrained Model Weight Performance on Target Dataset

HAM10000-pretrained models in Fold 4 demonstrated the challenge of data subset variability. The mel class showed the most significant degradation: 88/110 correct with 10 false negatives to nv, reflecting increased difficulty in this fold's lesion distribution. The bkl class achieved 33/50 correct with persistent confusion to nv (7 cases) and bcc (9 cases). Nevertheless, bcc maintained strong performance at 64/89 correct, and nv achieved 100/103 true positives, indicating the majority class remained well-learned even in challenging splits.

ImageNet-only initialization in Fold 4 showed mixed performance: bkl deteriorated to 28/50 correct with heavy confusion to nv (11 cases) and bcc (8 cases); mel achieved 92/110 correct, surprisingly slightly better than HAM10000 in this fold; bcc maintained 66/89 correct. Cross-fold comparison reveals that while both approaches struggle with Fold 4's inherent difficulty, domain-specific pretraining maintains more stable minority class performance.

For comprehensive confusion matrices spanning all five cross-validation folds, refer to Appendix 20 (ImageNet1K-only initialization) and Appendix 21 (HAM10000-pretrained initialization).

4.4.3. ROC Analysis: Best vs Worst Performing Folds

Best-Case Performance - Fold 2 (ImageNet1K Weight vs HAM10000 Pretrained Model Weight):

The ROC analysis on the best performing fold is visualised in Fig. 4.17 (ImageNet1K-initialised model) and Fig. 4.18 (HAM10000-pretrained model).

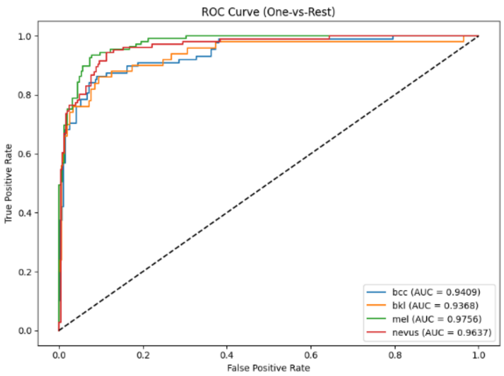

Fig. 4.17. Fold 2 - One-vs-Rest ROC Curves for ImageNet1K Weight Model Performance on Proprietary Multispectral Dataset

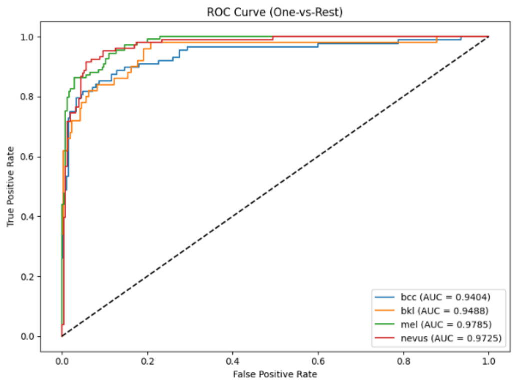

Fig. 4.18. Fold 2 - One-vs-Rest ROC Curves for HAM10000 Pretrained Weight Model Performance on Proprietary Multispectral Dataset

All classes showed excellent discrimination for models pre-trained with HAM10000 with mel scoring the highest (0.9785) followed by nv (0.9725), bkl (0.9488) and bcc (0.9404). Importantly, the clinically critical class mel performed best indicating better learned features for melanoma detection. All classes scored AUC > 0.94 denoting strong discriminative capability.

ImageNet-only initialization also resulted in comparable, but slightly lower performance: mel (0.9756), nv (0.9637), bkl (0.9368), bcc (0.9409). The performance differences underscore the advantages of HAM10000 with improvements in nv discrimination by +0.0087, bkl by +0.0120 and mel by +0.0029. Interestingly, bcc had practically identical performance (-0.0005) suggesting this unique lesion type may not benefit much from domain-specific pretraining. Both initialization methods displayed robust performance for mel (>0.97 AUC), suggesting effective sensitivity for melanoma detection regardless of the pretraining source.

Worst-Case Performance - Fold 4 (ImageNet1K Weight vs HAM10000 Pretrained Model Weight):

The corresponding ROC curves on the worst performing fold (Fold 4) are displayed in Fig. 4.19 for the ImageNet1K-initialised model and Fig. 4.20 for the HAM10000-pretrained model.

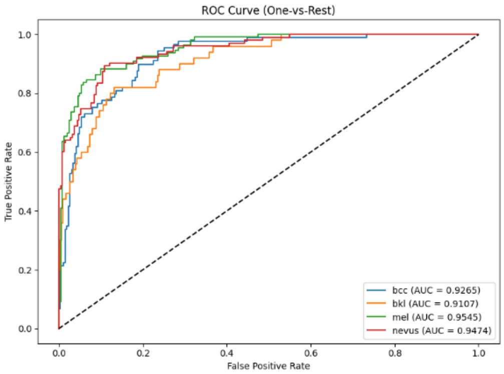

Fig. 4.19. Fold 4 - One-vs-Rest ROC Curves for ImageNet1K Weight Model Performance on Proprietary Multispectral Dataset

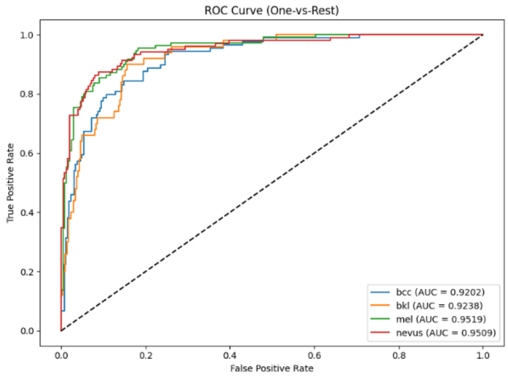

Fig. 4.20. Fold 4 - One-vs-Rest ROC Curves for HAM10000-Pretrained Weight Model Performance on Proprietary Multispectral Dataset

HAM10000-pretrained models faced reduced discrimination in this challenging fold: mel (0.9519), nv (0.9509), bkl (0.9238), bcc (0.9202). The largest degradation occurred in mel (-0.0266 from Fold 2) and nv (-0.0216), while bkl declined by -0.0250 and bcc by -0.0202.

ImageNet-only initialization exhibited similar degradation patterns: mel (0.9545), nv (0.9474), bkl (0.9107), bcc (0.9265). Surprisingly, ImageNet-only achieved slightly higher mel AUC (+0.0026), bcc AUC (+0.0063) in this fold, though HAM10000 maintained advantages in bkl (+0.0131) discrimination, nv (+0.0035) discrimination compared to ImageNet-only. The mel class showed nearly identical AUC for both approaches (difference <0.003), indicating this fold's melanoma cases posed equivalent difficulty regardless of initialization. The most notable gap remains in bkl discrimination, where HAM10000's +0.0131 advantage demonstrates domain-specific features better handle visually ambiguous benign lesions even in challenging data subsets.

The complete set of ROC curves for all five folds is presented in Appendix 22 (ImageNet1K-only initialization) and Appendix 23 (HAM10000-pretrained initialization).

4.4.4. Per-class Metrics Analysis: Best vs Worst Performing Folds

Best-Case Performance - Fold 2 (ImageNet1K Weight vs HAM10000 Pretrained Model Weight):

A detailed breakdown of the per-class metrics on the best performing fold (Fold 2) can be found in Fig. 4.21 for the ImageNet1K-initialised model and Fig. 4.22 for the HAM10000-pretrained model.

Fold 2 – Per-class metrics:

	precision	recall	f1-score	support
bcc	0.8462	0.7500	0.7952	88
bkl	0.7917	0.7600	0.7755	50
mel	0.8857	0.8532	0.8692	109
nevus	0.7951	0.9151	0.8509	106
accuracy			0.8329	353
macro avg	0.8297	0.8196	0.8227	353
weighted avg	0.8353	0.8329	0.8320	353

Fig. 4.21. Fold 2 - Per-class Classification Report showing ImageNet1K Weight Model Performance on Proprietary Multispectral Dataset

Fold 2 – Per-class metrics:

	precision	recall	f1-score	support
bcc	0.8642	0.7955	0.8284	88
bkl	0.7407	0.8000	0.7692	50
mel	0.9293	0.8440	0.8846	109
nevus	0.8403	0.9434	0.8889	106
accuracy			0.8555	353
macro avg	0.8436	0.8457	0.8428	353
weighted avg	0.8596	0.8555	0.8555	353

Fig. 4.22. Fold 2 - Per-class Classification Report showing HAM10000 Pretrained Weight Model Performance on Proprietary Multispectral Dataset

HAM10000-pretrained models achieved strong discrimination across all classes: nv (melanocytic nevi) reached the highest F1-score (0.8889) with balanced precision (0.8403) and recall (0.9434), benefiting from its majority representation. mel (melanoma) showed robust performance with F1-score of 0.8846, demonstrating balanced precision-recall (0.9293 precision, 0.8440 recall), indicating the model correctly identifies melanoma while maintaining low false positive rates. bcc achieved F1-score of 0.8284 with high recall (0.7955) but lower precision (0.8642), suggesting some false positives. bkl exhibited moderate performance (F1: 0.7692) with balanced precision-recall (0.7407 precision, 0.8000 recall).

ImageNet-only initialization deteriorated the performance for most classes. nv (majority class) had a high F1-score (0.8509) but was lower than HAM10000 (-0.0380). mel was reduced to F1-score of 0.8692, which was mainly due to the lower precision (0.8857 vs. 0.9293 for HAM10000), while the recall was marginally higher (0.8532 vs. 0.8440). This suggests that ImageNet-only initialization produced more false positives for melanoma, reducing precision despite slightly better sensitivity. bkl collapsed to F1-score of 0.7755 with imbalanced precision-recall (0.7917 precision, 0.7600 recall, both lower than HAM10000). bcc achieved F1-score of 0.7952, lower than

HAM10000's 0.8284. The performance gap is most evident in mel precision, where HAM10000 pretraining improved by +0.0436, demonstrating superior feature discrimination for this clinically critical class.

Worst-Case Performance - Fold 4 (ImageNet1K Weight vs HAM10000 Pretrained Model Weight):

Conversely, a detailed breakdown of the per-class metrics on the worst performing fold (Fold 4) can be found in Fig. 4.23 for the ImageNet1K-initialised model and Fig. 4.24 for the HAM10000-pretrained model.

Fold 4 – Per-class metrics:

	precision	recall	f1-score	support
bcc	0.7529	0.7191	0.7356	89
bkl	0.7021	0.6600	0.6804	50
mel	0.8627	0.8000	0.8302	110
nevus	0.7712	0.8835	0.8235	103
accuracy			0.7841	352
macro avg	0.7723	0.7656	0.7674	352
weighted avg	0.7854	0.7841	0.7831	352

Fig. 4.23. Fold 4 - Per-class Classification Report showing ImageNet1K Weight Model Performance on Proprietary Multispectral Dataset

Fold 4 – Per-class metrics:

	precision	recall	f1-score	support
bcc	0.7857	0.7416	0.7630	89
bkl	0.7000	0.5600	0.6222	50
mel	0.8598	0.8364	0.8479	110
nevus	0.7603	0.8932	0.8214	103
accuracy			0.7898	352
macro avg	0.7765	0.7578	0.7636	352
weighted avg	0.7893	0.7898	0.7866	352

Fig. 4.24. Fold 4 - Per-class Classification Report showing HAM10000-Pretrained Weight Model Performance on Proprietary Multispectral Dataset

HAM10000-pretrained models faced reduced discrimination in this challenging fold: mel maintained relatively strong F1-score (0.8479) despite -0.0367 decline from Fold 2, with balanced precision-recall (0.8598 precision, 0.8364 recall). nv achieved F1-score of 0.8214 with precision (0.7603) and high recall (0.8932), showing -0.0675 decline from F1-score Fold 2. bcc declined to F1-score of 0.7630 with lower precision-recall (0.7857 precision, 0.7416 recall). bkl showed the largest degradation to F1-score of 0.6222 (-0.1470 from Fold 2) with severely imbalanced metrics (0.7000 precision but only 0.5600 recall), indicating the model misses many actual bkl cases in this challenging fold.

ImageNet-only initialization exhibited similar degradation patterns but with worse absolute performance: mel achieved F1-score of 0.8302 (-0.0390 from Fold 2) with slightly lower precision-recall (0.8627 precision, 0.8000 recall). nv maintained F1-score of 0.8235 with precision (0.7712) and recall (0.8835). bcc improved slightly to

F1-score of 0.7356 with balanced metrics (0.7529 precision, 0.7191 recall). bkl declined to F1-score of 0.6804 with extremely imbalanced performance (0.7021 precision, 0.6600 recall). Notably, Fold 4 exhibits a reversal in bkl performance (ImageNet: 0.6804 vs HAM10000: 0.6222), though HAM10000 retains advantages in mel (+0.0177) discrimination. This underscores that domain pretraining benefits are class- and fold-dependent rather than universal.

Full per-class performance metrics across all five folds are documented in Appendix 24 (ImageNet1K-only initialization) and Appendix 25 (HAM10000-pretrained initialization).

RESULTS AND CONCLUSIONS

This bachelor thesis investigated how knowledge learned from large-scale public RGB dermoscopic datasets can be transferred to a smaller proprietary multispectral skin imaging dataset, with the objective of improving diagnostic performance on the four-class classification task. The central research question, whether knowledge learned from a large RGB skin dataset such as HAM10000 can be effectively transferred to a smaller multispectral and multimodal dataset, has been addressed through the development, training, and evaluation of a dedicated deep learning framework, MultiModalSkinNet, combined with a multistage transfer learning strategy.

The first contribution is the design of MultiModalSkinNet, a deep learning architecture that combines an EfficientNet-B0 image backbone with a lightweight metadata branch processing patient age, sex, and lesion localisation. The 5-fold cross-validation on HAM10000 demonstrated that this multimodal architecture outperforms the image-only counterpart by +6.43% on macro F1-score (0.7491 ± 0.0120 vs. 0.6848 ± 0.0183) and +4.68% on accuracy (0.8517 ± 0.0098 vs. 0.8049 ± 0.0156), while simultaneously reducing the cross-fold variance by approximately 34-37%. The smaller accuracy-F1 gap of the multimodal model (10.26% vs. 12.01%) further indicates that the metadata branch disproportionately helps minority classes, which is the most relevant clinical scenario. The result confirms that structured clinical information complements visual features rather than duplicating them, and supports the design choice of treating metadata as a first-class model input rather than as a secondary auxiliary signal.

The second contribution is the multistage transfer learning pipeline that progressively adapts the backbone from generic ImageNet features, through dermatology-specific features learned on HAM10000, to the target proprietary multispectral dataset. The comparison between HAM10000-pretrained and ImageNet-only initialisation on the multispectral target showed a mean improvement of +2.51% on F1-score (0.7978 ± 0.0274 vs. 0.7727 ± 0.0286) and +2.39% on accuracy (0.8150 ± 0.0246 vs. 0.7911 ± 0.0238), with consistent gains across 4 of the 5 folds. This validates the central research question of the thesis: domain-relevant intermediate pretraining is an effective and generalisable strategy to transfer knowledge from large RGB skin datasets to a smaller multispectral dataset.

Beyond the individual numerical gains, the author draws three main conclusions from this research. First, the combination of multimodal fusion and multistage transfer learning is a viable and practically useful recipe for skin lesion classification in resource-constrained multispectral settings: the reported improvements are obtained without any additional data collection and with a single, lightweight backbone, which makes the approach attractive for clinical research groups operating with small proprietary datasets. Second, the value of metadata is large enough to justify the operational cost of collecting structured clinical attributes prospectively whenever a new multispectral acquisition campaign is planned; the +6.43% F1 gain observed on HAM10000 is a strong argument for treating metadata as a first-class input rather than as an optional add-on.

The experimental results and the limitations identified in the previous chapter point to four concrete directions for further research, ordered from the most immediate to the most exploratory.

The most immediate next step is extending the model input to the autofluorescence channel. The current pipeline uses only the three reflectance bands of the multispectral acquisition device including green (526 nm), red (663 nm), and near-infrared (964 nm), and treats them as independent grayscale inputs to a shared backbone. The same device, however, also captures a sequence of autofluorescence frames under 405 nm UV excitation, recording the fluorescence decay of skin tissue over time, as well as a white-light reference image. These channels were excluded from the present work for scope reasons but encode complementary biological information that is qualitatively different from the reflectance bands already used: autofluorescence in particular is known to differ between benign and malignant lesions because of underlying differences in fluorophore distribution and concentration. A natural next step is therefore to extend the model input to include the autofluorescence frames, either as additional input channels or through a dedicated temporal branch that models the decay sequence, together with the white-light reference image.

A second direction is to investigate Vision Transformers and hybrid CNN-Transformer designs, which can model cross-channel and long-range dependencies more naturally than a shared CNN backbone with late fusion. Recent multimodal skin classification literature (Maqbool et al., 2024; Surendren & Rajesh, 2026) reports promising results with attention mechanisms combined with metadata fusion, and integrating these components into the proposed multistage pipeline is a logical extension. The expected benefit is twofold: better cross-channel reasoning on multispectral inputs, and a more flexible fusion of image and metadata features through cross-modal attention rather than simple concatenation.

A third direction concerns the multimodal fusion at the target task. Once the proprietary multispectral dataset is enriched with structured clinical metadata, the same fusion mechanism that proved effective on HAM10000 can be applied directly to the target task, which is expected to close part of the gap between the intermediate and final stages of the pipeline. This direction does not require any new modelling work, only a prospective data collection effort, which makes it especially attractive as a low-risk extension and a strong candidate for the immediate continuation of the host research project.

Finally, the current architecture processes the three decomposed channels through a single shared EfficientNet-B0 backbone. This choice keeps the parameter budget low but is also the source of the gradient conflicts identified in the limitations: when the R, G, and B channels carry conflicting diagnostic signals, the shared weights have to compromise between them, and minority classes whose signature depends on inter-channel differences are the most affected. A natural extension is to replace the shared backbone with three independent channel-specific backbones, one dedicated branch per input channel, fused at the feature level before the metadata branch. This design gives each channel its own optimisation trajectory and allows the backbone to specialise on the visual cues that are most informative for that wavelength, which is particularly relevant for the multispectral target where the green (526 nm), red (663 nm), near-infrared (964 nm), and autofluorescence (405 nm) bands encode physically distinct information about different skin layers. The trade-off is a roughly three-fold increase in parameter count and GPU memory, which is the same constraint that previously forced the choice of EfficientNet-B0 over B3 in the current pipeline; therefore this direction also depends on access to larger computational resources than those available during this work, and is best treated as a medium-term extension once such resources are secured.

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APPENDICES

Appendix 1

HAM10000 Dataset Metadata Descriptions

This appendix contains detailed descriptions of all metadata attributes in the HAM10000 dataset, which were used to pre-train the skin lesion classification model. Each field in the metadata file of the dataset has a particular role in organizing the clinical information.

Table A.1.1

HAM10000 Dataset Attribute Definitions

Attribute	Meaning
lesion_id	Unique lesion identifier in format HAM_XXXXXXX. Multiple images may share the same lesion_id if captured from different angles or zoom levels
image_id	Unique image identifier in format ISIC_XXXXXXX, used to locate the image file in the dataset directory
dx	Diagnostic label using acronym-based naming: akiec (actinic keratosis), bcc (basal cell carcinoma), bkl (benign keratosis), df (dermatofibroma), mel (melanoma), nv (melanocytic nevus), vasc (vascular lesion)
dx_type	Diagnosis verification method: histopathology (dermatopathological examination), confocal microscopy (non-invasive near-cellular imaging), follow-up (longitudinal confirmation for benign cases), or consensus (expert panel agreement)
age	Patient age at time of image capture
sex	Patient biological sex (male/female)
localization	Anatomical location of the lesion (e.g., back, hand, scalp, face, trunk, extremities)

Appendix 2

Distribution analysis of HAM10000 dataset

In this appendix, the author provides detailed analyses on distribution of the HAM10000 dataset before and after filtering of classes. As mentioned in Section 2.1, the original HAM10000 dataset with seven diagnostic classes was filtered to obtain only four relevant classes (bcc, bkl, mel, nv) for the skin lesion classification task, reducing the dataset from 10,015 to 9,431 images.

The below visualizations show demographic and anatomical distribution comparisons between the original and filtered subsets. Figure A.2.1 shows the localization patterns across 15 anatomical sites, reaffirming the back and lower extremities as the most common lesion locations in both versions. Age and gender distributions are shown in Figures A.2.2 and A.2.3 respectively, reaffirming the filtering process does not change the original demographic distributions. A stratified perspective of age distribution by gender is provided in Figure A.2.4, demonstrating consistent patterns among male and female subgroups before and after filtering.

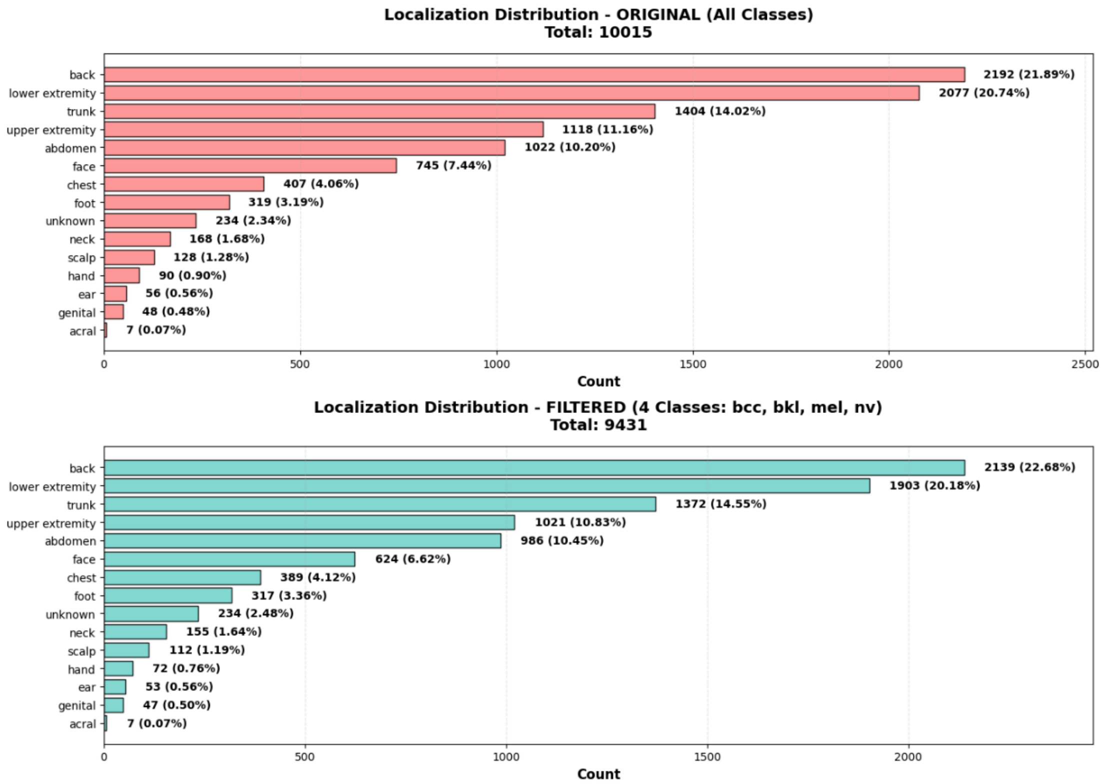

Fig. A.2.1. Distribution of Lesion Localization Before and After Class Filtering in the HAM10000 Dataset

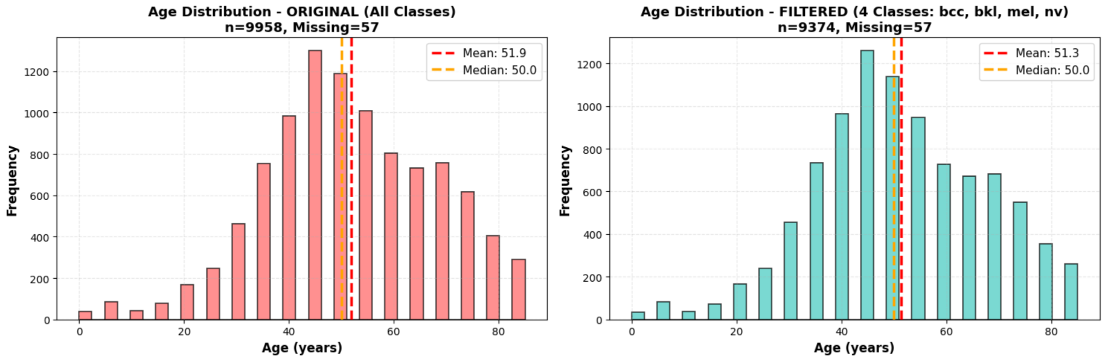

Fig. A.2.2. Age Distribution Before and After Class Filtering in the HAM10000 Dataset

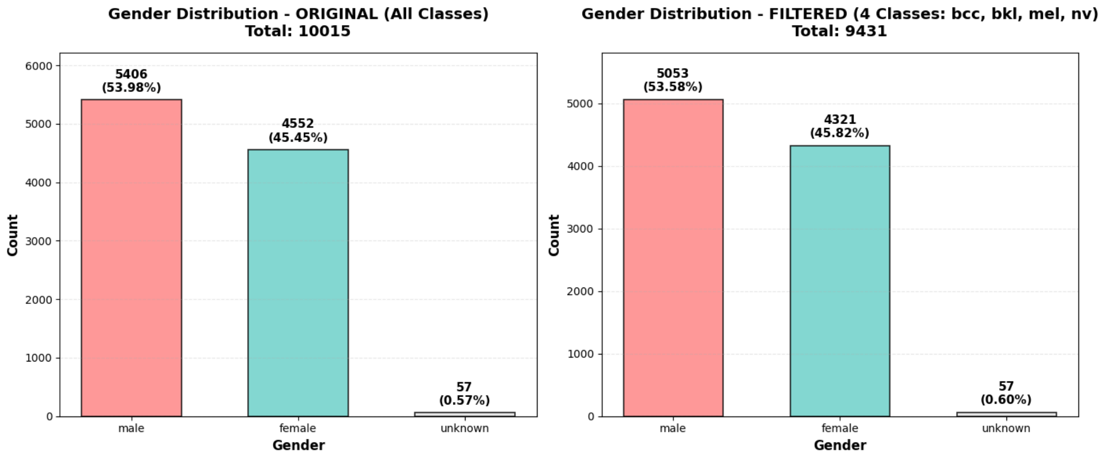

Fig. A.2.3. Gender Distribution Before and After Class Filtering in the HAM10000 Dataset

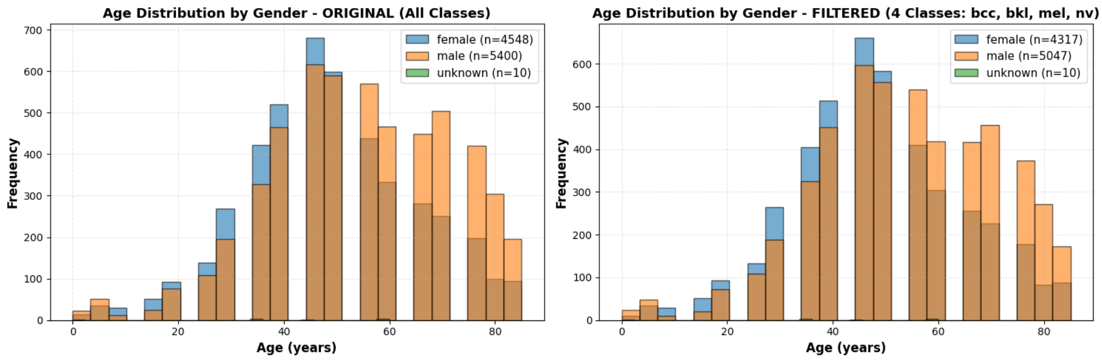

Fig. A.2.4. Age Distribution by Gender Before and After Class Filtering in the HAM10000 Dataset

Appendix 3

Multispectral Imaging Modalities - Proprietary Dataset

This appendix provides visual examples of the multispectral imaging modalities described in Section 2.2.1. Figure A.3.1 shows the five basic spectral bands recorded from one melanoma lesion (C43).

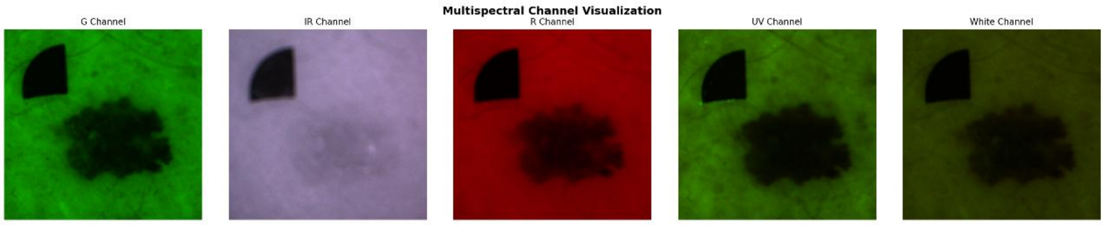

Fig. A.3.1. Multispectral channel visualization of a single skin lesion (C43 - Melanoma) shows Green (G), Infrared (IR), Red (R), Ultraviolet (UV), and White light channels

Figure A.3.2 shows a fluorescence decay sequence recorded over 20 consecutive frames of UV exposure with 200ms spacing. This temporal sequence captures the autofluorescence decay dynamics of melanoma tissue after 405 nm excitation and highlights time-dependent photophysical properties, which are different in benign and malignant lesions. The slow decay in intensity throughout the UV10-UV29 sequence adds further discriminative features for melanoma detection beyond the static spectral information.

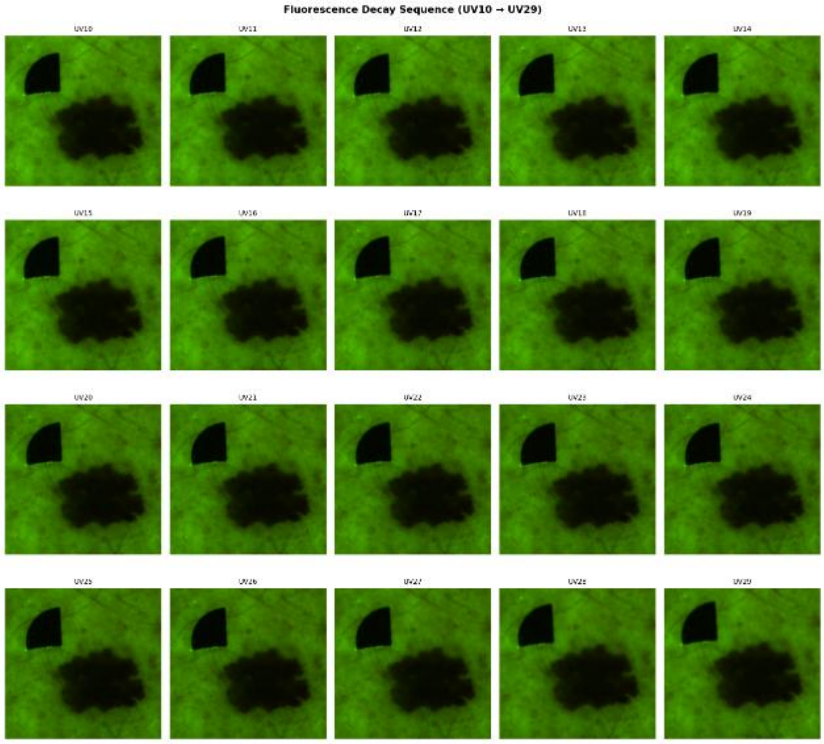

Figure A.3.2. Fluorescence decay image sequence of a Melanoma lesion (C43), captured across 20 consecutive UV exposure frames (UV10-UV29) at 200ms intervals

Appendix 4

Image Quality Issues and Exclusion Examples - Proprietary Dataset

This appendix provides visual examples of the severe image quality defects identified during manual inspection (Section 2.2.3) that led to the exclusion of three lesions from the training set.

Figure A.4.1 shows a BCC lesion with near-complete Red channel signal loss. Figure A.4.2 - melanoma lesion with catastrophic Green channel failure producing complete black G image. Figure A.4.3 - nevus lesion with similar Green channel dropout for which IR and R channels are still interpretable.

Fig. A.4.1. Example of lesion excluded due to severe channel dropout (bcc, lesion ID: 2_146_15_in_1286_1287)

Fig. A.4.2. Example of lesion excluded due to extreme Green channel failure (mel, lesion ID: 2_893_44_in_7506)

Fig. A.4.3. Example of lesion excluded due to poor Green channel quality (NEVUS, lesion ID: LOC_2017_10_18_3_2)

Appendix 5

Proprietary Dataset Metadata Documentation

This appendix describes the metadata schema and provides a completeness analysis for the proprietary multispectral skin imaging dataset described in Section 2.2. Clinical metadata for each lesion is stored in a nested folder structure with different levels of available data.

The overall composition of the dataset, as shown in Table A.5.1 , includes the distribution of 1,774 lesions across four diagnostic classes (bcc, bkl, mel, nevus) and the availability of complete R/G/IR channel triplets for each lesion.

Table A.5.1.

Overview of diagnostic classes with total lesions, lesions with all R/G/IR channels, and training lesions

Class	**Meaning**	**Number of Lesions (%)**	**Number of Lesions with Complete R/G/IR Channels (%)**	**Number of Training-Eligible Lesions (%)**
bcc	Basal cell carcinoma	452 (25.48%)	450 (25.50%)	449 (25.48%)
bkl	Benign keratosis (solar lentigines / seborrheic keratoses and lichen-planus like keratoses)	251 (14.15%)	250 (14.16%)	250 (14.19%)
mel	Melanoma	539 (30.38%)	538 (30.48%)	537 (30.48%)
nevus	Melanocytic nevus	532 (29.99%)	527 (29.86%)	526 (29.85%)
Total		1774	1765	1762

The availability of metadata files (metadata.csv, search-resolved.csv) is summarized in Table A.5.2 , indicating that 77.28% of lesions (1371/1774) have corresponding clinical metadata stored in CSV files.

Table A.5.2.

Distribution of Metadata Folders, Search-Resolved Files, and Images by Lesion Class

Class	Number of Lesions with Metadata Folder (%)	Number of Lesions with Search-Resolved Info Files (%)	Total Lesions (%)
bcc	376 (83.19%)	376 (83.19%)	452 (100.00%)
bkl	155 (61.75%)	155 (61.75%)	251 (100.00%)
mel	527 (97.77%)	527 (97.77%)	539 (100.00%)
nevus	313 (58.83%)	313 (58.83%)	532 (100.00%)
Total	1371 (77.28%)	1371 (77.28%)	1774 (100.00%)

The distribution pattern of metadata files is analyzed in Table A.5.3, showing that lesions may contain either one or two CSV files with class-specific variations.

Table A.5.3.

Distribution of Metadata CSV Files Across Lesion Classes

Class	Number of Lesions with 1 CSV file (%)	Number of Lesions with 2 CSV files (%)	Total Lesions with metadata folder (%)
bcc	188 (50.00%)	188 (50.00%)	376 (100.00%)
bkl	59 (38.06%)	96 (61.94%)	251 (100.00%)
mel	344 (65.28%)	183 (34.72%)	527 (100.00%)
nevus	28 (8.95%)	285 (91.05%)	313 (100.00%)
Total	619	752	1371

Tables A.5.4 and A.5.5 provide a complete field-level analysis of the fields in the metadata schema. Table A.5.4 describes the fields in the merged search-resolved information file along with the missingness rate for each field. Table A.5.5 describes the full metadata schema from both metadata.csv and metadata2.csv files, including 27 fields.

Table A.5.4.

Search-Resolved Information Fields and Missing Value Analysis

Field	Description	Missing Values (%)
codes	main disease/category code: C44: malignant neoplasms C43: malignant melanoma D22: melanocytic naevi L81: Other disorders of pigmentation	0/1371 (0%)
new name	full lesion/sample identifier	0/1371 (0%)
exp type in md1	a structured metadata field stored as text	0/1371 (0%)
exp type in md2	a structured metadata field stored as text	1371/1371 (100%)
typo	a quality-control or text-normalization flag	0/1371 (0%)
analyze doc comm	final extracted diagnostic note from document analysis from two sources	118/1371 (8.61%)
patient desc	free-text lesion description from two sources	321/1371 (23.41%)
clinical diag	clinical diagnosis text from two sources	941/1371 (68.64%)
clinical code	clinical diagnosis code(s) from two sources	1102/1371 (80.38%)

hist code	histology code(s) from two sources	1128/1371 (82.28%)
hist comm	histology comment(s) from two sources	1135/1371 (82.79%)
analyze diag comm	post-processed diagnosis comment from two sources	1369/1371 (99.85%)

Table A.5.5.

Metadata Field Descriptions and Missing Value Analysis

Column	**Meaning**	**Missing Value in metadata.csv (Percentage)**	**Missing Value in metadata2.csv (Percentage)**
measurementId	Unique identifier for each measurement/scan session	0/1371 (0%)	0/752 (0%)
measurementSetId	ID grouping multiple related measurements (e.g., same session or batch)	0/1371 (0%)	0/752 (0%)
measurementDateCreated	Timestamp when the measurement was created	0/1371 (0%)	0/752 (0%)
measurementDateFinished	Timestamp when the measurement was completed	0/1371 (0%)	0/752 (0%)
receivedImagesCount	Number of images captured in this measurement	0/1371 (0%)	0/752 (0%)
deviceId	ID of the scanning device used	0/1371 (0%)	0/752 (0%)
deviceName	Name of the device	0/1371 (0%)	0/752 (0%)
analyzeId	Unique identifier for the analysis performed on this measurement	76/1371 (5.54%)	0/752 (0%)

numberInSequence	Sequential number of this analysis (e.g., follow-ups)	76/1371 (5.54%)	0/752 (0%)
analyzeDateCreated	Timestamp when the analysis was performed	76/1371 (5.54%)	0/752 (0%)
analyzePatientId	Patient ID from the analysis record	76/1371 (5.54%)	0/752 (0%)
patientId	Patient ID from the measurement record	76/1371 (5.54%)	0/752 (0%)
analyzeScannedBodyPartId	ID of the specific body part/lesion being analyzed	110/1371 (8.02%)	1/752 (0.13%)
analyzeDoctorComment	Doctor's comments during analysis	121/1371 (8.83%)	11/752 (1.46%)
patientDescription	Description or diagnosis summary for the patient	321/1371 (23.41%)	230/752 (30.59%)
clinicalDiagnosis	Clinical diagnosis given before histology	941/1371 (68.64%)	426/752 (56.65%)
clinicalCode	Clinical diagnostic code	1103/1371 (80.45%)	487/752 (64.76%)
histologyCode	Histopathology diagnostic code from biopsy	1129/1371 (82.35%)	731/752 (97.21%)
histologyComment	Additional histology comments	1135/1371 (82.79%)	731/752 (97.21%)
processingScriptId	ID of the image processing algorithm used	1218/1371 (88.84%)	752/752 (100%)
processingScript	Name/description of the processing script	1218/1371 (88.84%)	752/752 (100%)
processingScriptVersion	Version number of the processing script	1218/1371 (88.84%)	752/752 (100%)
clark	Clark's level (depth of invasion in skin layers)	1260/1371 (91.90%)	732/752 (97.34%)
staging	Cancer stage (I, II, III, IV)	1261/1371 (91.98%)	733/752 (97.47%)

breslow	Breslow thickness (tumor thickness in mm)	1264/1371 (92.20%)	736/752 (97.87%)
analyzeDiagnosisComment	Additional diagnostic comments or notes	1369/1371 (99.85%)	750/752 (99.73%)
measurementComment	Optional notes or comments about the measurement	1370/1371 (99.93%)	752/752 (100%)

Appendix 6

HAM10000 Dataset Training Set Distributions Across Cross-Validation Folds

This appendix documents the class distributions in the training sets across all five folds of the stratified cross-validation experiments conducted on the HAM10000 pretraining dataset (Section 3.1.2 and 3.1.3). Each fold's training set contains approximately 7,500 samples after withholding the corresponding validation set.

Tables A.6.1 through A.6.5 present the training set composition for folds 1 through 5, showing both absolute counts and percentages for the four retained diagnostic classes (bcc, bkl, mel, nevus). The distributions confirm that stratified splitting preserved the inherent class imbalance across all training folds.

Table A.6.1.

HAM10000 Training Set Distribution (Fold 1)

Training Set	bcc: 0	bkl: 1	mel: 2	nv: 3	Total
Count	413	921	896	5355	7585
Percentage	5.44	12.14	11.81	70.61	100.00

Table A.6.2.

HAM10000 Training Set Distribution (Fold 2)

Training Set	bcc: 0	bkl: 1	mel: 2	nv: 3	Total
Count	423	888	856	5340	7507
Percentage	5.64	11.83	11.40	71.13	100.00

Table A.6.3.

HAM10000 Training Set Distribution (Fold 3)

Training Set	bcc: 0	bkl: 1	mel: 2	nv: 3	Total
Count	397	838	909	5387	7531

Percentage	5.27	11.13	12.07	71.53	100

Table A.6.4.

HAM10000 Training Set Distribution (Fold 4)

Training Set	**bcc: 0**	**bkl: 1**	**mel: 2**	**nv: 3**	**Total**
Count	420	881	895	5364	7560
Percentage	5.56	11.65	11.84	70.95	100

Table A.6.5.

HAM10000 Training Set Distribution (Fold 5)

Training Set	**bcc: 0**	**bkl: 1**	**mel: 2**	**nv: 3**	**Total**
Count	403	868	896	5374	7541
Percentage	5.35	11.51	11.88	71.26	100

Appendix 7

HAM10000 Dataset Validation Set Distributions Across Cross-Validation Folds

This appendix documents the class distributions in the validation sets across all five folds of the stratified cross-validation experiments conducted on the HAM10000 pretraining dataset (Section 3.1.2). Each fold's validation set contains approximately 1,900 samples (20% of the total 9,431 filtered samples).

Tables A.7.1-A.7.5 present the composition of the validation set for fold 1-5, with absolute counts and percentages for the four diagnostic classes that were retained (bcc, bkl, mel, nevus).

Table A.7.1.

HAM10000 Validation Set Distribution (Fold 1)

Validation Set	bcc: 0	bkl: 1	mel: 2	nv: 3	Total
Count	101	178	217	1350	1846
Percentage	5.47	9.64	11.76	73.13	100.00

Table A.7.2.

HAM10000 Validation Set Distribution (Fold 2)

Validation Set	bcc: 0	bkl: 1	mel: 2	nv: 3	Total
Count	91	211	257	1365	1924
Percentage	4.73	10.97	13.36	70.94	100.00

Table A.7.3.

HAM10000 Validation Set Distribution (Fold 3)

Validation Set	bcc: 0	bkl: 1	mel: 2	nv: 3	Total
Count	117	261	204	1318	1900
Percentage	6.16	13.73	10.74	69.37	100.00

Table A.7.4.

HAM10000 Validation Set Distribution (Fold 4)

Validation Set	bcc: 0	bkl: 1	mel: 2	nv: 3	Total
Count	94	218	218	1341	1871
Percentage	5.03	11.65	11.65	71.67	100

Table A.7.5.

HAM10000 Validation Set Distribution (Fold 5)

Validation Set	bcc: 0	bkl: 1	mel: 2	nv: 3	Total
Count	111	231	217	1331	1890
Percentage	5.87	12.22	11.48	70.43	100.00

Appendix 8

HAM10000 Training Distribution Rebalancing with WeightedRandomSampler

In this appendix, it is discussed the effectiveness of the WeightedRandomSampler strategy adopted in HAM10000 pretraining to address the extreme class imbalance (Section 3.1.3). As mentioned in Appendix 6 and Section 2.1, the original training sets suffer from severe imbalance. To avoid the model collapsing to majority class predictions, WeightedRandomSampler (Section 3.1.3) was used to oversample minority classes in each training epoch.

Tables A.8.1-A.8.5 present simulated class distributions for a single training epoch across folds 1-5, showcasing the rebalanced sampling outcomes. The WeightedRandomSampler succeeded to translate the original imbalanced distributions into a nearly uniform class proportion, where all the four classes (bcc, bkl, mel, nevus) are equally represented at the rate of around 24-26% and each class is assigned with 1,800-1,900 samples per epoch.

Table A.8.1.

HAM10000 Simulated Training Epoch Class Distribution with WeightedRandomSampler (Fold 1)

Training Set	bcc: 0	bkl: 1	mel: 2	nv: 3	Total
Count	1904	1890	1870	1920	7584
Percentage	25.10	24.92	24.66	25.32	100.00

Table A.8.2.

HAM10000 Simulated Training Epoch Class Distribution with WeightedRandomSampler (Fold 2)

Training Set	bcc: 0	bkl: 1	mel: 2	nv: 3	Total
Count	1869	1846	1859	1914	7488
Percentage	24.96	24.65	24.83	25.56	100.00

Table A.8.3.

HAM10000 Simulated Training Epoch Class Distribution with WeightedRandomSampler (Fold 3)

Training Set	bcc: 0	bkl: 1	mel: 2	nv: 3	Total
Count	1813	1931	1923	1853	7520
Percentage	24.11	25.68	25.57	24.64	100.00

Table A.8.4.

HAM10000 Simulated Training Epoch Class Distribution with WeightedRandomSampler (Fold 4)

Training Set	bcc: 0	bkl: 1	mel: 2	nv: 3	Total
Count	1878	1882	1941	1851	7552
Percentage	24.87	24.92	25.70	24.51	100.00

Table A.8.5.

HAM10000 Simulated Training Epoch Class Distribution with WeightedRandomSampler (Fold 5)

Training Set	bcc: 0	bkl: 1	mel: 2	nv: 3	Total
Count	1863	1833	1851	1973	7520
Percentage	24.77	24.38	24.61	26.24	100.00

Appendix 9

Hair Removal Preprocessing Results - HAM10000 Dataset

This appendix presents visual examples to demonstrate the effectiveness of the morphological hair removal preprocessing on the HAM10000 dataset (Section 3.1.3). Figures A.9.1 to A.9.4 show before-and-after comparisons with different severities of hair artifacts.

Fig. A.9.1. Effect of Hair Removal Preprocessing on Dermoscopic Image Without Hair Artifacts

Fig. A.9.2. Effect of Hair Removal on Dermoscopic Image with Thin Hair Filaments

Fig. A.9.3. Effect of Hair Removal on Dermoscopic Image with Sparse, Thick, and Curvilinear Hair Structures

Fig. A.9.4. Effect of Hair Removal on Dermoscopic Image with Long, Thin, and Curvilinear Hair Structures

Appendix 10

Proprietary Dataset Training Set Distributions Across Cross-Validation Folds

This appendix presents the class distributions in the training sets of the five folds of the stratified cross-validation experiments on the proprietary multispectral dataset (Section 3.2.2). Each fold's training set contains approximately 1,400-1,410 samples after withholding the corresponding validation set.

Tables A.10.1 through A.10.5 present the training set composition for folds 1 through 5, showing both absolute counts and percentages for the four diagnostic classes (bcc, bkl, mel, nevus).

Table A.10.1.

Training Set Class Distribution (Fold 1) - Proprietary Multispectral Dataset

Training Set	bcc: 0	bkl: 1	mel: 2	nv: 3	Total
Count	354	200	428	427	1409
Percentage	25.12	14.19	30.38	30.31	100

Table A.10.2.

Training Set Class Distribution (Fold 2) - Proprietary Multispectral Dataset

Training Set	bcc: 0	bkl: 1	mel: 2	nv: 3	Total
Count	361	200	428	420	1409
Percentage	25.62	14.19	30.38	29.81	100

Table A.10.3.

Training Set Class Distribution (Fold 3) - Proprietary Multispectral Dataset

Training Set	bcc: 0	bkl: 1	mel: 2	nv: 3	Total
Count	363	200	433	414	1410
Percentage	25.75	14.18	30.71	29.36	100.00

Table A.10.4.

Training Set Class Distribution (Fold 4) - Proprietary Multispectral Dataset

Training Set	bcc: 0	bkl: 1	mel: 2	nv: 3	Total
Count	360	200	427	423	1410
Percentage	25.53	14.19	30.28	30.00	100.00

Table A.10.5.

Training Set Class Distribution (Fold 5) - Proprietary Multispectral Dataset

Training Set	bcc: 0	bkl: 1	mel: 2	nv: 3	Total
Count	358	200	432	420	1410
Percentage	25.39	14.18	30.64	29.79	100.00

Appendix 11

Proprietary Dataset Validation Set Distributions Across Cross-Validation Folds

This appendix reports the class distributions for all five validation folds of the stratified cross-validation experiments performed on the proprietary multispectral dataset (Section 3.2.2). The stratification strategy was designed to keep the class ratios stable between folds and to prevent any lesion-level data leakage between the training and validation sets. Each fold's validation set contains approximately 350-360 samples (20% of the total 1,762 training-eligible lesions).

Tables A.11.1 to A.11.5 list the validation set composition for each fold of the proprietary multispectral dataset, including absolute numbers and percentages for the four diagnostic classes (bcc, bkl, mel, nevus).

Table A.11.1

Validation Set Class Distribution (Fold 1) - Proprietary Multispectral Dataset

Validation Set	bcc: 0	bkl: 1	mel: 2	nv: 3	Total
Count	95	50	109	99	353
Percentage	26.91	14.16	30.88	28.05	100.00

Table A.11.2

Validation Set Class Distribution (Fold 2) - Proprietary Multispectral Dataset

Validation Set	bcc: 0	bkl: 1	mel: 2	nv: 3	Total
Count	88	50	109	106	353
Percentage	24.93	14.16	30.88	30.03	100.00

Table A.11.3

Validation Set Class Distribution (Fold 3) - Proprietary Multispectral Dataset

Validation Set	bcc: 0	bkl: 1	mel: 2	nv: 3	Total
Count	86	50	104	112	352

Percentage	24.43	14.20	29.55	31.82	100.00

Table A.11.4

Validation Set Class Distribution (Fold 4) - Proprietary Multispectral Dataset

Validation Set	bcc: 0	bkl: 1	mel: 2	nv: 3	Total
Count	89	50	110	103	352
Percentage	25.28	14.21	31.25	29.26	100.00

Table A.11.5

Validation Set Class Distribution (Fold 5) - Proprietary Multispectral Dataset

Validation Set	bcc: 0	bkl: 1	mel: 2	nv: 3	Total
Count	91	50	105	106	352
Percentage	25.85	14.21	29.83	30.11	100.00

Appendix 12

Proprietary Dataset Training Distribution Rebalancing with WeightedRandomSampler

This appendix demonstrates the class distribution of training sets after applying WeightedRandomSampler during each training epoch for the proprietary multispectral dataset (Section 3.2.2).

Table A.12.1 presents the simulated class distribution for one training epoch across all five folds.

Table A.12.1

Simulated Training Epoch Class Distribution with WeightedRandomSampler (Folds 1-5) - Proprietary Dataset

Training Set	bcc: 0	bkl: 1	mel: 2	nv: 3	Total
Count	693	715	713	695	2816
Percentage	24.61	25.39	25.32	24.68	100.00

Appendix 13

Hair Removal Preprocessing Results - Proprietary Multispectral Dataset

This appendix provides visual examples demonstrating the effectiveness of the channel-specific hair removal preprocessing on the proprietary multispectral dataset (Section 3.2.3).

Figures A.13.1 to A.13.4 show before-and-after comparisons for R, G and IR channels on lesions with different severities of hair artifacts.

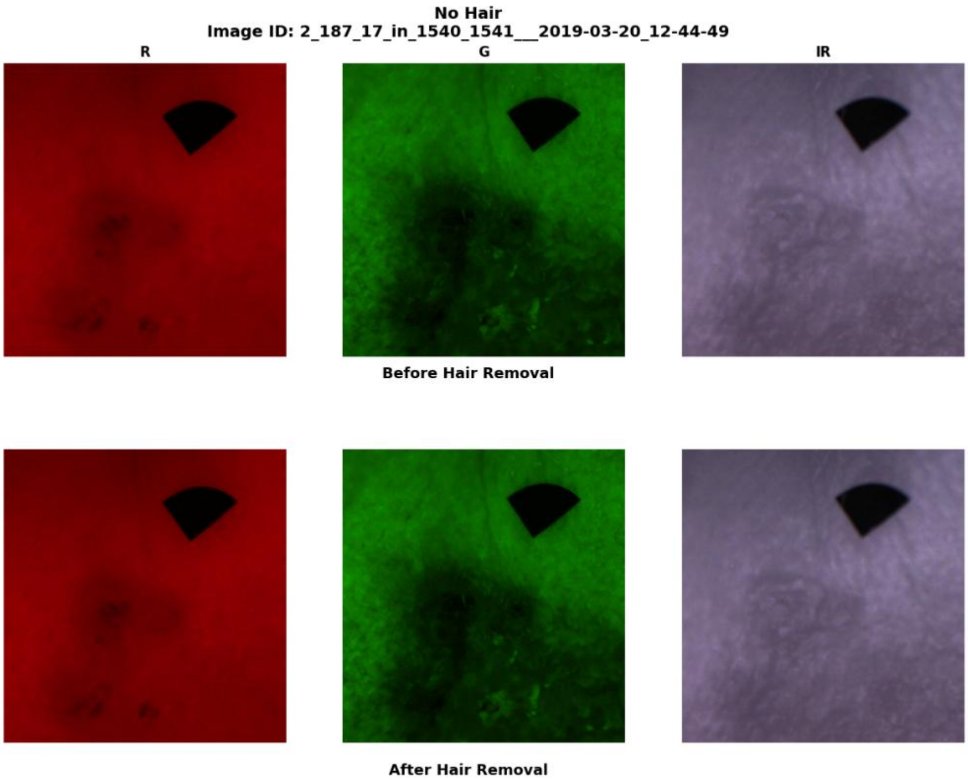

Fig. A.13.1. Effect of Hair Removal Preprocessing on Multispectral Images Without Hair Artifacts

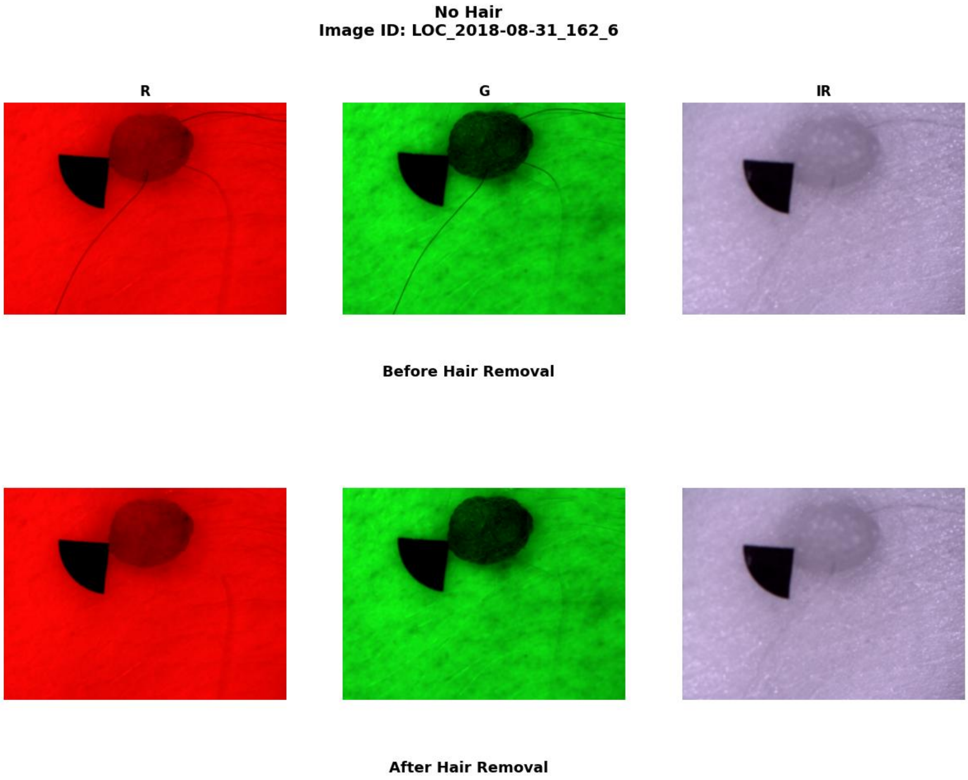

Fig. A.13.2. Effect of Hair Removal on Multispectral Images with Long, Thin, and Curvilinear Hair Structures

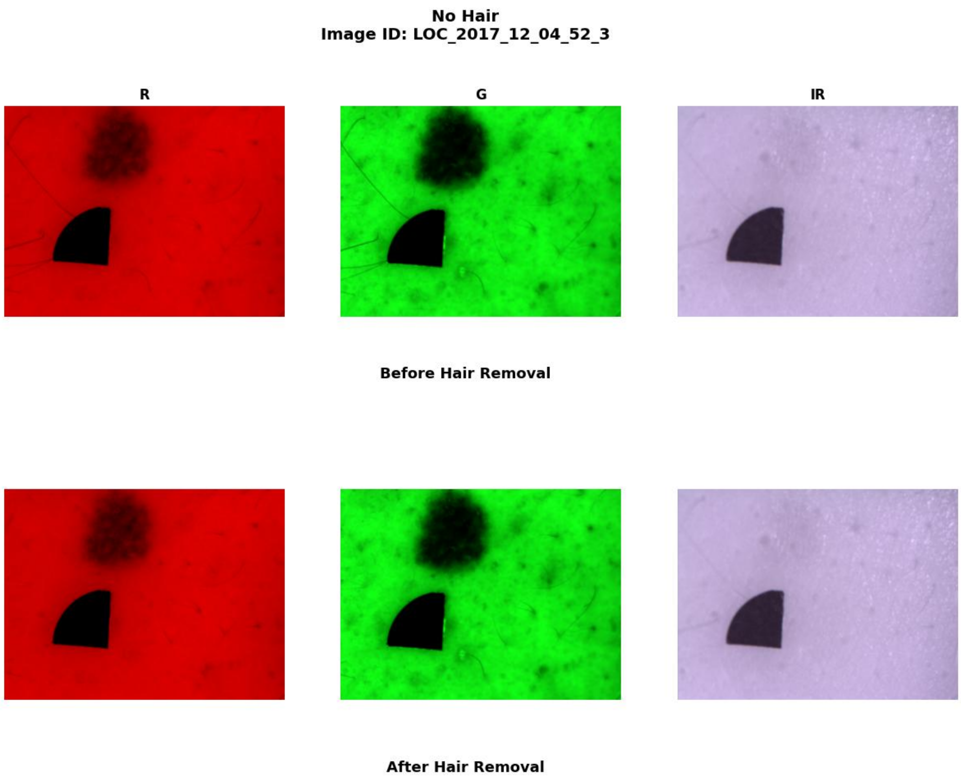

Fig. A.13.3. Effect of Hair Removal on Multispectral Images with Sparse, and Curvilinear Hair Structures

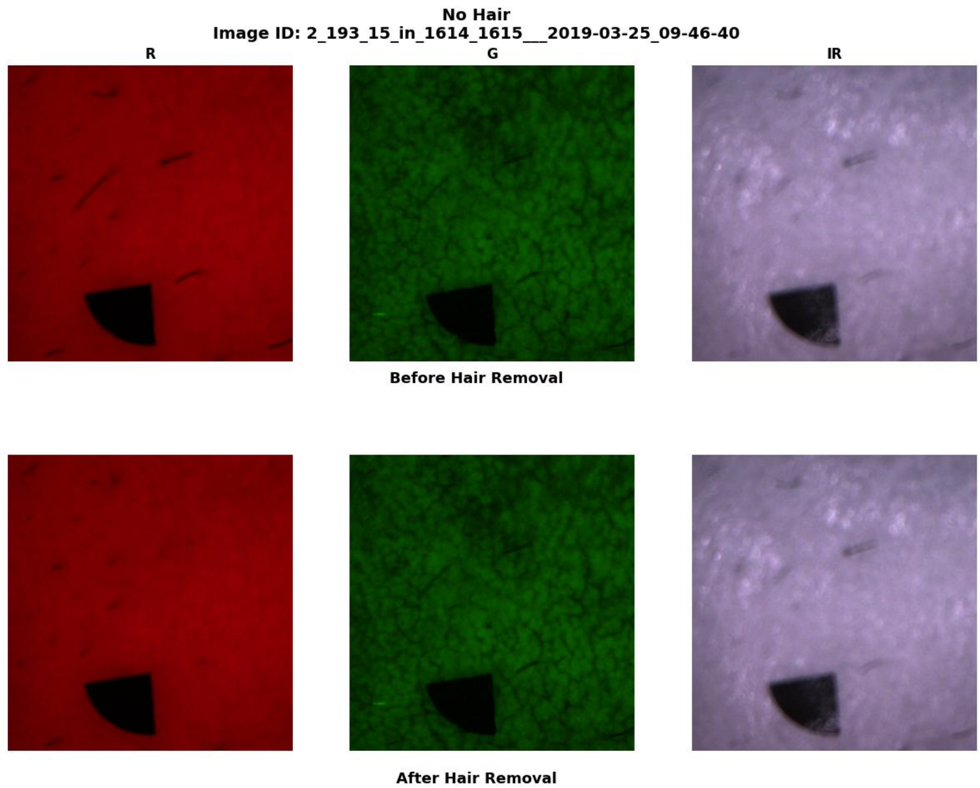

Fig. A.13.4. Effect of Hair Removal on Multispectral Images with Sparse, Thick, and Curvilinear Hair Structures

Appendix 14

Confusion Matrices - ImageOnlySkinNet on HAM10000 Dataset

This appendix shows detailed ImageOnlySkinNet baseline model confusion matrices for all 5 cross validation folds on the HAM10000 dataset (Section 4.2.2).

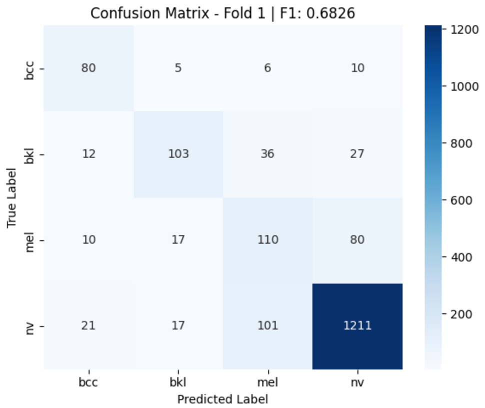

Fig. A.14.1. Confusion Matrix for ImageOnlySkinNet on HAM10000 Validation Set (Fold 1)

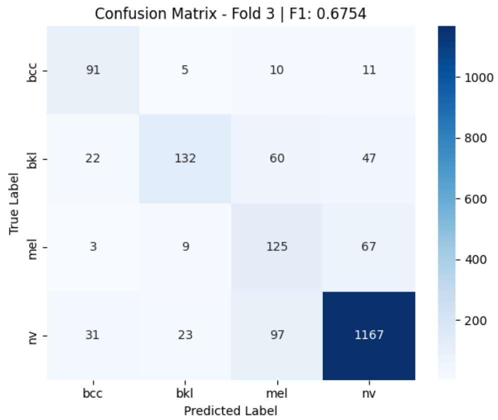

Fig. A.14.3. Confusion Matrix for ImageOnlySkinNet on HAM10000 Validation Set (Fold 3)

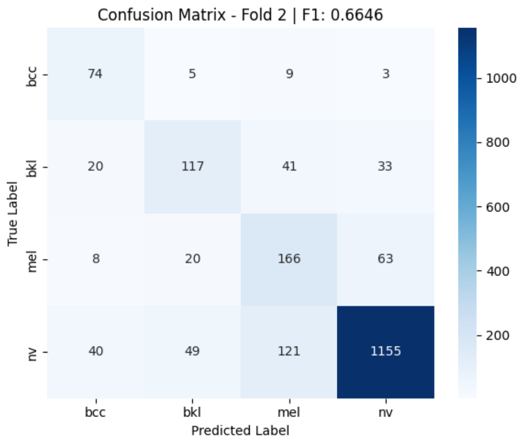

Fig. A.14.2. Confusion Matrix for ImageOnlySkinNet on HAM10000 Validation Set (Fold 2)

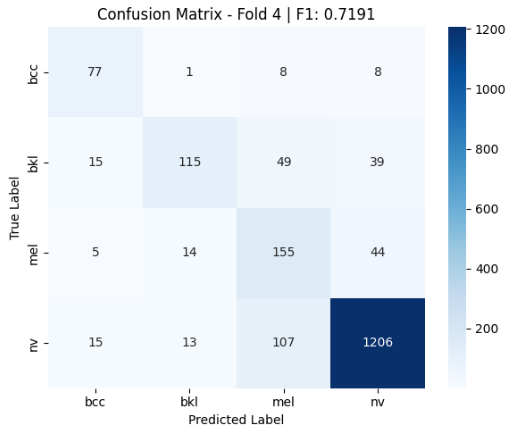

Fig. A.14.4. Confusion Matrix for ImageOnlySkinNet on HAM10000 Validation Set (Fold 4)

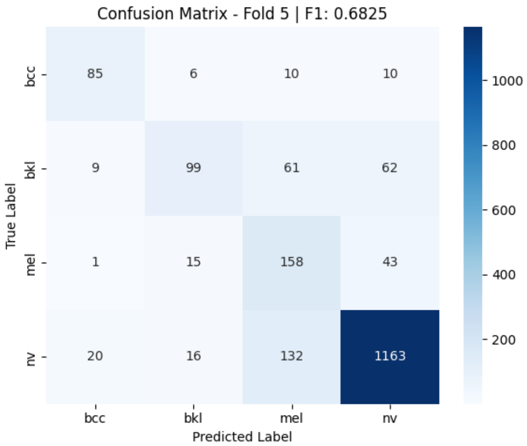

Fig. A.14.5. Confusion Matrix for ImageOnlySkinNet on HAM10000 Validation Set (Fold 5)

Appendix 15

Confusion Matrices - MultiModalSkinNet on HAM10000 Dataset

This appendix presents detailed confusion matrices for the MultiModalSkinNet multimodal fusion model across all five cross-validation folds on the HAM10000 dataset (Section 4.2.2).

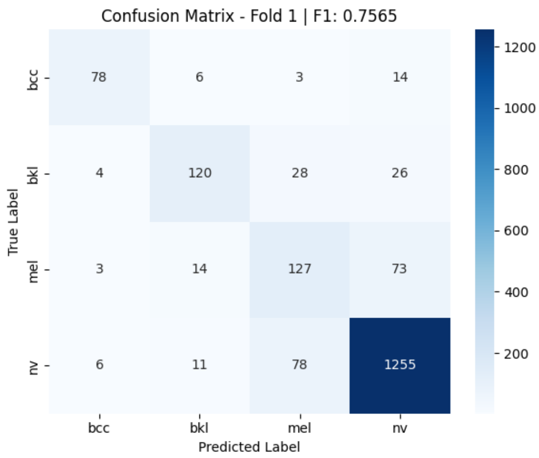

Fig. A.15.1. Confusion Matrix for MultiModalSkinNet on HAM10000 Validation Set (Fold 1)

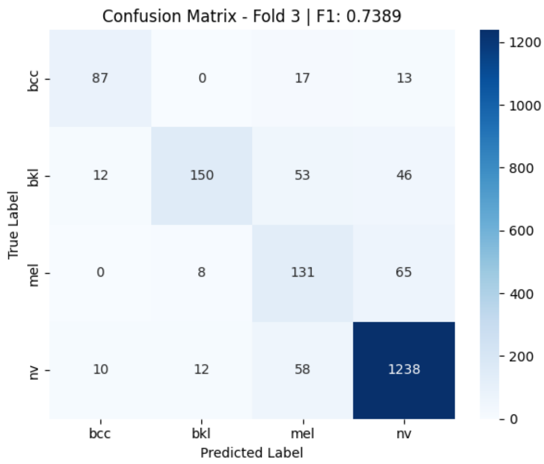

Fig. A.15.3. Confusion Matrix for MultiModalSkinNet on HAM10000 Validation Set (Fold 3)

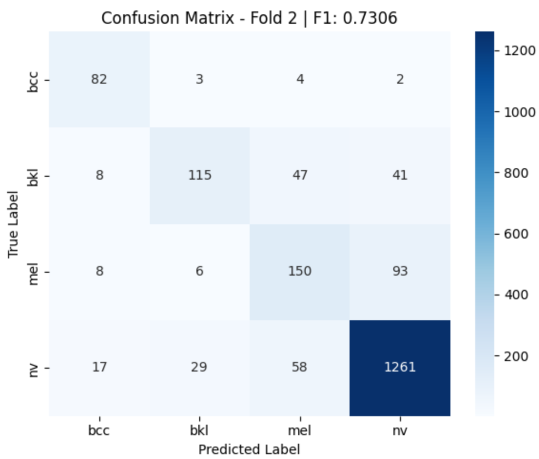

Fig. A.15.2. Confusion Matrix for MultiModalSkinNet on HAM10000 Validation Set (Fold 2)

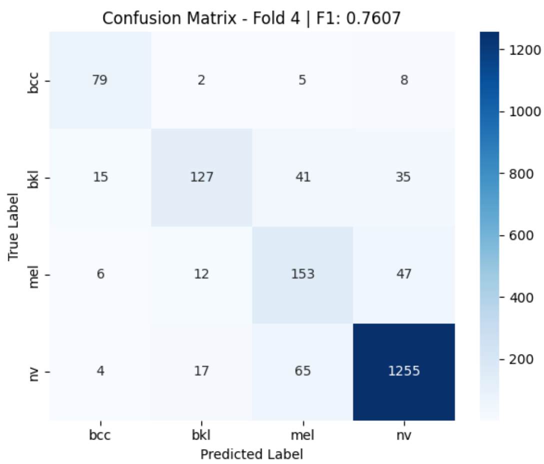

Fig. A.15.4. Confusion Matrix for MultiModalSkinNet on HAM10000 Validation Set (Fold 4)

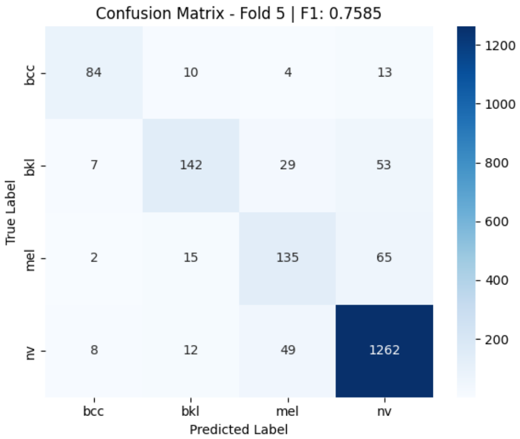

Fig. A.15.5. Confusion Matrix for MultiModalSkinNet on HAM10000 Validation Set (Fold 5)

Appendix 16

ROC Curves - ImageOnlySkinNet on HAM10000 Dataset

This appendix presents one-vs-rest ROC curves for the ImageOnlySkinNet baseline model across all five cross-validation folds on the HAM10000 dataset (Section 4.2.3).

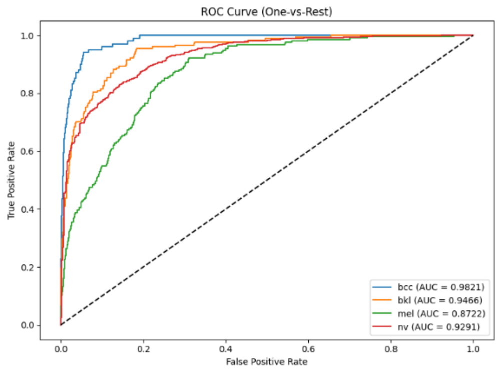

Fig. A.16.1. One-vs-Rest ROC Curves for ImageOnlySkinNet on HAM10000 Validation Set (Fold 1)

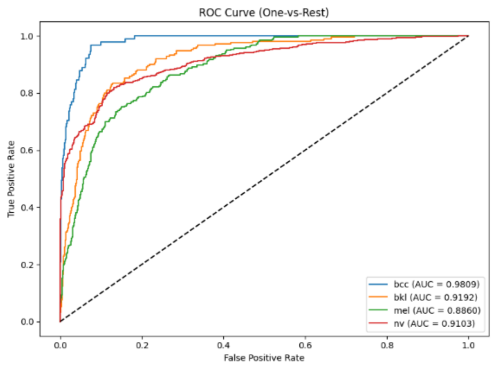

Fig. A.16.2. One-vs-Rest ROC Curves for ImageOnlySkinNet on HAM10000 Validation Set (Fold 2)

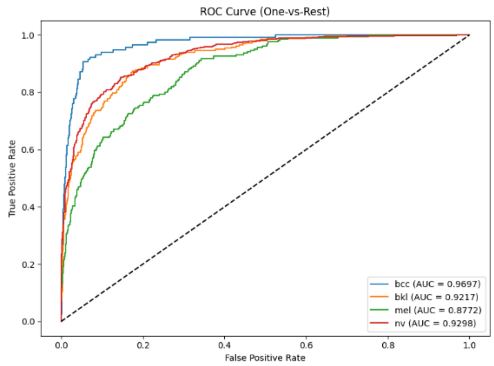

Fig. A.16.3. One-vs-Rest ROC Curves for ImageOnlySkinNet on HAM10000 Validation Set (Fold 3)

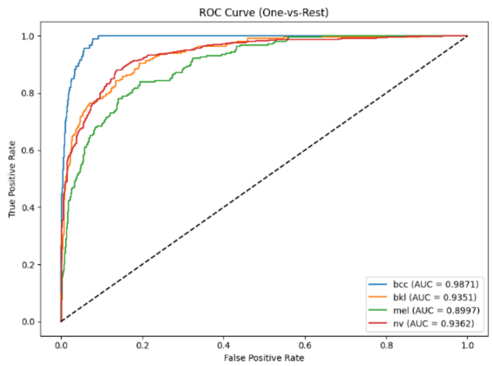

Fig. A.16.4. One-vs-Rest ROC Curves for ImageOnlySkinNet on HAM10000 Validation Set (Fold 4)

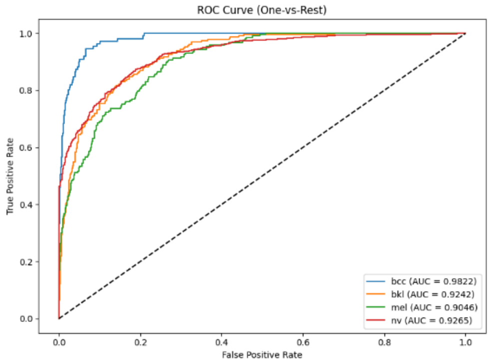

Fig. A.16.5. One-vs-Rest ROC Curves for ImageOnlySkinNet on HAM10000 Validation Set (Fold 5)

Appendix 17

ROC Curves - MultiModalSkinNet on HAM10000 Dataset

This appendix presents one-vs-rest ROC curves for the MultiModalSkinNet multimodal fusion model across all five cross-validation folds on the HAM10000 dataset (Section 4.2.3).

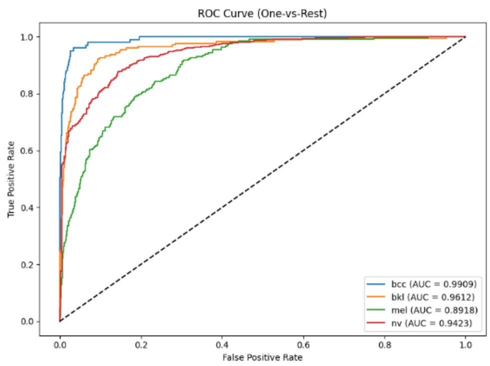

Fig. A.17.1. One-vs-Rest ROC Curves for MultiModalSkinNet on HAM10000 Validation Set (Fold 1)

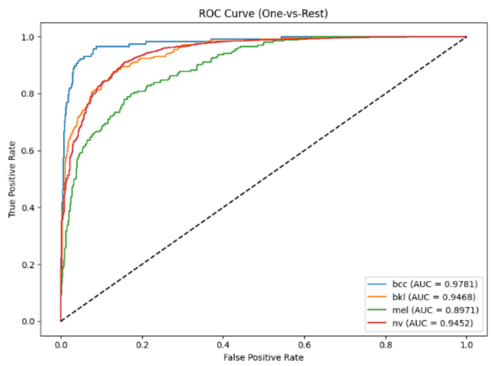

Fig. A.17.3. One-vs-Rest ROC Curves for MultiModalSkinNet on HAM10000 Validation Set (Fold 3)

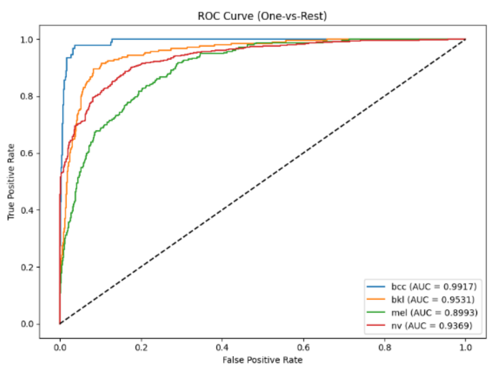

Fig. A.17.2. One-vs-Rest ROC Curves for MultiModalSkinNet on HAM10000 Validation Set (Fold 2)

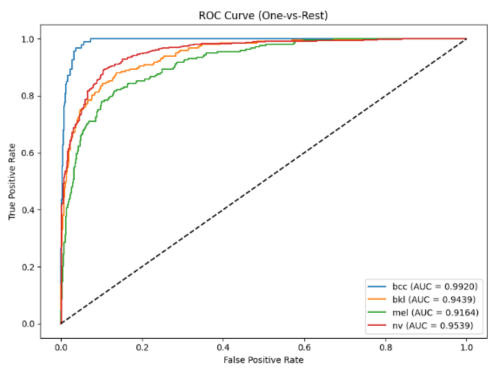

Fig. A.17.4. One-vs-Rest ROC Curves for MultiModalSkinNet on HAM10000 Validation Set (Fold 4)

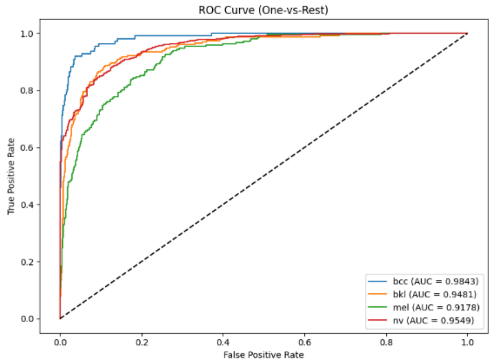

Fig. A.17.5. One-vs-Rest ROC Curves for MultiModalSkinNet on HAM10000 Validation Set (Fold 5)

Appendix 18

Per-Class Classification Reports - ImageOnlySkinNet on HAM10000 Dataset

This appendix provides the per-class precision, recall and F1-score metrics of the ImageOnlySkinNet baseline model for all five folds of the cross-validation on the HAM10000 dataset (Section 4.2.4).

Fold 1 – Per-class metrics:

	precision	recall	f1-score	support
bcc	0.6504	0.7921	0.7143	101
bkl	0.7254	0.5787	0.6438	178
mel	0.4348	0.5069	0.4681	217
nv	0.9119	0.8970	0.9044	1350
accuracy			0.8147	1846
macro avg	0.6806	0.6937	0.6826	1846
weighted avg	0.8235	0.8147	0.8176	1846

Fig. A.18.1. Per-Class Classification Report for ImageOnlySkinNet on HAM10000 Validation Set (Fold 1)

Fold 2 – Per-class metrics:

	precision	recall	f1-score	support
bcc	0.5211	0.8132	0.6352	91
bkl	0.6126	0.5545	0.5821	211
mel	0.4926	0.6459	0.5589	257
nv	0.9211	0.8462	0.8820	1365
accuracy			0.7859	1924
macro avg	0.6368	0.7149	0.6646	1924
weighted avg	0.8111	0.7859	0.7943	1924

Fig A.18.2. Per-Class Classification Report for ImageOnlySkinNet on HAM10000 Validation Set (Fold 2)

Fold 3 – Per-class metrics:

	precision	recall	f1-score	support
bcc	0.6190	0.7778	0.6894	117
bkl	0.7811	0.5057	0.6140	261
mel	0.4281	0.6127	0.5040	204
nv	0.9033	0.8854	0.8943	1318
accuracy			0.7974	1900
macro avg	0.6829	0.6954	0.6754	1900
weighted avg	0.8179	0.7974	0.8012	1900

Fig. A.18.3. Per-Class Classification Report for ImageOnlySkinNet on HAM10000 Validation Set (Fold 3)

Fold 4 – Per-class metrics:

	precision	recall	f1-score	support
bcc	0.6875	0.8191	0.7476	94
bkl	0.8042	0.5275	0.6371	218
mel	0.4859	0.7110	0.5773	218
nv	0.9298	0.8993	0.9143	1341
accuracy			0.8300	1871
macro avg	0.7269	0.7393	0.7191	1871
weighted avg	0.8513	0.8300	0.8344	1871

Fig. A.18.4. Per-Class Classification Report for ImageOnlySkinNet on HAM10000 Validation Set (Fold 4)

Fold 5 – Per-class metrics:

	precision	recall	f1-score	support
bcc	0.7391	0.7658	0.7522	111
bkl	0.7279	0.4286	0.5395	231
mel	0.4377	0.7281	0.5467	217
nv	0.9100	0.8738	0.8915	1331
accuracy			0.7963	1890
macro avg	0.7037	0.6991	0.6825	1890
weighted avg	0.8235	0.7963	0.8007	1890

Fig. A.18.5. Per-Class Classification Report for ImageOnlySkinNet on HAM10000 Validation Set (Fold 5)

Appendix 19

Per-Class Classification Reports - MultiModalSkinNet on HAM10000 Dataset

In this appendix, it is presented the detailed per-class precision, recall and F1-score metrics for the MultiModalSkinNet multimodal fusion model over all five folds of cross-validation on the HAM10000 dataset (Section 4.2.4).

Fold 1 – Per-class metrics:

	precision	recall	f1-score	support
bcc	0.8571	0.7723	0.8125	101
bkl	0.7947	0.6742	0.7295	178
mel	0.5381	0.5853	0.5607	217
nv	0.9174	0.9296	0.9235	1350
accuracy			0.8559	1846
macro avg	0.7768	0.7403	0.7565	1846
weighted avg	0.8577	0.8559	0.8561	1846

Fig. A.19.1. Per-Class Classification Report for MultiModalSkinNet on HAM10000 Validation Set (Fold 1)

Fold 2 – Per-class metrics:

	precision	recall	f1-score	support
bcc	0.7130	0.9011	0.7961	91
bkl	0.7516	0.5450	0.6319	211
mel	0.5792	0.5837	0.5814	257
nv	0.9026	0.9238	0.9131	1365
accuracy			0.8358	1924
macro avg	0.7366	0.7384	0.7306	1924
weighted avg	0.8339	0.8358	0.8324	1924

Fig. A.19.2. Per-Class Classification Report for MultiModalSkinNet on HAM10000 Validation Set (Fold 2)

Fold 3 – Per-class metrics:

	precision	recall	f1-score	support
bcc	0.7982	0.7436	0.7699	117
bkl	0.8824	0.5747	0.6961	261
mel	0.5058	0.6422	0.5659	204
nv	0.9090	0.9393	0.9239	1318
accuracy			0.8453	1900
macro avg	0.7738	0.7249	0.7389	1900
weighted avg	0.8552	0.8453	0.8447	1900

Fig. A.19.3. Per-Class Classification Report for MultiModalSkinNet on HAM10000 Validation Set (Fold 3)

Fold 4 – Per-class metrics:

	precision	recall	f1-score	support
bcc	0.7596	0.8404	0.7980	94
bkl	0.8038	0.5826	0.6755	218
mel	0.5795	0.7018	0.6349	218
nv	0.9331	0.9359	0.9345	1341
accuracy			0.8626	1871
macro avg	0.7690	0.7652	0.7607	1871
weighted avg	0.8681	0.8626	0.8625	1871

Fig. A.19.4. Per-Class Classification Report for MultiModalSkinNet on HAM10000 Validation Set (Fold 4)

Fold 5 – Per-class metrics:

	precision	recall	f1-score	support
bcc	0.8317	0.7568	0.7925	111
bkl	0.7933	0.6147	0.6927	231
mel	0.6221	0.6221	0.6221	217
nv	0.9060	0.9482	0.9266	1331
accuracy			0.8587	1890
macro avg	0.7883	0.7354	0.7585	1890
weighted avg	0.8552	0.8587	0.8552	1890

Fig. A.19.5. Per-Class Classification Report for MultiModalSkinNet on HAM10000 Validation Set (Fold 5)

Appendix 20

Confusion Matrices - EfficientNet-B0 with ImageNet1K Pretrained Weights on Proprietary Multispectral Dataset

This appendix presents detailed confusion matrices for the ImageNet1K-only pretrained baseline model across all five cross-validation folds on the proprietary multispectral dataset (Section 4.4.2).

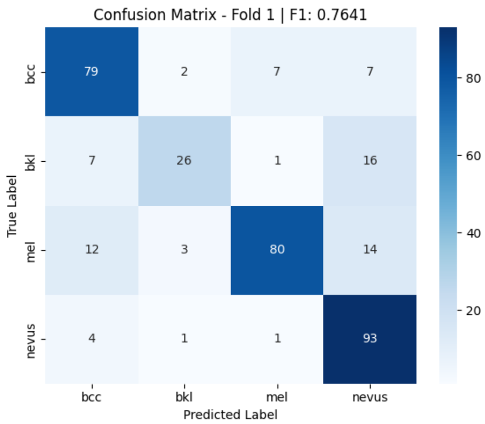

Fig. A.20.1. Confusion Matrix for EfficientNet-B0 (ImageNet1K Weights Only) on Proprietary Dataset Validation Set (Fold 1)

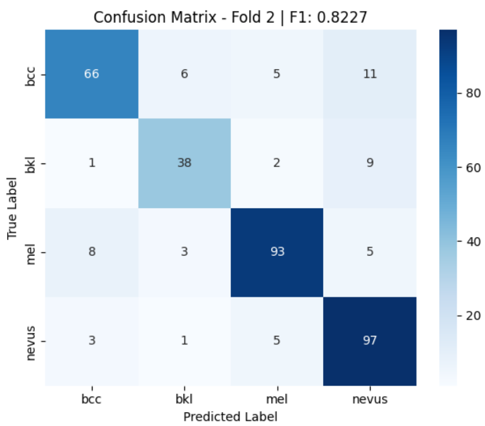

Fig. A.20.2. Confusion Matrix for EfficientNet-B0 (ImageNet1K Weights Only) on Proprietary Dataset Validation Set (Fold 2)

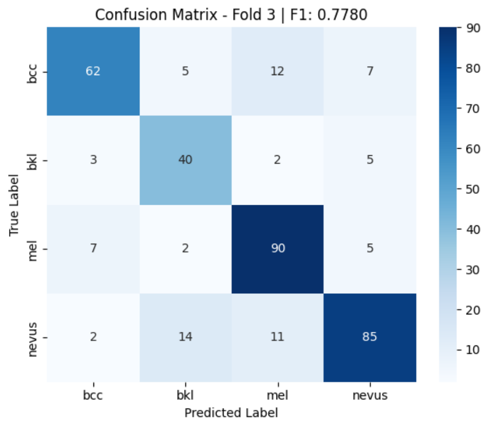

Fig. A.20.3. Confusion Matrix for EfficientNet-B0 (ImageNet1K Weights Only) on Proprietary Dataset Validation Set (Fold 3)

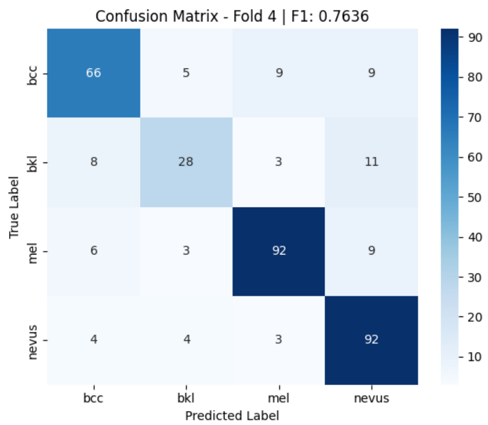

Fig. A.20.4. Confusion Matrix for EfficientNet-B0 (ImageNet1K Weights Only) on Proprietary Dataset Validation Set (Fold 4)

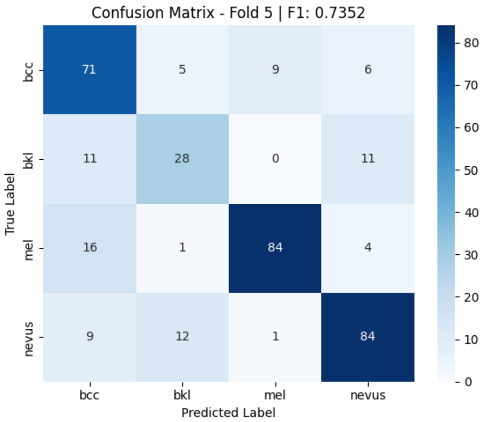

Fig. A.20.5. Confusion Matrix for EfficientNet-B0 (ImageNet1K Weights Only) on Proprietary Dataset Validation Set (Fold 5)

Appendix 21

Confusion Matrices - EfficientNet-B0 with HAM10000-Pretrained Weights on Proprietary Multispectral Dataset

This appendix presents detailed confusion matrices for the HAM10000-pretrained EfficientNet-B0 model across all five cross-validation folds on the proprietary multispectral dataset (Section 4.4.2).

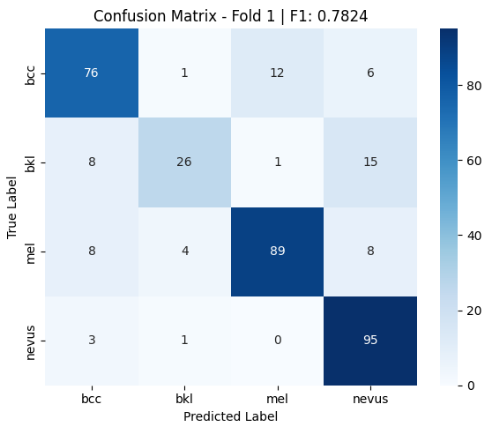

Fig. A.21.1. Confusion Matrix for EfficientNet-B0 with HAM10000-Pretrained Weights on Proprietary Dataset (Fold 1)

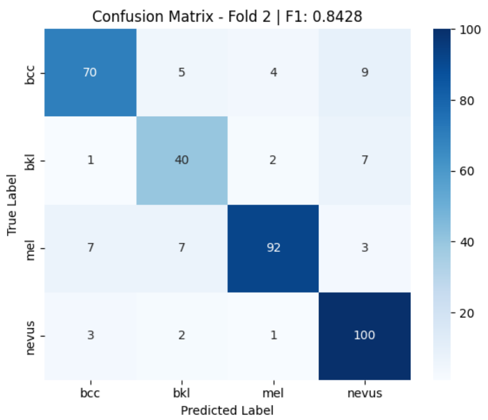

Fig. A.21.2. Confusion Matrix for EfficientNet-B0 with HAM10000-Pretrained Weights on Proprietary Dataset (Fold 2)

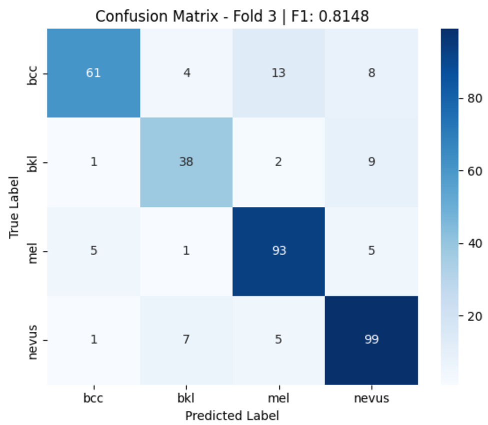

Fig. A.21.3. Confusion Matrix for EfficientNet-B0 with HAM10000-Pretrained Weights on Proprietary Dataset (Fold 3)

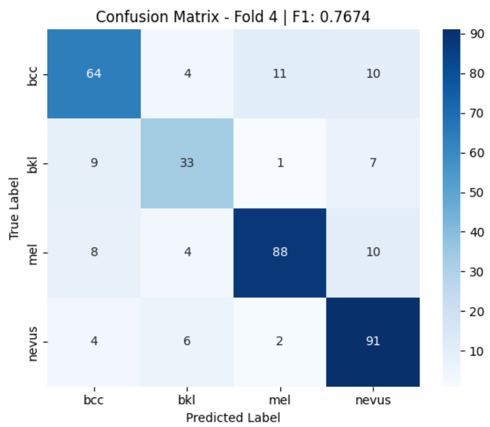

Fig. A.21.4. Confusion Matrix for EfficientNet-B0 with HAM10000-Pretrained Weights on Proprietary Dataset (Fold 4)

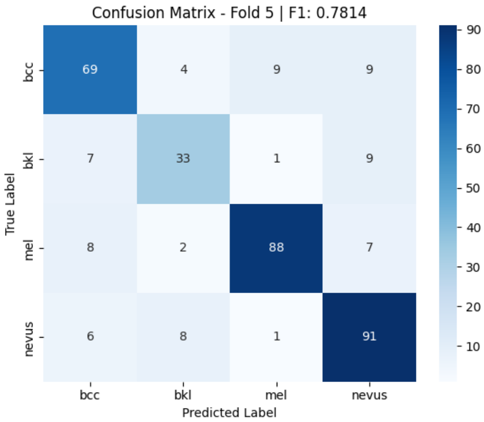

Fig. A.21.5. Confusion Matrix for EfficientNet-B0 with HAM10000-Pretrained Weights on Proprietary Dataset (Fold 5)

Appendix 22

ROC Curves - EfficientNet-B0 with ImageNet1K Pretrained Weights on Proprietary Multispectral Dataset

This appendix presents one-vs-rest ROC curves for the ImageNet1K-only pretrained baseline model across all five cross-validation folds on the proprietary multispectral dataset (Section 4.4.3).

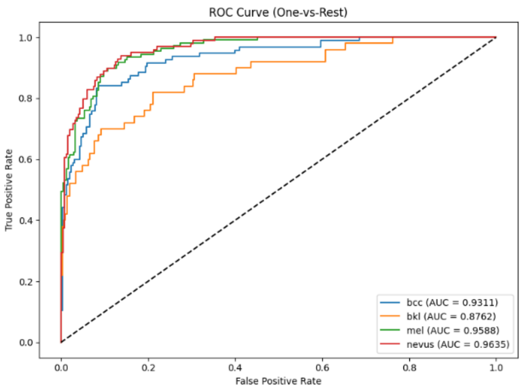

Fig. A.22.1. One-vs-Rest ROC Curves for ImageNet1K-Pretrained EfficientNet-B0 (Fold 1) - Proprietary Dataset

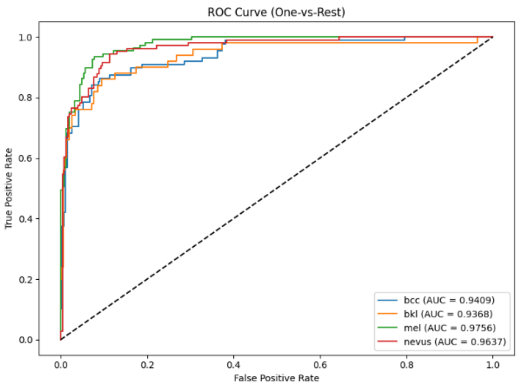

Fig. A.22.2. One-vs-Rest ROC Curves for ImageNet1K-Pretrained EfficientNet-B0 (Fold 2) - Proprietary Dataset

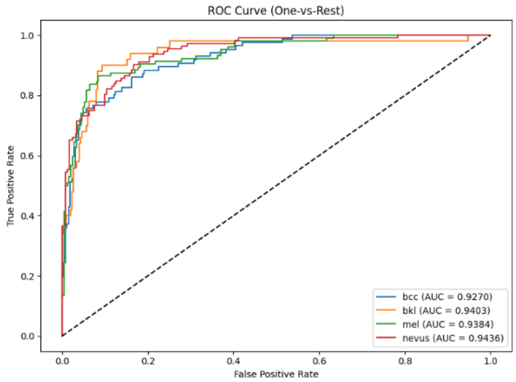

Fig. A.22.3. One-vs-Rest ROC Curves for ImageNet1K-Pretrained EfficientNet-B0 (Fold 3) - Proprietary Dataset

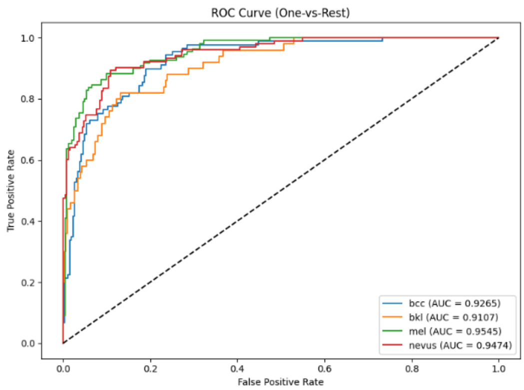

Fig. A.22.4. One-vs-Rest ROC Curves for ImageNet1K-Pretrained EfficientNet-B0 (Fold 4) - Proprietary Dataset

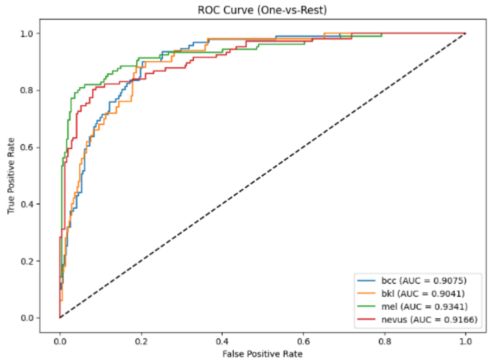

Fig. A.22.5. One-vs-Rest ROC Curves for ImageNet1K-Pretrained EfficientNet-B0 (Fold 5) - Proprietary Dataset

Appendix 23

ROC Curves - EfficientNet-B0 with HAM10000-Pretrained Weights on Proprietary Multispectral Dataset

This appendix presents one-vs-rest ROC curves for the HAM10000-pretrained EfficientNet-B0 model across all five cross-validation folds on the proprietary multispectral dataset (Section 4.4.3).

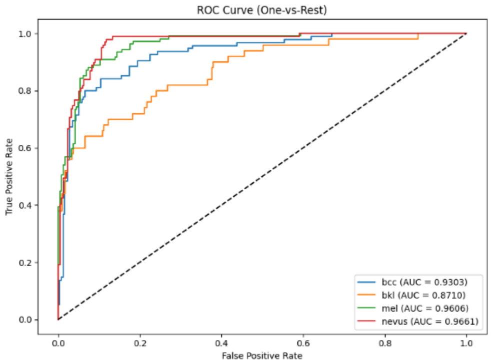

Fig. A.23.1. One-vs-Rest ROC Curves for EfficientNet-B0 with HAM10000-Pretrained Weights (Fold 1) - Proprietary Dataset

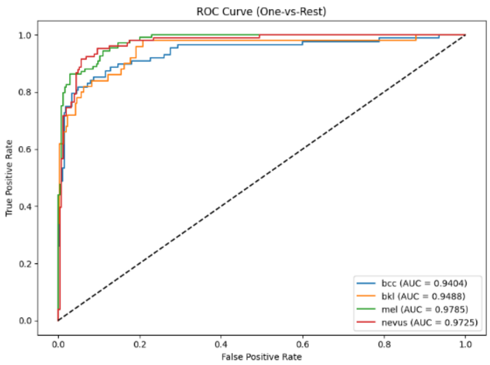

Fig. A.23.2. One-vs-Rest ROC Curves for EfficientNet-B0 with HAM10000-Pretrained Weights (Fold 2) - Proprietary Dataset

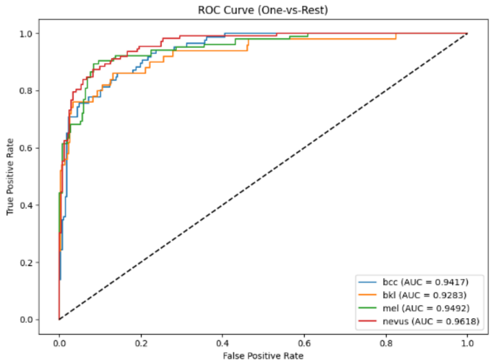

Fig. A.23.3. One-vs-Rest ROC Curves for EfficientNet-B0 with HAM10000-Pretrained Weights (Fold 3) - Proprietary Dataset

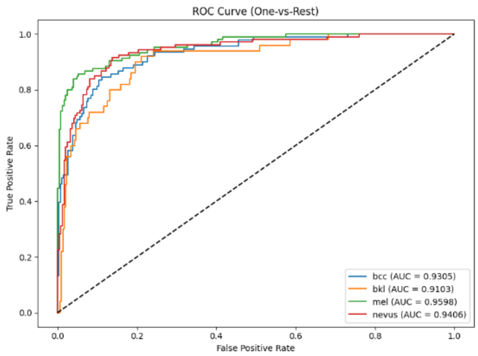

Fig. A.23.4. One-vs-Rest ROC Curves for EfficientNet-B0 with HAM10000-Pretrained Weights (Fold 4) - Proprietary Dataset

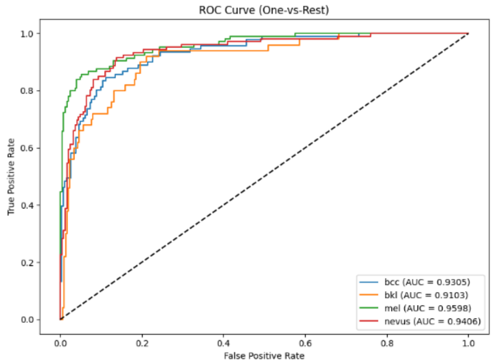

Fig. A.23.5. One-vs-Rest ROC Curves for EfficientNet-B0 with HAM10000-Pretrained Weights (Fold 5) - Proprietary Dataset

Appendix 24

Per-Class Classification Report for EfficientNet-B0 Initialized with ImageNet1K Weights on Proprietary Dataset

This appendix presents detailed per-class precision, recall, and F1-score metrics for the ImageNet1K-only pretrained baseline model across all five cross-validation folds on the proprietary multispectral dataset (Section 4.4.4).

Fold 1 – Per-class metrics:

	precision	recall	f1-score	support
bcc	0.7745	0.8316	0.8020	95
bkl	0.8125	0.5200	0.6341	50
mel	0.8989	0.7339	0.8081	109
nevus	0.7154	0.9394	0.8122	99
accuracy			0.7875	353
macro avg	0.8003	0.7562	0.7641	353
weighted avg	0.8017	0.7875	0.7830	353

Fig. A.24.1. Per-Class Classification Report for EfficientNet-B0 Initialized with ImageNet1K Weights on Proprietary Dataset (Fold 1)

Fold 2 – Per-class metrics:

	precision	recall	f1-score	support
bcc	0.8462	0.7500	0.7952	88
bkl	0.7917	0.7600	0.7755	50
mel	0.8857	0.8532	0.8692	109
nevus	0.7951	0.9151	0.8509	106
accuracy			0.8329	353
macro avg	0.8297	0.8196	0.8227	353
weighted avg	0.8353	0.8329	0.8320	353

Fig. A.24.2. Per-Class Classification Report for EfficientNet-B0 Initialized with ImageNet1K Weights on Proprietary Dataset (Fold 2)

Fold 3 – Per-class metrics:

	precision	recall	f1-score	support
bcc	0.8378	0.7209	0.7750	86
bkl	0.6557	0.8000	0.7207	50
mel	0.7826	0.8654	0.8219	104
nevus	0.8333	0.7589	0.7944	112
accuracy			0.7869	352
macro avg	0.7774	0.7863	0.7780	352
weighted avg	0.7942	0.7869	0.7873	352

Fig. A.24.3. Per-Class Classification Report for EfficientNet-B0 Initialized with ImageNet1K Weights on Proprietary Dataset (Fold 3)

Fold 4 – Per-class metrics:

	precision	recall	f1-score	support
bcc	0.7857	0.7416	0.7630	89
bkl	0.7000	0.5600	0.6222	50
mel	0.8598	0.8364	0.8479	110
nevus	0.7603	0.8932	0.8214	103
accuracy			0.7898	352
macro avg	0.7765	0.7578	0.7636	352
weighted avg	0.7893	0.7898	0.7866	352

Fig. A.24.4. Per-Class Classification Report for EfficientNet-B0 Initialized with ImageNet1K Weights on Proprietary Dataset (Fold 4)

Fold 5 – Per-class metrics:

	precision	recall	f1-score	support
bcc	0.6636	0.7802	0.7172	91
bkl	0.6087	0.5600	0.5833	50
mel	0.8936	0.8000	0.8442	105
nevus	0.8000	0.7925	0.7962	106
accuracy			0.7585	352
macro avg	0.7415	0.7332	0.7352	352
weighted avg	0.7655	0.7585	0.7599	352

Fig. A.24.5. Per-Class Classification Report for EfficientNet-B0 Initialized with ImageNet1K Weights on Proprietary Dataset (Fold 5)

Appendix 25

Per-Class Classification Reports - EfficientNet-B0 with HAM10000-Pretrained Weights on Proprietary Multispectral Dataset

This appendix presents detailed per-class precision, recall, and F1-score metrics for the EfficientNet-B0 model initialized with HAM10000-pretrained weights across all five cross-validation folds on the proprietary multispectral dataset (Section 4.4.4).

Fold 1 – Per-class metrics:

	precision	recall	f1-score	support
bcc	0.8000	0.8000	0.8000	95
bkl	0.8125	0.5200	0.6341	50
mel	0.8725	0.8165	0.8436	109
nevus	0.7661	0.9596	0.8520	99
accuracy			0.8102	353
macro avg	0.8128	0.7740	0.7824	353
weighted avg	0.8147	0.8102	0.8046	353

Fig. A.25.1. Per-Class Classification Report for EfficientNet-B0 with HAM10000-Pretrained Weights on Proprietary Dataset (Fold 1)

Fold 2 – Per-class metrics:

	precision	recall	f1-score	support
bcc	0.8642	0.7955	0.8284	88
bkl	0.7407	0.8000	0.7692	50
mel	0.9293	0.8440	0.8846	109
nevus	0.8403	0.9434	0.8889	106
accuracy			0.8555	353
macro avg	0.8436	0.8457	0.8428	353
weighted avg	0.8596	0.8555	0.8555	353

Fig. A.25.2. Per-Class Classification Report for EfficientNet-B0 with HAM10000-Pretrained Weights on Proprietary Dataset (Fold 2)

Fold 3 — Per-class metrics:

	precision	recall	f1-score	support
bcc	0.8971	0.7093	0.7922	86
bkl	0.7600	0.7600	0.7600	50
mel	0.8230	0.8942	0.8571	104
nevus	0.8182	0.8839	0.8498	112
accuracy			0.8267	352
macro avg	0.8246	0.8119	0.8148	352
weighted avg	0.8306	0.8267	0.8251	352

Fig. A.25.3. Per-Class Classification Report for EfficientNet-B0 with HAM10000-Pretrained Weights on Proprietary Dataset (Fold 3)

Fold 4 — Per-class metrics:

	precision	recall	f1-score	support
bcc	0.7529	0.7191	0.7356	89
bkl	0.7021	0.6600	0.6804	50
mel	0.8627	0.8000	0.8302	110
nevus	0.7712	0.8835	0.8235	103
accuracy			0.7841	352
macro avg	0.7723	0.7656	0.7674	352
weighted avg	0.7854	0.7841	0.7831	352

Fig. A.25.4. Per-Class Classification Report for EfficientNet-B0 with HAM10000-Pretrained Weights on Proprietary Dataset (Fold 4)

Fold 5 — Per-class metrics:

	precision	recall	f1-score	support
bcc	0.7667	0.7582	0.7624	91
bkl	0.7021	0.6600	0.6804	50
mel	0.8889	0.8381	0.8627	105
nevus	0.7845	0.8585	0.8198	106
accuracy			0.7983	352
macro avg	0.7855	0.7787	0.7814	352
weighted avg	0.7993	0.7983	0.7980	352

Fig. A.25.5. Per-Class Classification Report for EfficientNet-B0 with HAM10000-Pretrained Weights on Proprietary Dataset (Fold 5)

Appendix 26

CODE AVAILABILITY

The source code for this thesis is available at: https://github.com/Tramdanglamduoc/skin-lesion-thesis

The repository contains everything used in this work:

- The HAM10000 pretraining stage, including both MultiModalSkinNet (image + metadata) and ImageOnlySkinNet (image only).
- The transfer learning stage on the proprietary multispectral dataset, covering both initialization strategies: ImageNet1K-only weights and HAM10000-pretrained weights.
- Data preprocessing, fold splitting, training, and evaluation scripts for both stages.

Regarding the datasets: the HAM10000 dataset is public and can be downloaded from the Harvard Dataverse or Kaggle (links in Section 2.1).

The proprietary multispectral dataset is not shared, since it is clinical data and access is restricted. The code is still provided in full, so the pipeline can be inspected and reused on any equivalent multispectral acquisition.